Eidgenössische Technische Hochschule Zürich

Analysis II

Lectures by J. Serra – FS26

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Chapter 1

Metric spaces

1.1 Euclidean space $\mathbb{R}^n$

We all know the set $\mathbb{R}^n = \{x = (x_1, \ldots, x_n) \mid x_i \in \mathbb{R}\}$, all n-tuples of real numbers. We want a vector space structure on this

$$x + y = (x_1 + y_1, \ldots, x_n + y_n), \quad \lambda x = (\lambda x_1, \ldots, \lambda x_n),$$

but with a notion on distance.

Definition 1.1: Euclidean Structure of $\mathbb{R}^n$

Given $x, y \in \mathbb{R}^n$, we define the *scalar product*

$$x \cdot y = \langle x, y \rangle := \sum_{i=1}^{n} x_i y_i,$$

the *Euclidean norm*

$$\|x\| := \sqrt{x_1^2 + \cdots + x_n^2}$$

and *Euclidean distance*

$$d_{\text{Euclidean}}(x, y) := \|x - y\|.$$

Lemma 1.2: Cauchy-Schwartz

For all $x, y \in \mathbb{R}^n$, $x \cdot y \leq \|x\| \|y\|$.

Proof. If either $x = 0$ or $y = 0$, both sides are zero and the inequality is true. So assume wlog both $x, y \neq 0$. Let $\lambda > 0$ and using $2ab \leq a^2 + b^2$

$$2x \cdot y = 2 \sum_{i=1}^{n} \lambda x_i \frac{y_i}{\lambda} \leq \sum_{i=1}^{n} \lambda^2 x_i^2 + \frac{y_i^2}{\lambda} = \lambda^2 \|x\|^2 + \frac{1}{\lambda^2} \|y\|^2.$$

Take $\lambda = \dfrac{\|y\|}{\|x\|}$ and the equation becomes

$$\lambda^2 \|x\|^2 + \frac{1}{\lambda^2} \|y\|^2 = 2 \|x\| \|y\|,$$

which proves the inequality. $\square$

Lemma 1.3: Triangle inequality

For all $x, y, z \in \mathbb{R}^n$,

$$\|x - z\| \leq \|x - y\| + \|y - z\|.$$

Proof. We want to prove

$$\forall x, y \in \mathbb{R}^n : \|x + y\| \leq \|x\| + \|y\|.$$

This is equivalent by substitution. We square this to get

$$\|x + y\|^2 \leq \|x\|^2 + \|y\|^2 + 2\|x\|\|y\|.$$

But the LHS computes to

$$\sum_{i=1}^{n} (x_i + y_i)^2 = \sum_{i=1}^{n} x_i^2 + y_i^2 + 2x_i y_i = \|x\|^2 + \|y\|^2 + 2x \cdot y.$$

And the statement is equivalent to $x \cdot y \leq \|x\|\|y\|$, which holds by the Cauchy-Schwartz, Lemma 1.2. ☐

1.2 Definition of metric space

We will work with just three axioms for distance, staying general until we later apply what we learned to $\mathbb{R}^n$ specifically.

Definition 1.4: Metric space

A metric space is a pair (X, d), where X is a set and the *distance* $d : X \times X \to [0, \infty)$ satisfies

(1) $\forall x, y \in X : d(x, y) = 0 \iff x = y$.
(2) $\forall x, y \in X : d(x, y) = d(y, x)$.
(3) $\forall x, y, z \in X : d(x, z) \leq d(x, y) + d(y, z)$.

Example 1.5

$(\mathbb{R}^n, d_{\text{Euclidean}})$ is a metric space.
But for $\mathbb{R}^2$, there is also the distance $d(x, y) = d((x_1, x_2), (y_1, y_2)) = |x_1 - y_1| + |x_2 - y_2|$.
For a metric (X, d) and $Y \subset X$, (Y, d) is also a metric space.
On the sphere $X = \left\{x \in \mathbb{R}^3 \mid \|x\| = 1\right\}$, you can also either define the distance as the Euclidean distance or by going on a circular arc from one point to another. This also demonstrates that the most common metric isn't always the most natural/useful pick.
Let $X = C_0([a, b])$, the space of all real-valued continuos functions on $[a, b]$. Then

$$d_1(f, g) = \max_{[a,b]} |f - g|, \quad d_2(f, g) = \left(\int_a^b (f - g)^2 \, dx\right)^{\frac{1}{2}}$$

are both metrics.

1.3 Sequences in metric spaces

Like in the real numbers, we can define sequences in all metric spaces. And in fact, we need to understand sequences in real numbers to define sequences in metric spaces.

Definition 1.6: Sequence

For a set X, we call a sequence in X a map $x : \mathbb{N} \to X$ and write $x_n = x(n)$. For the entire sequence, we write $(x_n)_n^\infty \subset X$.

Definition 1.7: Convergence and limit

Let $(x_n)_n^\infty$ be a sequence in a metric space (X, d). We say $(x_n)_n^\infty$ converges to $x \in X$ if $d(x_n, x) \to 0$ as a sequence of real numbers.
Equivalently, $\forall \varepsilon > 0\, \exists N \in \mathbb{N}$ s.t. $\forall n \geq N : d(x_n, x) < \varepsilon$.
We write $\lim\limits_{n\to\infty} x_n = x$, or $x_n \to x$.

Lemma 1.8: Limit is unique

Let (X, d) be a metric space and $(x_n)_n^\infty \subset X$ a sequence with $x_n \to x, x_n \to y$. Then $x = y$.

Proof. Trivial. □

Definition 1.9: Subsequence

Let $(x_n)_n^\infty$ be a sequence in X. We define a subsequence as any sequence of the form $\left(x_{f(k)}\right)_k^\infty$ where $f : \mathbb{N} \to \mathbb{N}$ is increasing. We write $f(k) = n_k$, so the subsequence is denoted $(x_{n_k})_k^\infty$.

Definition 1.10: Accumulation point

Let (X, d) be a metric space.
(1) For a subset $Y \subset X$, we say that $y \in X$ is an accumulation point of Y if $\exists (y_n)_n^\infty \subset Y$ such that $y_n \to y$.
(2) For a sequence $(x_n)_n^\infty \subset X$, we say that $x \in X$ is an accumulation point if $\exists$ a subsequence $x_{n_k} \to x$.

Lemma 1.11: Subsequences and limit

A sequence $(x_n)_n \subset X$, for (X, d) metric space, converges to x if and only if every subsequence $(x_{n_k})_k^\infty$ converges to x.

Proof. $\Longleftarrow$: If every subsequence converges to x, since $(x_n)_n^\infty$ is also a subsequence, the limit is x.

$\Longrightarrow$: Let $(x_n)_n^\infty \to x$ and $(x_{n_k})_k^\infty$ any subsequence. For $\varepsilon > 0$ find $N \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon$ for $n \geq N$, then since $n_k \geq k$, we get $d(x_{n_k}, 0) < \varepsilon$, so $x_{n_k} \to x$ as well. □

Lemma 1.12: Subsequence and limit 2

Under the same assumptions, $x_n \to x$ if and only if every subsequence $(x_{n_k})_k^\infty$ has a sub-subsequence $x_{n_{k_\ell}} \to x$.

Proof. Exercise. □

Definition 1.13: Cauchy sequence

In a metric space, $(x_n)_n^\infty$ is Cauchy if

$$\forall \varepsilon > 0\, \exists N \in \mathbb{N} \text{ s.t. } \forall n, m \geq N : d(x_n, x_m) < \varepsilon.$$

Definition 1.14: Completeness

(X, d) is complete if every Cauchy sequence in it converges.

Example 1.15

Consider $(\mathbb{R}, d_{\text{Euclidean}})$. The space $A = (0, \infty)$ is *not* complete, but $A = [a, b]$ is. $\mathbb{Q} \subset \mathbb{R}$ is not complete, because you can have a sequence going to $\sqrt{2}$, which is not in $\mathbb{Q}$.

Remark 1.16

There is a conflict of notation in the following proofs, because for a sequence $(x_n)_n^\infty \subset \mathbb{R}^n$, x_n could either mean $x(n)$ or the n-th component of the vector x. So in the following two proofs, m will be the index in the sequence and i, j the coordinate index. And then $x_{m,i}$ means the i-th component of $x(m)$.

Lemma 1.17: Coordinate wise convergence

Let $(x_m)_m^\infty$ be a sequence. Then $x_m \to x$ if and only if $x_{m,i} \to x_i$ for every $1 \leq i \leq n$.

Proof. $\implies$: Assume $x_m \to x$. This means for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $\|x_m - x\| < \varepsilon$ if $m \geq N$. Given $1 \leq i \leq n$, notice

$$|x_{m,i} - x_i| \leq \sqrt{\sum_{j=1}^{n} |x_{m,j} - x_j|^2} = \|x_m - x\| < \varepsilon.$$

This way, $x_{m,i} \to x_i$.

$\impliedby$: Let $\varepsilon > 0$. By assumption for all $1 \leq i \leq n$, $\exists N_i \in \mathbb{N}$ s.t. $\forall m \geq N_i : |x_{m,i} - x_i| < \frac{\varepsilon}{\sqrt{n}}$. If we define $N = \max_{1 \leq i \leq n} N_i$, then for $m \geq N$,

$$\|x_m - x\| = \sqrt{\sum_{j=1}^{n} |x_{m,i} - x_i|^2} \leq \sqrt{n \frac{\varepsilon^2}{n}} = \varepsilon.$$

This way, $x_m \to x$ as a whole. $\square$

Theorem 1.18: Euclidean $\mathbb{R}^n$ is complete

The metric space $(\mathbb{R}^n, d_{\text{Euclidean}})$ with the Euclidean metric 1.1 is complete.

Proof. So let $(x_n)_n^\infty$ be a Cauchy sequence in $\mathbb{R}^n$. Because $|x_i - y_i| \leq \|x - y\|$ for $x, y \in \mathbb{R}^n$, we get that all the component-wise sequence are Cauchy. But those are in $\mathbb{R}$, so they converge. By 1.17, $x_n \to x$. $\square$

1.4 Open and closed sets

Continuing our quest to generalise everything, we extend the definition of open and closed sets in $\mathbb{R}$ to any metric space (X, d).

DEFINITION 1.19: OPEN BALL

Let (X, d) be a metric space. Define the open ball of radius $r > 0$ centered at $x \in X$ as

$$B_r(x) = B(x, r) = \{y \in X \mid d(y, x) < r\}.$$

DEFINITION 1.20: OPEN AND CLOSED SETS

For (X, d) a metric space, we say

- $U \subset X$ is *open* if $\forall x \in U\ \exists r > 0$ s.t. $B(x, r) \subset U$.
- $A \subset X$ is *closed* if $X \setminus A$ is open.

The collection of all open sets $\mathcal{T}_d = \{U \subset X \mid U \text{ is open}\}$ is called the *topology*.

LEMMA 1.21: UNION AND INTERSECTION OF OPEN SETS IS OPEN

Let (X, d) be a metric space and $\{U_i\}_{i \in I}$ a family of open sets in X.

- Arbitrary unions of open sets are open. So $\bigcup_{i \in I} U_i$ is open.
- Finite intersections of open sets are open. So if I is finite, $\bigcap_{i \in I} U_i$ is open.

Proof. Set

$$U = \bigcup_{i \in I} U_i.$$

If $x \in U$, Then there exists $i \in I$ with $x \in U_i$ and since U_i is open, there exists an $r > 0$ such that $B(x, r) \subset U_i$, so also contained in U. Thus, U is open.

For the second one, consider that if x is in all the open sets U_i, then for each i, there exists a small enough radius r_i such that $B(r_i, x) \subset U_i$. Then, out of all r_i's, pick the smallest one, call it r and $B(r, x)$ is contained in all the U_i, so in U. $\square$

LEMMA 1.22: UNION AND INTERSECTION OF CLOSED SETS

Let (X, d) be a metric space and $\{A_i\}_{i \in I}$ a family of closed sets.

- Arbitrary intersections of closed sets are closed. So $\bigcup_{i \in I} A_i$ is closed.
- Finite unions of closed sets are closed. So if I is finite, $\bigcap_{i \in I} A_i$ is closed.

Proof. Consider that

$$X \setminus \bigcup_{i \in I} A_i = \bigcap_{i \in I} (X \setminus A_i),$$
$$X \setminus \bigcap_{i \in I} A_i = \bigcup_{i \in I} (X \setminus A_i).$$

Using this, one applies Lemma 1.21 to get the statement. $\square$

EXAMPLE 1.23

The intersection of infinitely many open sets might not be open. In particular, in $(\mathbb{R}, d_{|\cdot|})$, one gets for $U_k = \left(-\frac{1}{k}, \frac{1}{k}\right)$ that the intersection over all $k \in \mathbb{N}$ is $\{0\}$, which is not open. Passing this into the complement also gives a counterexample to "infinitely many unions of closed sets are closed.

Definition 1.24: Interior, closure and boundary

Given $\Omega \subset X$, for (X, d) a metric space, we define:

- The interior $\Omega^\circ = \bigcup\{U \subset \Omega \mid U \text{ is open}\}$.
- The closure $\overline{\Omega} = \{x \in X \mid \exists (x_n)_n^\infty \subset \Omega \text{ s.t. } x_n \to x\} = \bigcap\{A \supset \Omega \mid A \text{ is closed}\}$[a].
- The (topological) boundary $\partial\Omega := \overline{\Omega} \setminus \Omega^\circ$.

One shows $\Omega^\circ, \partial\Omega$ are closed, while $\overline{\Omega}$ is open using 1.22 and 1.21.

[a] Provable with 1.27. Exercise.

Example 1.25

The set $[0,1) \times [0,1) \subset \mathbb{R}^2$ will have for instance $(0,0)$ in the closure, but also $(1,1)$. It will not contain $(0,0)$ in the interior.

Remark 1.26

We say a proposition $\mathcal{P}(x_n)$ holds "eventually" for a sequence $(x_n)_n^\infty$, when there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $\mathcal{P}(x_n)$ holds. We will use this terminology a lot, so it's good to get accustomed to it. Sometimes, we also say "for n large enough".

Lemma 1.27: Open and closed sets through sequences

Let (X, d) be a metric space.

- A subset $U \subset X$ is open if and only if for every sequence $(x_n)_n^\infty \subset X$ with $x_n \to x \in U$, x_n lies eventually in U.
- A subset $A \subset X$ is closed if and only if for every sequence $(x_n)_n^\infty \subset A : x_n \to x \in X \implies x \in A$.

Proof. First, we prove the statement about open sets.

$\implies$: Take $x \in U$. Since U is open, there exists $r > 0$ such that $B(x,r) \subset U$. Since $x_n \to x$, $\exists N$ s.t. $d(x_n, x) < r$, so $x_n \in B(r, x)$ for all $n \geq N$.

$\impliedby$: We will prove this by contraposition. Assume that U is not open. This means $\exists x \in U$ such that $\forall r > 0$, $B(x,r) \not\subset U$. This means there exists $x_r \in B(x,r) \cap (X \setminus U)$ for any $r > 0$. Taking $r = \frac{1}{n}$, we can produce a sequence of points $x_n \to x$, with none of the x_n in U.

Now we prove the one about closed sets.

$\implies$: Assume A is closed, so $X \setminus A$ is open. Assume that $x_n \to x$, but $x \in X \setminus A$. By what we just proved, x_n must eventually lie in $X \setminus A$, a contradiction to $(x_n)_n^\infty \subset A$.

$\impliedby$: By contraposition. Assume A is not closed. This means $X \setminus A$ is not open. By the statement about open sets we just proved, there exists a sequence $(x_n)_n^\infty \subset X$ with $x_n \to x \in X \setminus A$ that never lies in $X \setminus A$, so it lies entirely in A. $\square$

Exercise. Prove that if (X, d) is complete and $A \subset X$ is closed, then (A, d) is complete.

1.5 Continuity

This time we generalise for continuity. So we will consider maps $f : X \to Y$ for metric spaces (X, d_X) and (Y, d_Y), which might not be the same. This is a significant difference to the old continuity, which was defined between two sets that are both the reals, hence share the same metric.

DEFINITION 1.28: CONTINUITY

Let (X, d) be a metric space. Consider a function $f : X \to Y$. We say f is continuous if one of the following 3 equivalent properties hold.

(1) $\forall x \in X\ \forall \varepsilon > 0\ \exists \delta > 0$ s.t. $f(B_\delta(x)) \subset B_\varepsilon(f(x))$.

(2) For $(x_n)_n^\infty \subset X$ with $x_n \to x$, $(f(x_n))_n^\infty$ is a convergent sequence in Y with limit $f(x_n)$.

(3) $\forall V \subset Y$ open, $f^{-1}(V) \subset X$ is open.

We call them $\varepsilon - \delta$-continuity (1), sequential continuity (2) and topological continuity (3).

PROPOSITION 1.29: EQUIVALENT DEFINITIONS OF CONTINUITY

The three definitions of continuity are equivalent.

Proof. We will prove an implication cycle.

(1) $\implies$ (2): Assume $f : X \to Y$ is $\varepsilon - \delta$ continuous and let $x_n \to x$. Let $\varepsilon > 0$. I want to prove $d(f(x_n), f(x)) < \varepsilon$ eventually, or in other words $f(x_n) \in B_\varepsilon(f(x))$ eventually. By $\varepsilon - \delta$ continuity, there exists $\delta > 0$ such that $f(B_\delta(x)) \subset B_\varepsilon(f(x))$. But $x_n \in B_\delta(x)$ eventually, so $f(x_n) \in B_\varepsilon(f(x))$ eventually.

$\neg$(3) $\implies$ $\neg$(2): By $\neg$(3), we have $\exists V \subset Y$ open such that $f^{-1}(V)$ not open. By Lemma 1.27, $\exists x \in f^{-1}(V)$ such that there is a sequence $x_n \to x$ with $x_n \in X \setminus f^{-1}(V)$ for all n. In particular, $f(x_n) \in Y \setminus V$, but $f(x) \in V$. Because V is open, $f(x_n) \not\to f(x)$, by Lemma 1.27.

(3) $\implies$ (1): Let $x \in X$ and $\varepsilon > 0$. The set $V = B_\varepsilon(f(x))$ is open, so $f^{-1}(V)$ is also open. We have of course $x \in f^{-1}(V)$. In particular, there exists $\delta > 0$ such that $B_\delta(x) \subset f^{-1}(V)$. But then, for $x' \in B_\delta(x)$, we have $f(x') \in V$, so $f(x') \in B_\varepsilon(f(x))$. This proves $f(B_\delta(x)) \subset B_\varepsilon(f(x))$. $\square$

DEFINITION 1.30: UNIFORM / LIPSCHITZ CONTINUITY

Let $f : X \to Y$ be a map between metric spaces.

(1) f is uniformly continuous if $\forall \varepsilon > 0\ \exists \delta > 0$ s.t. $\forall x \in X : f(B_\delta(x)) \subset B_\varepsilon(f(x))$.

(2) f is L-Lipschitz for $L > 0$ if $\forall x, y \in X : d_Y(f(x), f(y)) \leq L \cdot d_X(x, y)$.

EXERCISE. Prove that any Lipschitz function is also uniformly continuous, hence also continuous.

EXAMPLE 1.31

The distance is a continuous map: Let (X, d) be a metric space and $x_0 \in X$. Define $f(x) = d(x, x_0)$, mapping to the space $Y = [0, \infty)$ with the Euclidean distance (from now on implied, unless specified otherwise). Then f is 1-Lipschitz. Indeed,

$$d_Y(f(x), f(y)) = |f(x) - f(y)| = |d(x, x_0) - d(y, x_0)| \leq d(x, y),$$

using triangle inequality.

There is this very useful and fundamental theorem about Lipschitz maps in complete spaces.

THEOREM 1.32: BANACH FIXED POINT THEOREM

Let (X, d) be a complete metric space and $T : X \to X$ be a λ-Lipschitz map for $\lambda \in (0, 1)$. Then T has a unique fixed point $x \in X$ such that $T(x) = x$.

Proof. Fix $x_0 \in X$. Define $x_{n+1} = T(x_n)$. Then,

$$d(x_{n+1}, x_n) = d(T(x_n), T(x_{n-1})) \leq \lambda d(x_n, x_{n-1}),$$

using that T is λ-Lipschitz. We claim that $(x_n)_n^\infty$ is Cauchy. We have

$$d(x_{n+1}, x_n) \leq \lambda d(x_n, x_{n-1}) \leq \lambda^2 d(x_{n-1}, x_{n-2}) \leq \cdots \leq \lambda^n d(x_1, x_0).$$

Hence for wlog $m < n$,

$$d(x_n, x_m) \leq \sum_{k=m}^{n-1} d(x_{k+1}, x_k) \leq \sum_{k=m}^{n-1} \lambda^k d(x_1, x_0) = d(x_1, x_0)\lambda^m \frac{1}{1-\lambda}.$$

As $\lambda^m \to 0$ with $m \to \infty$, the sequence is Cauchy and converges to $x = \lim_{n\to\infty} x_n$, because X is complete. For this point, we thus have, using sequential continuity,

$$T(x) = \lim_{n\to\infty} T(x_n) = \lim_{n\to\infty} x_{n+1} = \lim_{n\to\infty} x_n = x.$$

If x, y are two fixed points, then

$$d(x, y) = d(T(x), T(y)) \leq \lambda d(x, y)),$$

which is possible only if $d(x, y) = 0$, so $x = y$. $\square$

1.6 Compactness

Compact is *not* just bounded and closed, like in $\mathbb{R}$.

Definition 1.33: Cover and subcover

Given a set X and $E \subset X$, we say $\mathcal{U} = \{U_i\}_{i\in I}$, a family of subsets of X, is a cover of E if

$$E \subset \bigcup \mathcal{U} = \bigcup_{i\in I} U_i.$$

If $\mathcal{V} \subset \mathcal{U}$ is still a cover, we call $\mathcal{V}$ a subcover. If $\mathcal{U}$ is a collection of open sets, we call it an open cover.

Definition 1.34: Compactness

Let (X, d) be a metric space. A set $K \subset X$ is called

(1) sequentially compact if $\forall (x_n)_n^\infty \subset K$, $\exists$ a subsequence s.t. $\lim_{k\to\infty} x_{n_k} \in K$.

(2) topologically compact if $\forall$ open covers $\mathcal{U}$ of K, $\exists$ a finite subcover.

Example 1.35

In $\mathbb{R}$, the intervals $[a, b]$ are compact by Bolzano-Weierstrass, while $\mathbb{Q} \cap [0, 1]$ is for instance not compact.
Consider $X = (0, 1) \cap \mathbb{Q}$. Enumerate $\mathbb{Q} \subset \{x_n \mid n \geq 0\}$. If we let the cover $\mathcal{U} = \left\{B_{2^{-n-10^3}}(x_n)\right\}$, then the total "length" of the intervals in this cover will be much less than 1. But this cover by definition includes all rationals in $(0, 1)$, but it doesn't even cover the length of the interval $(0, 1)$.

PROPOSITION 1.36: COMPACTNESS

The two definitions of compactness are equivalent.

Proof. (1) $\implies$ (2): Assume $K \subset X$ is sequentially compact. Let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover. This means $\forall x \in K \ \exists U_i$ open s.t. $x \in U_i$. For $x \in K$, we let

$$r(x) = \min \Big\{ \sup \{r > 0 \mid B_r(x) \subset U_j \text{ for some } U_j \in \mathcal{U}\}, 1 \Big\}.$$

Using this, given $x \in K$, select $U_{i(x)} \in \mathcal{U}$ s.t.

$$B_{r(x)/2}(x) \subset U_{i(x)}.$$

In a way, this is an open set containing a (half) maximally big neighborhood of x. With this in mind, we can construct a finite subcover for K.

Pick any $x_0 \in K$. Define

$$\mathcal{V} = \{U_0, U_1, \dots\},$$

by setting $U_0 = U_{i(x_0)}$ and always picking x_n so that it's in none of the previous open sets $U_0, \dots, U_{n-1}$, and letting $U_n = U_{i(x_n)}$. If this is not possible, then we're done. We will prove that assuming this goes on forever leads to a contradiction.

Formally, unless this stops, we'll produce a sequence $(x_n)_n^\infty$ with

$$x_n \in K \setminus \bigcup_{k=0}^{n-1} U_{i(x_k)}.$$

By sequential compactness, $(x_n)_n^\infty$ has a converging subsequence $(x_{n_\ell})_\ell^\infty$ with $x = \lim_{\ell \to \infty} x_{n_\ell} \in K$. Note that $x \notin U_{n_\ell}$ for all ℓ, because the complement of the union of all the open sets U that came before U_{n_ℓ} is closed (see Lemma 1.22 and 1.27) and the sequence elements that come afterward and the limit must hence remain in the complement. In particular, $r(x_{n_\ell}) \to 0$, otherwise x would have to lie in one of the U_{n_ℓ} at some point.

Since $B_{r(x)/2}(x)$ is open, x_{n_ℓ} must eventually lie inside it as $x_{n_\ell} \to x$. And since $r(x_{n_\ell}) \to 0$, it must eventually contain

$$B_{2r(x_{n_\ell})}(x_{n_\ell}) \subset B_{r(x)/2}(x) \subset U_{i(x)},$$

which is impossible, because this would imply $2r(x_{n_\ell})(x) \leq r(x_{n_\ell})$, as $r(x_{n_\ell})$ was supposed to be the supremum of all the radii such that $B_r(x)$ is contained in any open set in the cover.

(2) $\implies$ (1): Let $(x_n)_n^\infty \subset K$ be a sequence. We need to show that there exists a convergent subsequence. So assume by contradiction that $\forall x \in K$, x is not an accumulation point of x_n. But this means for any $x \in K$ there exists $\varepsilon(x) > 0$, such that eventually, $x_n \in K \setminus B_{\varepsilon(x)}(x)$ (quick contraposition).

Define $\mathcal{U} = \big\{B_{\varepsilon(x)}(x) \mid x \in K\big\}$. This is an open cover of K, so it admits a finite subcover, such that

$$K \subset \bigcup_{j=1}^{n} B_{\varepsilon(y_j)}(y_j).$$

But then $(x_n)_n^\infty$ can only have finitely many terms, by the definition of $\varepsilon(x)$, a contradiction. $\square$

Corollary 1.37

The last proposition gives us the following corollaries:

(1) $K \subset X$ compact $\implies$ K closed.

(2) $K \subset X$ compact $\implies$ K complete.

(3) $K \subset X$, $A \subset X$ closed $\implies$ $K \cap A$ compact.

Proof. (1) is simple, we can use sequential closedness 1.27. If a sequence converges, then it must converge to the limit of any of its subsequences. (2) also follows since if a Cauchy sequence has a convergent subsequence, then it also converges as a whole, and converges into K, because it is closed by (1). (3) also follows from sequential compactness and the fact that the limit of the subsequence must be in A. $\square$

Proposition 1.38: Continuous image of a compact set is compact

Let (X, d_X) and (Y, d_Y) be metric spaces and $K \subset X$ be compact. If $f : X \to Y$ is continuous, then $f(K)$ is compact in Y.

Proof. The goal is to show that $f(K)$ is topologically compact, so for $\mathcal{V}$ an open cover of $f(K)$, I want to find a finite subcover. So let $\mathcal{U} = \left\{f^{-1}(V) \mid V \in \mathcal{V}\right\}$, which is an open (by continuity) cover of K. Extract a finite subcover of K and bring it back to the codomain. $\square$

Theorem 1.39: Extreme value theorem

Let (X, d) be a metric space and $f : K \to \mathbb{R}$ be continuous and $K \subset X$ compact. Then both $\sup\{f(x) \mid x \in K\}$ and $\inf\{f(x) \mid x \in K\}$ are attained.

Proof. We will prove it for the supremum only. By definition of the supremum $s = \sup_{x\in K}\{f(x)\}$, there exists a sequence $(x_n)_n^\infty \subset K$ such that $f(x_n) \to s$. By compactness, this has a convergent subsequence $x_{n_k} \to \bar{x}$. By sequential continuity, $f(\bar{x}) = s$. $\square$

Remark 1.40

We take a subset $K \subset \mathbb{R}^n$ to be bounded w.r.t. the Euclidean distance if there exists $N \in \mathbb{N}$ such that $\|x\| < N$ for all $x \in K$, equivalently $K \subset B_N(0)$.

Theorem 1.41: Heine-Borel

For the metric space $\mathbb{R}^n$ with the Euclidean distance, $K \subset \mathbb{R}^n$ is compact if and only if it is bounded and closed.

Proof. $\implies$: K is closed by Corollary 1.37. If K is unbounded, $\exists (x_n)_n^\infty \subset K$ such that $\|x_n\| \geq N$ for all $N \in \mathbb{N}$. Any subsequence will hence diverge. Indeed, $x_{n_k} \to x$ would imply $\|x_{n_k} - x\| \to 0$, and by triangle inequality, 1.3

$$\|x_{n_k}\| \leq \|x\| + \|x_{n_k} - x\|,$$

a contradiction to x_{n_k} being arbitrarily large.

$\impliedby$: It suffices to show that for $N \in \mathbb{N}$, $[-N, N]^n \subset \mathbb{R}^n$ is compact, assuming that $K \subset \mathbb{R}^n$ is closed and bounded by $B_N(0)$. This is because $K \subset B_N(0) \subset [-N, N]^n$, and a closed subset of a compact set is compact, so K would be compact, by Corollary 1.37.

The proof will have to use the Bolzano-Weierstrass Theorem in $\mathbb{R}$. Given a sequence $(x_k)_k^\infty$, we can write this component-wise

$$x_k = (x_{k,1}, \dots, x_{k,n}).$$

Consider the sequences $(x_{k,i})_{k=1}^\infty \subset [-N, N]$ for $1 \le i \le n$. By Bolzano–Weierstrass, there exists a strictly increasing sequence of indices $(k_m^{(1)})_{m=1}^\infty$ such that

$$x_{k_m^{(1)},1} \to \ell_1 \in [-N, N] \quad \text{as } m \to \infty.$$

Now consider the sequence $(x_{k_m^{(1)},2})_{m=1}^\infty \subset [-N, N]$. Again by Bolzano–Weierstrass, there exists a strictly increasing sequence $(k_m^{(2)})_{m=1}^\infty$ which is a subsequence of $(k_m^{(1)})_{m=1}^\infty$ such that

$$x_{k_m^{(2)},2} \to \ell_2 \in [-N, N] \quad \text{as } m \to \infty.$$

Since $(k_m^{(2)})$ is a subsequence of $(k_m^{(1)})$, we still have

$$x_{k_m^{(2)},1} \to \ell_1.$$

Proceeding inductively, for each $j = 1, \dots, n$ we obtain a strictly increasing sequence $(k_m^{(j)})_{m=1}^\infty$ which is a subsequence of $(k_m^{(j-1)})_{m=1}^\infty$ and such that

$$x_{k_m^{(j)},j} \to \ell_j \in [-N, N] \quad \text{as } m \to \infty,$$

and for all $i \le j$,

$$x_{k_m^{(j)},i} \to \ell_i.$$

After n steps we obtain $(k_m^{(n)})_{m=1}^\infty$ such that

$$x_{k_m^{(n)},i} \to \ell_i \quad \text{for all } i = 1, \dots, n.$$

Define $x := (\ell_1, \dots, \ell_n) \in [-N, N]^n$. Then $x_{k_m^{(n)}} \to x$ componentwise, hence $x_{k_m^{(n)}} \to x$ in $\mathbb{R}^n$ by Lemma 1.17. $\square$

Example 1.42

Consider $X = \mathbb{R}$ and $A = (0, 1]$. Why is this not compact? Because

$$A \subset \bigcup_{n=1}^{\infty} \left(\frac{1}{n}, 2 \right)$$

gives us a cover of A. But any finite cover of this will definitely miss some numbers of A.

Proposition 1.43: Uniform continuity

Let $f : X \to Y$ be a continuous map between metric spaces. If $K \subset X$ is compact, then $f|_K$ is uniformly continuous.

Proof. The goal is to show that given $\varepsilon > 0$, there exists $\delta > 0$ such that $\forall x \in K : f(B_\delta(x)) \subset B_\varepsilon(f(x))$. Using continuity, $\forall x \in K$, there exists $\delta(x) > 0$ such that $f(B_{\delta(x)}(x)) \subset B_{\varepsilon/2}(f(x))$. Define

$$\mathcal{U} = \left\{ B_{\delta(x)/2}(x) \mid x \in K \right\}.$$

This is an open cover of K. So it has a finite subcover. So

$$K \subset \bigcup_{i=1}^{n} B_{\delta(x_i)/2}(x_i)$$

and we define $\delta = \frac{1}{2} \min \{\delta(x_i)\}$.

If $x \in K$ and $y \in B_\delta(x)$, then $x \in B_{\delta(x_i)/2}(x_i)$ for some i. Consider that

$$d(x,y) < \delta \leq \frac{\delta(x_i)}{2}, \quad d(x_i,y) < \frac{\delta(x_i)}{2} + \frac{\delta(x_i)}{2} < \delta(x_i).$$

That means

$$d_Y(f(x), f(y)) \leq d_Y(f(x), f(x_i)) + d_Y(f(x_i), f(y)) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

$\square$

1.7 Connectedness

In any metric space, $X, \emptyset$ are always both open and closed (clopen). Are there other clopen sets and how many? Or, consider the following question. For the sphere $X = \mathbb{S}^2 = \left\{v \in \mathbb{R}^3 \mid \|v\| = 1\right\}$ and the Euclidean distance, is it possible to write $\mathbb{S} = U \cup V$ with $U \cap V = \emptyset$ such that U, V are both open and non-empty? Intuitively, it seems quite improbable. One can ask the same question for $(0,1) \subset \mathbb{R}$. We want to explore the concept of connected space in this section and end up answering these seemingly unrelated questions.

DEFINITION 1.44: CONNECTEDNESS

Let (X,d) be a metric space and $A \subset X$ a nonempty subset.
X is *disconnected* if $\exists U, V \subset A$ open, disjoint and nonempty such that

$$A \subset U \cup V \text{ and } A \cap U \neq \emptyset, A \cap V \neq \emptyset.$$

Any subset A that is not disconnected is called *connected*.

EXAMPLE 1.45

In $\mathbb{R}$, the set $A = (0,1) \cup (2,3)$ is immediately seen to be disconnected, as is $A = \{0,1\}$.

REMARK 1.46

A subset I of $\mathbb{R}$ is an interval if and only if $\forall x_1, x_2 \in I$ with $x_1 \leq x_2$, $[x_1, x_2] \subset I$.

PROPOSITION 1.47: CONNECTED SUBSETS OF $\mathbb{R}$

For the metric space $(\mathbb{R}, d_{\text{Eucl}})$, $E \subset \mathbb{R}$ is connected if and only if E is an interval.

Proof. $\implies$: By contraposition. Assume E is not an interval. That means there exist points $x_1, x_2 \in E$ and $y \in \mathbb{R} \setminus E$ such that $x_1 < y < x_2$. This gives us the disjoint open cover $(-\infty, y)$ and (y, ∞). Hence, E is disconnected.

$\implies$ Again, by contraposition. Assume E is disconnected. This means $\exists U_1, U_2$ open s.t. $E \subset U_1 \cup U_2$ and $U_1 \cap U_2 = \emptyset$. There exist $x_1 \in E \cap U_1$ and $x_2 \in E \cap U_2$. Assume wlog that $x_1 < x_2$. We define

$$y = \sup \{t \geq x_1 \mid [x_1, t) \subset U_1\} \in \mathbb{R}.$$

This exists because the set is clearly bounded by x_2. y is the supremum, hence the limit of points in $\mathbb{R} \setminus U_2$, which is closed, so itself in $\mathbb{R} \setminus U_2$. Furthermore, $y \in \mathbb{R} \setminus U_1$. Because if $y \in U_1$, there exists a ball $(y - \varepsilon, y + \varepsilon) \subset U_1$ for $\varepsilon > 0$, but then that would give us $[x_1, y + \varepsilon) \subset U_1$, a contradiction. As a consequence, $y \in \mathbb{R} \setminus (U_1 \cup U_2) \subset \mathbb{R} \setminus E$. This gives us x_1, x_2 and $y \in (x_1, x_2)$ with $y \notin E$. □

PROPOSITION 1.48: CONNECTEDNESS UNDER CONTINUOUS MAPS

Let $f : X \to Y$ be a continuous map between metric spaces and $E \subset X$ connected. Then $f(E)$ is connected.

Proof. By contraposition. Assume that $f(E)$ is disconnected. This means

$$\exists V_1, V_2 \subset Y \text{ open disjoint, with } f(E) \subset V_1 \cup V_2 \text{ and } V_i \cap f(E) \neq \emptyset.$$

We define $U_1 = f^{-1}(V_1)$ and $U_2 = f^{-1}(V_2)$. Then $E \subset f^{-1}(V_1) \cup f^{-1}(V_2) = U_1 \cup U_2$. Since the images were disjoint, we have

$$U_1 \cap U_2 = f^{-1}(V_1) \cap f^{-1}(V_2) = f^{-1}(V_1 \cap V_2) = f^{-1}(\emptyset) = \emptyset.$$

U_1, U_2 are also open since f is continuous and the intersection with E is also non-empty, so U_1, U_2 split E, so it is disconnected. □

COROLLARY 1.49: INTERMEDIATE VALUE THEOREM

Let (X, d) be a connected metric space and $f : X \to \mathbb{R}$ continuous, with $f(x) = a, f(y) = b$ and $a \leq b$ for some $x, y \in X$. Then $\exists z \in X$ s.t. $f(z) = c$ for all $c \in [a, b]$.

Proof. Because X is connected, $f(X)$ is connected. But in $\mathbb{R}$, by Proposition 1.48, the only connected sets are the intervals, immediately proving the corollary. □

DEFINITION 1.50: CURVE

Let (X, d) be a metric space. A curve or path in X is a map $\gamma : [0, 1] \to X$ that is continuous. We call $\gamma(0)$ the starting point and $\gamma(1)$ the end point.
A path with $\gamma(0) = \gamma(1)$ is called a closed curve, or a loop.

DEFINITION 1.51: PATH-CONNECTED

Let (X, d) be a metric space. A set $E \subset X$ is called path connected if $\forall x, y \in E$, there exists a path $\gamma : [0, 1] \to E$ joining x and y, i.e. $\gamma(0) = x$ and $\gamma(1) = y$.

PROPOSITION 1.52: PATH CONNECTED IMPLIES CONNECTED

Let (X, d) be a metric space and $E \subset X$. If E is path connected, then E is connected.

Proof. By contraposition. Assume E is disconnected, so there exist U_1, U_2 open, disjoint s.t. $E \subset U_1 \cup U_2$ and $\exists x_i \in U_i \cap E$. Assume further that there exists a path $\gamma : [0, 1] \to E$ joining x_1 and x_2. Define $V_i = \gamma^{-1}(U_i) \subset [0, 1]$. These are open disjoint nonempty ($0 \in V_1$ and $1 \in V_2$) and covering $[0, 1]$. Hence $[0, 1]$ is disconnected, which is impossible by 1.47. □

EXAMPLE 1.53: TOPOLIGSTS' CURVE

Consider $(\mathbb{R}^2, d_{\text{Eucl}})$. Define $E = (\{0\} \times [0,1]) \cup \{(t, \sin(1/t)) \mid t > 0\}$. One can prove this is connected as an exercise, but not path connected, creating sequences that converge to some point in $[0, y]$, which would make a path connection from $[0, y]$ to any point on the sine curve impossible. So connected does not imply path connected, at least not in general.

DEFINITION 1.54: COMPOSED AND REVERSED PATHS

Let $\gamma_1 : [0,1] \to X$ with $\gamma_1(0) = x$ and $\gamma_1(1) = y$ and $\gamma_2 : [0,1] \to X$ with $\gamma_2(0) = y$ and $\gamma_2(1) = z$. We define $\gamma_1^*(t) = \gamma_1(1-t)$, the reverse path. Since $1-t$ is continuous, γ_1^* is a continuous map as well and represents the reversion of γ_1.
We also define the composition

$$\gamma_3(t) = \begin{cases} \gamma_1(2t) & t \in \left[0, \frac{1}{2}\right] \\ \gamma_2(2t-1) & t \in \left[\frac{1}{2}, 1\right]. \end{cases}$$

This is also a curve (check that it's continuous, best with $\varepsilon - \delta$).

Finally, we are ready to show that in Euclidean space, path connected and connected mean the same thing, also revealing for instance that $\mathbb{S}^2$ is connected (because it is path connected), so can not be decomposed into open sets. And I'll throw in a remark about clopen sets. With this and the work leading up to this, all of the questions posted at the beginning of the section should be answered.

REMARK 1.55: CLOPEN SETS

It turns out that the answer to the question about clopen sets was right in front of us the whole time. Assume X is connected and let $A \subset X$ be clopen. Then $X \setminus A$ and A give a separation of X, so one of them must be empty. Conversely, if X is disconnected, then $X = U \cup V$ for $U, V \subset X$ a separation. But then $U \cap V = \emptyset$, so we're left with $V = X \setminus U$. But these sets are by assumption non-empty, giving a non-trivial clopen set V.

THEOREM 1.56: CONNECTED IFF PATH CONNECTED IN $\mathbb{R}^n$

In Euclidean $\mathbb{R}^n$, an open subset $U \subset \mathbb{R}^n$ is connected if and only if it is path connected.

Proof. One direction we already proved in Proposition 1.52. The goal is to fix an $x_0 \in U$, and then prove we can join it with any other point $x \in U$ by a path. Then, any two points $x, y \in U$ can be joined by composing the path from x to x_0 and the path from x_0 to y.

Define the set $G \subset U$, with

$$G = \{x \in U \mid \exists \text{ path } \gamma : [0,1] \to U \text{ with } \gamma(0) = x_0, \gamma(1) = x\}.$$

We will prove that (1) G is open and (2) $U \setminus G$ is open. But $x_0 \in G$, and since U is connected, $U \setminus G$ will have to be empty, leaving us with $G = U$.

The key observation is that, using openness of U, $\forall x \in U\ \exists B_r(x) \subset U$, and that for $y \in B_r(x)$, $y \in G \iff x \in G$. Why? Because the map $t \in [0,1] \mapsto (1-t)x + ty \in \mathbb{R}^n$ is continuous and lives only inside $B_r(x) \subset U$, as

$$\|(1-t)x + ty - x\| = \|t(y-x)\| = t\|y-x\| < r,$$

connecting y and x, hence if $y \in G$, then x can be connected to x_0 via y. And conversely, if $x \in G$, then $y \in G$ by the same argument.

But this just shows that if $x \in G$, then there exists a small enough ball $B_r(x) \subset G$, showing G is open, but also that if $x \notin G$, that there exists another small ball $B_r(x)$ that has *no* elements in G, so $B_r(x) \subset X \setminus G$. But this proves $X \setminus G$ is open and we're done. $\square$

1.8 Normed vector spaces

Having tackled metric spaces in general, we still have to consider vector spaces with a special operation, the norm. We need this because $\mathbb{R}^n$ with the Euclidean distance is exactly one such space. We consider normed vector spaces over $\mathbb{R}$, but one can also do it over $\mathbb{C}$ analogously.

Definition 1.57: Normed vector space

Let V be a vector space over $\mathbb{R}$. A map $\|\cdot\| : V \to [0, \infty)$ is called a norm if

(1) $\|v\| = 0 \iff v = 0$.

(2) $\forall v \in V, \alpha \in \mathbb{R} : \|\alpha v\| = |\alpha| \, \|v\|$.

(3) $\forall v, w \in V : \|v + w\| \leq \|v\| + \|w\|$.

Then, V is called a normed vector space (over $\mathbb{R}$).

Example 1.58

We have of course $\mathbb{R}^n$ with $\|\cdot\|_{\text{Euclid}}$, which we will denote with $|\cdot|$ from now on, because it is so common.

For $p \in \mathbb{N}$, we have the p-norm on $\mathbb{R}^n$,

$$\|v\|_p = \left(\sum_{i=1}^{n} |v_i|^p \right)^{\frac{1}{p}}.$$

One shows as an exercise that

$$\|v\|_\infty = \lim_{p \to \infty} \|v\|_p = \max_{1 \leq i \leq n} |v_i|$$

also gives a norm.

Definition 1.59: Hilbert-Schmidt norm

Consider the space $V = \operatorname{Hom}(\mathbb{R}^n, \mathbb{R}^m)$. For $M \in V$, we define

$$\|M\|_2 = \sqrt{\operatorname{tr}(M^* M)} = \sqrt{\sum_{i=1}^{n} |Me_i|^2},$$

where e_i is the canonical basis on $\mathbb{R}^n$. One checks this is a norm.

Lemma 1.60: Hilbert-Schmidt norm

For every $x \in \mathbb{R}^n$ and $M : \mathbb{R}^n \to \mathbb{R}^m$ linear, $|Mx| \leq \|M\|_2 \, |x|$. In particular, any linear map is Lipschitz.

Proof. We have

$$\begin{aligned}|Mx|^2 = \left| M\left(\sum_{i=1}^n x_i e_i\right)\right|^2 &= \left|\sum_{i=1}^n x_i M e_i\right|^2 \leq \left(\sum_{i=1}^n |x_i M e_i|\right)^2 \\ &= \left(\sum_{i=1}^n |x_i|\,|Me_i|\right)^2 \leq |x|^2 \sum_{i=1}^n |Me_i|^2 = |x|^2 \,\|M\|_2^2,\end{aligned}$$

invoking Cauchy-Schwartz 1.2 and the triangle inequality. □

Example 1.61

Let's consider more examples. Let V be the continuous real-valued functions on $[0,1]$. Define the p-norm for $f \in V$:

$$\|f\|_p = \left(\int_0^1 |f|^p \, dx\right)^{\frac{1}{p}}.$$

There is also the analogous ∞-norm, or supremum norm

$$\|f\|_\infty = \max_{x\in[0,1]} |f|.$$

To prove the p-norm is a norm is tricky but the infinity norm is easier to prove, left as an exercise.

Proposition 1.62: Norm induces metric

Every normed vector space is a metric space with $d(v,w) = \|v-w\|$.

Definition 1.63: Inner product vector space

Let V be a vector space over $\mathbb{R}$. An inner product is a map $\langle\cdot,\cdot\rangle : V \times V \to \mathbb{R}$ that satisfies

(1) $\forall v, w \in V : \langle v, w\rangle = \langle w, v\rangle$.

(2) $\forall \alpha, \beta \in \mathbb{R}, u, v, w \in V : \langle \alpha v + \beta u, w\rangle = \alpha\langle v, w\rangle + \beta\langle u, w\rangle$.

(3) $\forall v \in V : \langle v, v\rangle \geq 0$ and $\langle v, v\rangle = 0 \iff v = 0$.

Proposition 1.64: Inner product induces norm

Every inner product space is a normed vector space with the norm $\|v\| = \sqrt{\langle v, v\rangle}$.

Proof. Exercise on problem sheet 3. We will also do all of this in the Linear Algebra Course. □

Remark 1.65

From now on, $\mathbb{R}^n$ will be the vector space over $\mathbb{R}$ with the Euclidean norm denoted $|\cdot|$. It is also an inner product space, with the inner product as in 1.1. It is also a metric space with $d(x,y) = |x-y|$.

If we say that $x_n \to x$ in $\mathbb{R}^n$, then we mean $|x_n - x| \to 0$. Recall that convergence is equivalent to component-wise convergence $|x_{i,n} - x_i| \to 0$ for all $1 \leq i \leq n$.

Also, if $U \subset \mathbb{R}^n$ is an open set and $f : U \to \mathbb{R}^m$ a function, we say

$$y = \lim_{x\to x_0} f(x)$$

if and only if for every sequence $(x_k)_k^\infty$ with $x_k \to x_0$, we have $f(x_k) \to y$.

Chapter 2

Multidimensional differential calculus

2.1 The differential

This will be the most important definition of the course.

Definition 2.1: Differential

Let $f : U \to \mathbb{R}^m$ with $U \subset \mathbb{R}^n$ open be a function. We say that $L : \mathbb{R}^n \to \mathbb{R}^m$, a linear map, is the differential of f at $x_0 \in U$ if

$$\lim_{x \to 0} \frac{f(x_0 + x) - f(x) - L(x)}{|x|} = 0.$$

Then, we call f differentiable at x_0.
One can prove that L is unique and we write $L = Df_{x_0}$.

Remark 2.2

Clearly, for $m = n = 1$, $L(x) = f'(x_0)x$. Indeed

$$0 = \lim_{x \to 0} \frac{f(x_0 + x) - f(x_0) - f'(x_0)x}{x} \implies f'(x_0) = \lim_{x \to 0} \frac{f(x_0 + x) - f(x_0)}{x}.$$

In particular, for $f : \mathbb{R} \to \mathbb{R}$, we get that $Df_{x_0}(x) = f'(x_0) \cdot x$.

Lemma 2.3: Differential component-wise in the codomain

Let $U \subset \mathbb{R}^n$ open, $f : U \to \mathbb{R}^m$ and $x_0 \in U$. Then f is differentiable at x_0 if and only if f_j is differentiable at x_0 for every $1 \leq j \leq m$. In that case,

$$Df_{x_0}(x) = (Df_{1,x_0}(x), \ldots, Df_{m,x_0}(x)).$$

Proof.

$$\lim_{x \to 0} \frac{f(x_0 + x) - f(x_0) - L(x)}{|x|} = 0$$

exists and equals zero iff the limit exists and equals zero component-wise, by Lemma 1.17. So the above limit holds iff

$$\lim_{x \to 0} \frac{f_j(x_0 + x) - f_j(x_0) - (L(x))_j}{|x|} = 0,$$

for every $1 \le j \le m$, hence the statement. □

Definition 2.4: $\mathcal{O}$ Notation

Let $U \subset \mathbb{R}^n$ be open and $f : U \to \mathbb{R}^m$. We say that $f(x) = \mathcal{O}(|g(x)|)$ as $x \to x_0$ if

$$\limsup_{x \to x_0} \frac{|f(x)|}{|g(x)|} < \infty.$$

Here, $\limsup_{x \to x_0}$ means the supremum over all sequences $x_n \to x_0$, consistent with the definition of limit in Remark 1.65.

Definition 2.5: $\mathcal{o}$ Notation

Let $U \subset \mathbb{R}^n$ be open and $f : U \to \mathbb{R}^m$. We say that $f(x) = \mathcal{o}(|g(x)|)$ as $x \to x_0$ if

$$\lim_{x \to x_0} \frac{|f(x)|}{|g(x)|} = 0.$$

Remark 2.6

Clearly, $f : U \to \mathbb{R}^m$ is differentiable at $x_0 \in U$ if and only if $f(x_0 + x) - f(x_0) - L(x) = \mathcal{o}(x)$.

Definition 2.7: Directional derivative

Let $U \subset \mathbb{R}^n$ open and $f : U \to \mathbb{R}^m$, $x_0 \in U$. Given $v \in \mathbb{R}^n$, we define the directional derivative at x_0 (along v) as

$$\partial_v f(x_0) = \lim_{\mathbb{R} \ni t \to 0} \frac{f(x_0 + tv) - f(x_0)}{t}$$

provided the limit exists.

Proposition 2.8: Differentiable implies directionally differentiable

Let $U \subset \mathbb{R}^n$ be open and $f : U \to \mathbb{R}^m$. If f is differentiable at $x_0 \in U$, then $\partial_v f(x_0)$ exists for all $v \in \mathbb{R}^n$ and

$$\partial_v f(x_0) = Df_{x_0}(v).$$

Proof. Because f is differentiable,

$$f(x_0 + x) - f(x) - Df_{x_0}(x) = \mathcal{o}(|x|)$$

as $x \to 0$. But setting $x = tv$ and letting $t \to 0$, we get

$$\lim_{t \to 0} \frac{|f(x_0 + tv) - f(x_0) - Df_{x_0}(tv)|}{|tv|} = 0,$$

and using linearity of Df_{x_0}, multiplying by $|v|$ and dropping the absolute value (because it goes to zero),

$$\lim_{t \to 0} \frac{f(x_0 + tv) - f(x_0) - t \cdot Df_{x_0}(v)}{t} = 0,$$

and so

$$\lim_{t \to 0} \frac{f(x_0 + tv) - f(x_0)}{|t|} = Df_{x_0}(v).$$

□

2.2 Partial derivatives

Definition 2.9: Partial derivative

Let $\{e_i\}_{1\leq i\leq n}$ be the canonical basis for $\mathbb{R}^n$. We define the i-th partial derivative w.r.t. x_i at x_0 as

$$\frac{\partial f}{\partial x_i}(x_0) = \partial_i f(x_0) := \partial_{e_i} f(x_0),$$

assuming it exists. This equals $Df_{x_0}(e_i)$ assuming f is differentiable at x_0, by Proposition 2.8.

Example 2.10: Computing partial derivatives

Define the function $f(x_1, x_2, x_3) = \exp(x_2)[1+x_1x_3]$. How do we find $\partial_1 f(x_1, x_2, x_3)$? We compute

$$\lim_{t\to 0} \frac{f(x_1+t, x_2, x_3) - f(x_1, x_2, x_3)}{t}.$$

But this is as if we were computing the derivative of the function $g(t) = f(x_1+t, x_2, x_3)$ and so x_2 and x_3 are just like constants. Hence, using simple one dimensional differentiation,

$$\begin{aligned}
\partial_1 f(x_1, x_2, x_3) &= \exp(x_2)x_3 \\
\partial_2 f(x_1, x_2, x_3) &= \exp(x_2)(1+x_1x_3) \\
\partial_3 f(x_1, x_2, x_3) &= \exp(x_2)x_1.
\end{aligned}$$

More generally, $\partial_v f(x_0) = g'(0)$ for the function $g(t) = f(x_0+tv)$. So partial derivative are extremely easy to compute.

Remark 2.11

If $f : U \to \mathbb{R}^m$ is differentiable on all of U, then the partial derivatives define functions

$$\partial_i f = \frac{\partial f}{\partial x_i} : U \to \mathbb{R}^m : x \mapsto \frac{\partial f}{\partial x_i}(x).$$

Theorem 2.12: Sufficient condition for differentiability

Let $U \subset \mathbb{R}^n$ be open and $f : U \to \mathbb{R}^m$. If the functions $\partial_i f$ exist and are continuous for every $1 \leq i \leq n$, then f is differentiable at every $x_0 \in U$. Furthermore,

$$Df_{x_0}(x) = (\partial_1 f(x_0), \ldots, \partial_n f(x_0)) \cdot x = \begin{pmatrix} \partial_1 f_1(x_0) & \ldots & \partial_n f_1(x_0) \\ \vdots & \ddots & \vdots \\ \partial_1 f_m(x_0) & \ldots & \partial_n f_m(x_0) \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

is the matrix representation of Df_{x_0} in the standard basis.

Proof. The idea is to fix $x_0 \in U$ and take for $\varepsilon > 0$ the set $\{x \in \mathbb{R}^n \mid |x_i - x_{0,i}| < \varepsilon\} \subset U$. This is the n dimensional cube of "radius" ε around x_0. Then, we will use the partial derivatives to take steps in each of the n directions, ultimately giving us the differential.

Assume without loss of generality that $m = 1$, using Lemma 2.3. So from now on, $f : \mathbb{R}^n \to \mathbb{R}$. Fix

$x_0 \in U$ and take some $x \in U$. For $1 \leq k \leq n$, define $x^{(k)} = x_0 + \sum_{i=1}^{k} x_i e_i$. Then

$$f(x_0 + x) - f(x_0) = \sum_{k=1}^{n} f(x^{(k)}) - f(x^{(k-1)}),$$

by telescoping. We claim this is equal to

$$\sum_{k=1}^{n} \partial_k f(y^{(k)}) \cdot x_k,$$

for certain $y^{(k)}$ we are yet to pick. Define the functions $g_k : \mathbb{R} \to \mathbb{R}$ by

$$g_k(t) = f(x^{(k-1)} + te_k).$$

These satisfy

$$g_k(x_k) = f(x^{(k-1)} + x_k e_k) = f(x^{(k)}) \tag{1}$$
$$g_k'(t) = \lim_{h \to 0} \frac{f(x^{(k-1)} + (t+h)e_k) - f(x^{(k-1)} + te_k)}{h} = \partial_k f(x^{(k-1)} + te_k) \tag{2}$$

Applying (1), we get

$$f(x^{(k)}) - f(x^{(k-1)}) = g_k(x_k) - g_k(0).$$

By the mean value theorem, which we can apply because g is differentiable based on (2), this equals

$$f(x^{(k)}) - f(x^{(k-1)}) = g_k'(\xi_k)(x_k - 0) \text{ with } \xi_k \in (0, x_k),$$

which in turn equals

$$g_k'(\xi_k) \cdot x_k = \partial_k f(x^{(k-1)} + \xi_k e_k) \cdot x_k,$$

and so it stands to reason to define the points $y^{(k)} = x^{(k-1)} + \xi_k e_k = x_0 + \sum_{i=1}^{k-1} x_i e_i + \xi_k e_k$. And since we're taking $x \to 0$, and $|\xi_k| < |x_k|$, we ultimately get that $y^{(k)} \to x_0$. We summarize the term $\sum_{i=1}^{k-1} x_i e_i + \xi_k e_k = o(1)$, just meaning that it goes to zero and write $y^{(k)} = x_0 + o(1)$ as $x \to 0$.

Let's go back to the sum

$$f(x_0 + x) - f(x_0) = \sum_{k=1}^{n} \partial_k f(y^{(k)}) x_k = \sum_{k=1}^{n} \partial_k f(x_0 + o(1)) x_k.$$

Using the crucial fact that the partial derivatives are also continuous, we get that $\partial_k f(x_0 + o(1)) = \partial_k f(x_0) + o(1)$ and write

$$f(x_0 + x) - f(x_0) = \sum_{k=1}^{n} (\partial_k f(x_0) + o(1)) x_k = \sum_{k=1}^{n} \partial_k f(x_0) x_k + o(x)$$

since $|o(1) \cdot x_i| \leq |x|$ as $x \to x_0$. And so we define the function

$$Df_{x_0}(x) = \sum_{k=1}^{n} \partial_k f(x_0) \cdot x_k$$

and get

$$f(x_0 + x) - f(x_0) - Df_{x_0}(x) = o(x),$$

meaning that f is differentiable at x_0, with the proposed matrix values for Df_{x_0}. $\square$

Definition 2.13: C_1-functions

Given $U \subset \mathbb{R}^n$ open, we define

$$C^1(U, \mathbb{R}^m) := \{f : U \to \mathbb{R}^m \mid f \text{ is continuously differentiable in } U\},$$

where continuously differentiable means that all partial derivatives exist and are continuous as in Theorem 2.12. If $m = 1$, we simplify this to $C^1(U)$.

Definition 2.14: Jacobian Matrix

For $U \subset \mathbb{R}^n$, open $f : U \to \mathbb{R}^m$ is differentiable at x_0 we define the Jacobi matrix at x_0:

$$Jf(x_0) = (\partial_1 f(x_0), \dots, \partial_n f(x_0)) = \begin{pmatrix} \partial_1 f_1(x_0) & \dots & \partial_n f_1(x_0) \\ \vdots & \ddots & \vdots \\ \partial_1 f_m(x_0) & \dots & \partial_n f_m(x_0) \end{pmatrix}.$$

It is also sometimes denoted $Df(x_0)$, which is not to be confused with the differential Df_{x_0}, which is the linear map to which $Jf(x_0) = Df(x_0)$ is the matrix.

Remark 2.15: Distinction between differential and Jacobi matrix

Observe that for some $v = \sum_{i=1}^n v_i e_i \in \mathbb{R}^n$, we have

$$\partial_v f(x_0) = Df_{x_0}(v) = \sum_{i=1}^n v_i Df_{x_0}(e_i) = \sum_{i=1}^n v_i \frac{\partial f}{\partial x_i}(x_0) = Jf(x_0) \cdot v,$$

confirming the connection between Df_{x_0}, the linear map and $Jf(x_0)$, the matrix of that map in the canonical basis. In the linear algebra course, we write

$$Jf(x_0) = [Df_{x_0}]^{\mathcal{E}}_{\mathcal{E}},$$

where $\mathcal{E}$ is the standard canonical basis.

2.3 Differentiation theorems

We will now prove the chain rule in multiple dimensions.

Theorem 2.16: Chain rule

Let $U \subset \mathbb{R}^n$ be open, $V \subset \mathbb{R}^m$ open and $f : U \to V, g : V \to \mathbb{R}^k$. Assume that f is differentiable at $x_0 \in U$ and g is differentiable at $y_0 = f(x_0) \in V$. Then the function $g \circ f : U \to \mathbb{R}^k$ is differentiable at x_0 and

$$D(g \circ f)_{x_0} = Dg_{f(x_0)} \circ Df_{x_0}.$$

Proof. This is the intuition: We have that $f(x_0 + x) - f(x_0) \approx L(x)$ for some linear map $L(x)$ and $g(y_0 + y) - g(y_0) \approx M(y)$ for a linear M. Then we would like to substitute $y = f(x_0 + x) - f(x_0)$ to get

$$g(f(x_0 + x)) - g(f(x_0)) = g(y_0 + f(x_0 + x) - f(x_0)) - g(y_0) \approx M(f(x_0 + x) - f(x_0)) \approx M(L(x)).$$

Now comes the actual proof. We have that

$$f(x_0 + x) - f(x) = L(x) + R(x),$$

for a map R such that $\lim_{x\to 0} \frac{R(x)}{|x|} = 0$. Same for g, we have

$$g(y_0 + y) - g(y_0) = M(y) + S(y), \quad \lim_{y\to 0} \frac{S(y)}{|y|} = 0.$$

Hence we compute, setting $y = f(x_0 + x) - f(x_0)$,

$$\begin{aligned} g(f(x_0+x)) - g(f(x_0)) &= g(y_0 + f(x_0+x) - f(x_0)) - g(y_0) \\ &= M(f(x_0+x) - f(x_0)) + S(f(x_0+x) - f(x_0)) \\ &= M(L(x) + R(x)) + S(L(x) + R(x))n \\ &= ML(x) + \underbrace{M(R(x)) + S(L(x) + R(x))}_{T(x)}, \end{aligned}$$

and all we have to show now is that $T(x) = o(x)$, so $\lim_{x\to 0} \frac{T(x)}{|x|} = 0$. By the inequality for the Hilbert-Schmidt Norm, Proposition 1.60, we get

$$\lim_{x\to 0} \frac{|M(R(x))|}{|x|} \le \lim_{x\to 0} \frac{\|M\|_2 \, |R(x)|}{|x|} = 0,$$

as $R(x) = o(x)$. We also get

$$\frac{|S(L(x) + R(x))|}{|x|} = \frac{|S(L(x) + R(x))|}{|L(x) + R(x)| + |x|^2} \frac{|L(x) + R(x)| + |x|^2}{|x|}.$$

As $x \to 0$, $L(x) \to 0$ and $R(x) \to 0$, hence[1]

$$\frac{|S(L(x) + R(x))|}{|L(x) + R(x)| + |x|^2} \le \frac{|S(L(x) + R(x))|}{|L(x) + R(x)|} \to 0,$$

since $S(y) = o(y)$ and finally,

$$\frac{|L(x) + R(x)| + |x^2|}{|x|} \le \frac{|L(x)|}{|x|} + \frac{|R(x)|}{|x|} + |x| \le \|L\|_2 + \frac{o(x)}{|x|} + |x|$$

which is bounded, so the entire error term $T(x)$ goes to zero. □

Remark 2.17: Composition is multiplication of Jacobi matrices

This is useful because it allows us to compute the Jacobi matrix of the composition,

$$J(g \circ f)(x_0) = Jg(f(x_0)) \cdot Jf(x_0).$$

Component-wise, for $1 \le i \le n$, this becomes the $\mathbb{R}^k$ vector

$$\frac{\partial(g \circ f)}{\partial x_i}(x) = \sum_{j=1}^{m} \frac{\partial g}{\partial y_j}(f(x)) \cdot \frac{\partial f_j}{\partial x_i}(x) = \partial_i (g \circ f)(x) = \sum_{j=1}^{m} \partial_j g(f(x)) \cdot \partial_i f_j(x),$$

where y_j are the components in $\mathbb{R}^m$ and x_i are in $\mathbb{R}^n$.

[1] Notice how we added $|x|^2$ to the denominator. This is to prevent dividing by zero such that if it happens for some sequence testing the limit that $L(x) + R(x) = 0$, the following inequality still holds for all the parts of the sequence where there is no division by zero.

Theorem 2.18: Generalised mean value theorem

Let $U \subset \mathbb{R}^n$ open and $f \in C^1(U)$. Assume that $x_0 \in U$ and $h \in \mathbb{R}^n$ such that $x_0 + th \in U$ for all $t \in [0, 1]$. Then

$$f(x_0 + h) - f(x_0) = Df_\xi(h) = \partial_h f(\xi),$$

where $\xi = x_0 + \theta h$ for some $\theta \in (0, 1)$.

Proof. For $t \in [0, 1]$, define $g : \mathbb{R} \to \mathbb{R}$ by $g(t) = f(x_0 + th)$. Then

$$f(x_0 + h) - f(x_0) = g(1) - g(0) = g'(\theta)$$

for some $\theta \in (0, 1)$, by the MVT. But $g(t) = (f \circ \gamma)(t)$ for the curve $\gamma(t) = x_0 + th$, which has differential

$$D\gamma_t(s) = sh$$

for any t, as can be quickly verified. Then we can apply the chain rule to get

$$Dg_t = D(f \circ \gamma)_t = Df_{\gamma(t)} \circ D\gamma_t.$$

Since $g : \mathbb{R} \to \mathbb{R}$, we have $g'(t) = Dg_t(1)$, by Remark 2.2. Hence

$$g'(t) = Dg_t(1) = Df_{\gamma(t)}\big(D\gamma_t(1)\big) = Df_{\gamma(t)}(h),$$

and taking $t = \theta$ we get the statement. $\square$

Definition 2.19: Convex sets

A set $A \subset \mathbb{R}^n$ is called convex if

$$\forall x, y \in A : \forall t \in [0, 1] : tx + (1 - t)y \in A.$$

In intuitive terms, one can connect any two points in A by a straight line in n-dimensional space.

Definition 2.20: Gradient

For $U \subset \mathbb{R}^n$ open and $f \in C^1(U)$, we define the gradient of f at $x \in U$ as the vector

$$\nabla f(x) = Jf(x)^\top.$$

Notice that technically, $Jf(x) = (\partial_1 f(x), \ldots, \partial_n f(x))$ is a row vector and we want the gradient to be column, so we take the transpose. We will be very loose with whether $\nabla f(x)$ is a row or column vector, it never matters.

Proposition 2.21: Bounded gradient is Lipschitz

Let $U \subset \mathbb{R}^n$ open and $f \in C^1(U)$. Assume that U is convex and

$$\sup_{x \in U} |\nabla f(x)| \leq M < \infty.$$

Then f is M-Lipschitz.

Proof. Let $x, y \in U$, $h = x - y$ to write $x = y + h$ and $y = x_0$, then, applying Cauchy-Schwartz and by the generalised Mean Value Theorem 2.18, there exists a $\xi \in U$ on the line between x and y such that

$$\begin{aligned} |f(x) - f(y)| = |f(x_0 + h) - f(x_0)| &= |Df_\xi(h)| = |Jf(\xi) \cdot h| = |h \cdot \nabla f(\xi)| \\ &\leq |\nabla f(\xi)| \, |h| \leq M \, |h| = M \, |y - x| \, . \end{aligned}$$

□

Theorem 2.22: Constant function

Let $U \subset \mathbb{R}^n$ be open and connected. Let $f \in C^1(U)$ and assume that $\forall x \in U : \nabla f(x) = 0$. Then f is constant on U.

Proof. Let $x_0 \in U$. We define

$$G = \{x \in U \mid f(x) = f(x_0)\} \, .$$

We will show that G is open and $U \setminus G$ is open, forcing $G = U$, as it is connected.

So assume that $x \in U$. Then, by openness of U, there exists a ball $B_r(x)$ such that $B_r(x) \subset U$. Now take any $y \in B_r(x)$. Because the ball is convex (exercise), I can apply the Mean Value Theorem 2.18 to get

$$f(y) - f(x) = Df_\xi(h) = Jf(\xi) \cdot h = h \cdot \nabla f(\xi) = 0,$$

for $h = y - x$ and $\xi = x + \theta h \in B_r(x)$ with $\theta \in (0, 1)$. But that just means $f(x) = f(y)$.

Now, if $x \in G$, then the ball $B_r(x)$ must be a subset of G, as for any $y \in B_r(x)$, $f(y) = f(x) = f(x_0)$. But conversely, if $x \notin G$, then certainly $f(y) = f(x) \neq f(x_0)$ for any $y \in B_r(x)$. This shows both G and $U \setminus G$ are open.

And so finally, as $x_0 \in G$, it is non-empty, which forces $U = G$, by connectedness. □

2.4 Higher order derivatives

The next logical step is to take partial derivatives of partial derivatives of course.

Definition 2.23: Class C^k

Let $U \subset \mathbb{R}^n$ be open. We define

$$C^0(U, \mathbb{R}^m) = \{f : U \to \mathbb{R}^m \mid f \text{ continuous}\} \, .$$

Then, inductively we let

$$C^k(U, \mathbb{R}^m) = \left\{f : U \to \mathbb{R}^m \mid \partial_i f \in C^{k-1}(U, \mathbb{R}^m) \text{ for each } 1 \leq i \leq n\right\} .$$

If $f \in C^k(U, \mathbb{R}^m)$, we say that f is k-times continuously differentiable. Once again $C^k(U) := C^k(U, \mathbb{R})$ is a shorthand. If U is clear from the context, we just write C^k.

Proposition 2.24: Sum, product, composition of C^k functions

Let $f, g \in C^k(U)$, where $U \subset \mathbb{R}^n$ is open. Let $V \subset \mathbb{R}^m$ and $h : V \to \mathbb{R}^n$ such that $h(V) \subset U$ and $h \in C^k(V, \mathbb{R}^n)$. Then

(1) $f + g \in C^k(U)$ and $f \cdot g \in C^k(U)$.

(2) $f \circ h \in C^k(V)$.

Proof. We prove (1) by induction over k. The base case $k = 0$ is left as an exercise, as in Analysis 1. we will assume that (1) holds for $k-1$. But then, for $f, g \in C^k(U)$,

$$\partial_i(f+g) = \partial_i f + \partial_i g.$$

Since $f, g \in C^k(U)$, we have $\partial_i f, \partial_i g \in C^{k-1}$ by assumption. But by induction hypothesis, their sum is C_{k-1} and hence $f + g \in C^k(U)$.

Similarly, for the product, one verifies the base case. Then assume that $f, g \in C^k(U)$. By expansion,

$$\partial_i(fg) = \partial_i f \cdot g + f \cdot \partial_i g.$$

Once again, all functions on the RHS are $C^{k-1}(U)$, by assumption the products and sums are as well and so $\partial_i(fg) \in C^{k-1}(U)$, proving $fg \in C^k(U)$.

(2) we also prove by induction. Indeed, the case $k = 0$ is easily proven, as composition of continuous maps is continuous. Now assume the statement holds for $k-1$. By the Chain Rule, Theorem 2.16

$$\partial_j(f \circ h) = \sum_{i=1}^{n} [(\partial_i f) \circ h] \cdot \partial_j h_i.$$

By induction hypothesis, $(\partial_i f) \circ h \in C^{k-1}$ and so the sum runs over products of C^{k-1} functions, so is C^{k-1} itself. $\square$

Theorem 2.25: Schwarz

Let $f \in C^2(U)$, with $U \subset \mathbb{R}^n$ open. For every, $1 \leq i, j \leq n$,

$$\partial_j \partial_i f = \partial_i \partial_j f.$$

Proof. Intuition behind the proof: We will assume wlog $f : \mathbb{R}^2 \to \mathbb{R}$ is of class C^2 and consider wlog the statement at the origin, then cleverly use that to prove the general case. Roughly speaking,

$$\begin{aligned}
\partial_1 f(0,0) &\approx \frac{f(h,0) - f(0,0)}{h}, \quad & \partial_1 f(0,h) &\approx \frac{f(h,h) - f(0,h)}{h} \\
\partial_2 f(0,0) &\approx \frac{f(0,h) - f(0,0)}{h}, \quad & \partial_2 f(h,0) &\approx \frac{f(h,h) - f(h,0)}{h}.
\end{aligned}$$

And with that,

$$\begin{aligned}
\partial_2(\partial_1 f)(0,0) = \frac{\partial_1 f(0,h) - \partial_1 f(0,0)}{h} &\approx \frac{\frac{f(h,h)-f(0,h)}{h} - \frac{f(h,0)-f(0,0)}{h}}{h} \\
&= \frac{f(h,h) - f(0,h) - f(h,0) + f(0,0)}{h^2} \\
\partial_1(\partial_2 f)(0,0) = \frac{\partial_2 f(h,0) - \partial_2 f(0,0)}{h} &\approx \frac{\frac{f(h,h)-f(h,0)}{h} - \frac{f(0,h)-f(0,0)}{h}}{h} \\
&= \frac{f(h,h) - f(0,h) - f(h,0) + f(0,0)}{h^2}.
\end{aligned}$$

Actual proof: We will do first the case $n = 2, m = 1$, $U \subset \mathbb{R}^2$ is open with $(0,0) \in U$ and use it later to conclude the general case. So the goal is to show

$$\partial_2 \partial_1 f(0,0) = \partial_1 \partial_2 f(0,0).$$

We will define $F : \mathbb{R} \to \mathbb{R}$ by

$$F(h) = f(h,h) - f(h,0) - f(0,h) + f(0,0).$$

As U is open, for $h > 0$ small enough, $F(h) \in U$. We will only consider such h. We define another function $\varphi : [0,1] \to \mathbb{R}$ by

$$\varphi(t) = f(th,h) - f(th,0) \implies F(h) = \varphi(1) - \varphi(0) = \varphi'(\xi_1)$$

for some $\xi_1 \in (0,1)$, by the mean value theorem. But we have by definitions and the Chain Rule 2.16

$$\varphi'(\xi_1) = [\partial_1 f(\xi_1 h, h) - \partial_1 f(\xi_1 h, 0)] \cdot h = [\psi(1) - \psi(0)] \cdot h = \psi'(\xi_2) \cdot h,$$

for some $\xi_2 \in (0,1)$, for the function $\psi : [0,1] \to \mathbb{R}$ given by

$$\psi(s) = \partial_1 f(\xi_1 h, sh),$$

and applying the MVT. Again, by the Chain Rule,

$$\psi'(\xi_2) \cdot h = \partial_2 \partial_1 f(\xi_1 h, \xi_2 h) \cdot h^2.$$

We can do the same and define

$$\tilde{\varphi}(t) = f(h,th) - f(0,th) \implies F(h) = \tilde{\varphi}(1) - \tilde{\varphi}(0) = \tilde{\varphi}'(\tilde{\xi}_2) = [\partial_2 f(h, \tilde{\xi}_2 h) - \partial_2 f(0, \tilde{\xi}_2 h)] \cdot h.$$

If we let $\tilde{\psi}(s) = \partial_1 f(sh, \tilde{\xi}_2)$, we get by the Chain Rule 2.16

$$[\partial_2 f(h, \tilde{\xi}_2 h) - \partial_2 f(0, \tilde{\xi}_2 h)] \cdot h = [\tilde{\psi}(1) - \tilde{\psi}(0)] \cdot h = \tilde{\psi}'(\tilde{\xi}_1) = \partial_1 \partial_2 f(\tilde{\xi}_1 h, \tilde{\xi}_2 h) \cdot h^2.$$

So in essence, what we proved is that

$$\partial_2 \partial_1 f(\xi_1 j, \xi_2 h) = \partial_1 \partial_2 f(\tilde{\xi}_1 h, \tilde{\xi}_2 h).$$

Since $\xi_i, \tilde{\xi}_i$ are bounded in $(0,1)$, letting $h \to 0$ and using that $f \in C^2(U)$, so both partials are continuous, we get $\partial_2 \partial_1 f(0,0) = \partial_1 \partial_2 f(0,0)$.

From this, we can infer the general case. We let $f : U \to \mathbb{R}^m$ with $U \subset \mathbb{R}^n$ open and $y \in U$. Write $f = f(y_1, \ldots, y_n)$ and assume wlog $i < j$. We claim that

$$\partial_i \partial_j f_k(y) = \partial_j \partial_i f_k(y)$$

for any $1 \leq k \leq m$. Define the function

$$\tilde{f}(x_1, x_2) = f_k(y + x_1 e_i + x_2 e_j) = f_k(y_1, y_2, \ldots, y_i + x_1, \ldots, y_j + x_2, \ldots, y_n).$$

We get by the special case we just proved that

$$\partial_2 \partial_1 f_k(y) = \partial_2 \partial_1 \tilde{f}(0,0) = \partial_1 \partial_2 \tilde{f}(0,0) = \partial_1 \partial_2 f_k(y).$$

Doing this for all k, we get the general statement. ☐

Corollary 2.26: Permutation of partials

Let $U \subset \mathbb{R}^n$ be open and $f \in C^k(U)$. Let $i_1, \ldots, i_k$ be indices in $\{1, \ldots, n\}$ and $\sigma \in S_n$. Then

$$\partial_{i_1} \partial_{i_2} \cdots \partial_{i_k} f = \partial_{\sigma(i_1)} \partial_{\sigma(i_2)} \cdots \partial_{\sigma(i_k)} f.$$

Proof. Follows from Schwarz Theorem 2.25, as any permutation can be written as a product of transpositions. ☐

Definition 2.27: Multi-index notation

For $\alpha \in \mathbb{N}^n$, i.e. $\alpha = (\alpha_1, \dots, \alpha_n)$, we write

$$|\alpha| = \sum_{k=1}^{n} \alpha_i.$$

With the Schwarz corollary 2.26, we can define (well) the expressions

$$\partial^\alpha f = \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n} f$$

where ∂_i^k means taking the i-th partial derivative k times, and for $x \in \mathbb{R}^n$

$$x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}.$$

Furthermore, we define the multi-factorial

$$\alpha! = \alpha_1 \cdots \alpha_n.$$

This is very useful notation for dealing with multidimensional derivatives.

2.5 Taylor theorem and analyticity

The Taylor theorem in $\mathbb{R}^n$ will in essence be the one in $\mathbb{R}$ but applied to the function $f \circ \gamma = \varphi$, so basically we will be able to express the value of f around some point x_0 by approaching it with the path $\gamma = x_0 + th$. That in turn will tell us that for small h the Taylor polynomial at h is approaches $f(x_0 + h)$.

Lemma 2.28: Derivative in terms of partials

Let $U \subset \mathbb{R}^n$ be open. Assume $f \in C^m(U)$, $x_0 \in U$, $h \in \mathbb{R}^n$ and $x_0 + th \in U$ for all $t \in (-\varepsilon, 1 + \varepsilon)$ with some $\varepsilon > 0$. Let $\gamma(t) = x_0 + ht$ and define $\varphi(t) = f \circ \gamma(t)$. Then

$$\frac{\varphi^{(m)}(t)}{m!} = \sum_{|\alpha|=m} \frac{\partial^\alpha f(x_0 + th) h^\alpha}{\alpha!}.$$

Proof. Induction over m. For $m = 1$, the component-wise Chain Rule 2.17 gives

$$D\varphi_t(1) = \varphi'(t) = \sum_{j=1}^{n} \partial_j f(x_0 + th) \cdot h_j = \sum_{|\alpha|=1} \partial^\alpha f(x_0 + th) \frac{h^\alpha}{\alpha!},$$

since $\partial_1 \gamma(t) = \frac{d}{dt}\gamma(t) = h = (h_1, \dots, h_n)$.

Now assume the equality holds for m. Differentiating the induction assumption with the Chain Rule 2.17, we get

$$\frac{\varphi^{(m+1)}(t)}{m!} = \sum_{|\alpha|=m} \sum_{i=1}^{n} \frac{1}{\alpha!} \partial_i \partial^\alpha f(x_0 + th) \cdot h^\alpha h_i = \sum_{|\alpha|=m} \sum_{i=1}^{n} \frac{\partial^{\alpha+e_i} f(x_0 + th) h^{\alpha+e_i}}{\alpha!}.$$

Define for every i the multi-index $\beta = \alpha + e_i$ such that $\frac{1}{\alpha!} = \frac{1}{\beta!}\beta_i$. Then

$$\frac{\varphi^{(m+1)}(t)}{m!} = \sum_{|\beta|=m+1} \sum_{\beta_i>0} \frac{\partial^\beta f(x_0 + th) h^\beta}{\beta!} \beta_i = \sum_{|\beta|=m+1} \frac{\partial^\beta f(x_0 + th) h^\beta}{\beta!} \underbrace{\sum_{i=1}^{n} \beta_i}_{|\beta|=m+1}.$$

$\square$

THEOREM 2.29: TAYLOR THEOREM

Let $f \in C^{k+1}(U)$ with $U \subset \mathbb{R}^n$ open. Let $x_0 \in U$, $h \in \mathbb{R}^n$ and assume that $x_0 + th \in U$ for all $t \in [0,1]$. Then

$$f(x_0 + h) = \sum_{|\alpha|=0}^{k} \frac{\partial^\alpha f(x_0) h^\alpha}{\alpha!} + R_{k+1} f(h),$$

where

$$R_{k+1} f(h) = \int_0^1 (1-t)^k (k+1) \left(\sum_{|\alpha|=k+1} \frac{\partial^\alpha f(x_0 + th)}{\alpha!} h^\alpha \right) dt.$$

Furthermore, $R_{k+1} f(h) = \mathcal{O}(|h|^{k+1})$ as $h \to 0$.

Proof. By one dimensional Taylor theorem of φ evaluated at 1, expanded around 0, up to degree k,

$$f(x_0 + h) = \varphi(1) = \sum_{m=0}^{k} \frac{\varphi^{(m)}(0)}{m!} + \int_0^1 (k+1)(1-t)^k \frac{\varphi^{(k+1)}(t)}{(k+1)!} \, dt.$$

Plugging in the expression for $\frac{\varphi^{(m)}(t)}{m!}$ from the previous Lemma with $\varepsilon > 0$ small enough that we're contained in U, we get the equality.

Finally, for the asymptotic, consider that

$$|h^\alpha| = |h_1^{\alpha_1} \cdots h_n^{\alpha_n}| \leq |h|^{\alpha_1} \cdots |h|^{\alpha_n} = |h|^{|\alpha|} = |h|^{k+1}.$$

So pick a small enough ball such that $\overline{B_r(x_0)} \subset U$. For h with $|h| < r$, we have by the Extreme Value Theorem 1.39 that all $\partial^\alpha f$ are bounded (because continuous) on this ball. After that, with the triangle inequality for integrals, it's not particularly hard to estimate $R_{k+1} f$ from above by a constant multiple of $|h|^{k+1}$, and we're done. $\square$

PROPOSITION 2.30: POLYNOMIAL EXPANSIONS

Let $f \in C^{k+1}(U)$ with $U \subset \mathbb{R}^n$ open and $x_0 \in U$. Assume that

$$|f(x_0 + h) - P(h)| = o(|h|^k) \quad \text{as } h \to 0$$

for some polynomial P of degree $\leq k$. Then,

$$\partial^\alpha f(x_0) = \partial^\alpha P(0) \quad \forall \alpha \in \mathbb{N}^n \text{ with } |\alpha| \leq k.$$

Proof. Denote by $P_{x_0,k}$ the Taylor polynomial up to degree k at x_0. Then by Taylor Theorem 2.29,

$$|f(x_0 + h) - P_{x_0,k}(h)| \leq \mathcal{O}(|x|^{k+1}) = o(|x|^k).$$

By assumption, $|f(x_0 + h) - P(h)| = o(|h|^k)$. By triangle inequality, we get

$$|P_{x_0,k}(h) - P(h)| = o(|h|^k).$$

It will be an exercise in the problem sheet that from this follows that $P_{x_0,k}(h) - P(h)$ is the zero polynomial. And so of course the derivatives align as claimed, since the Taylor polynomial coefficients are determined by the partial derivatives of f. $\square$

Example 2.31

With this corollary, we can avoid always writing out the massive original Taylor formula. We want to compute the Taylor expansion of $e^{x_1^2-x_2+x_3^4}$. Remember from analysis 1 that

$$e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \mathcal{O}(t^4) \quad \text{as } t \to 0.$$

Hence

$$e^{x_1^2-x_2+x_3^4} = 1 + (x_1^2 - x_2 + x_4^4) + \frac{(x_1^2 - x_2 + x_4^4)^2}{2} + \frac{(x_1^2 - x_2 + x_4^4)^3}{6} + \mathcal{O}(|x|^4).$$

This can be reduced to

$$1 + x_1^2 - x_2 + \frac{2x_1^2 x_2 + x_2^2}{2} - \frac{1}{6}x_2^3 + \underbrace{\mathcal{O}(|x|^4)}_{o(|x|^3)}.$$

And so by Corollary 2.30, this must be the same as the Taylor expansion around $x = 0$, up to degree 3.

Definition 2.32: Class C^∞

A function $f : U \to \mathbb{R}^m$ is class $C^\infty(U, \mathbb{R}^m)$ if $f \in C^k(U, \mathbb{R}^m)$ for all $k \in \mathbb{N}$. We call this *infinitely differentiable.*

Definition 2.33: Real analytic functions (via estimate)

We say that $f \in C^\infty(U)$ for $U \subset \mathbb{R}^n$ open *satisfies analytic estimates* in U if $\forall x_0 \in U\ \exists C, \rho > 0$ such that

$$\max_{B_\rho(x_0)} |\partial^\alpha f| \le C \cdot |\alpha|! \cdot (n\rho)^{-|\alpha|} \quad \forall \alpha \in \mathbb{N}^n.$$

Theorem 2.34: Analytic expansion

Let $U \subset \mathbb{R}^n$ be open. Assume that $f \in C^\infty(U)$ satisfies analytic estimates. Let $x_0 \in U, \rho > 0$ and $C > 0$ be as in the previous definition. Let $P_{x_0,k}(h) = \sum_{|\alpha|=0}^k c_\alpha h^\alpha$ be the Taylor polynomial of degree k. Then for $r \in (0, \rho)$, $P_{x_0,k}(h)$ is absolutely convergent, i.e.

$$\forall \ell > k : |h| < r \implies \sum_{|\alpha|=k}^{\ell} |c_\alpha h^\alpha| \le \frac{C(r/\rho)^k}{1 - (r/\rho)}.$$

In particular $P_{x_0,\infty}(h) := \lim_{k\to\infty} P_{x_0,k}(h)$ is defined for all $h \in B_r(0)$. Moreover, $P_{x_0,\infty}(h)$ represents f in $B_\rho(x_0)$ i.e.

$$f(x_0 + h) = P_{x_0,\infty}(h) \quad \forall h \in B_\rho(0).$$

Proof. Fix $r \in (0,\rho)$. We prove the series is absolutely convergent. The key observation is that for $x \in B_r(x_0)$ and $h \in B_r(0)$

$$\begin{aligned}\sum_{|\alpha|=k} \left| \frac{\partial^\alpha f(x)}{\alpha!} h^\alpha \right| &\le \sum_{|\alpha|=k} \left| \frac{\partial^\alpha f(x)}{\alpha!} \right| \cdot |h|^{|\alpha|} \le \sum_{|\alpha|=k} \frac{|\partial^\alpha f(x)|}{\alpha!} \cdot r^k \\ &\le \sum_{|\alpha|=k} \frac{C\rho^{-k} n^{-k} r^k}{\alpha!} |\alpha|! = C\,(r/\rho)^k\, n^{-k} \underbrace{\sum_{|\alpha|=k} \frac{|\alpha|!}{\alpha!}}_{=1}.\end{aligned}$$

The identity $1 = \sum_{|\alpha|=k} \frac{|\alpha|!}{\alpha!}$ can be verified by induction or the multinomial theorem. Then, the Taylor series is bounded by

$$\sum_{|\alpha|=k}^{\infty} \left| \frac{\partial^\alpha f(x_0)}{x} h^\alpha \right| \le \sum_{m=k}^{\infty} C\,(r/\rho)^m = C \frac{(r/\rho)^k}{1 - r/\rho}$$

because $r/\rho < 1$ and that is a geometric series. Hence the limit

$$P_{x_0,\infty}(h) = \lim_{k\to\infty} P_{x_0,k}(h)$$

exists for any $h \in B_\rho(0)$.

Now we have to estimate the error. Let $h \in B_r(0)$. By Taylor Theorem 2.34

$$f(x_0 + h) = P_{x_0,k}(h) + Rf_{x_0,k+1}(h),$$

but we found

$$\begin{aligned}|R_{x_0,k+1} f(h)| &= \left| \int_0^1 (k+1)(1-t)^k \sum_{|\alpha|=k+1} \frac{\partial^\alpha f(x_0+th)}{\alpha!} h^\alpha \, dt \right| \\ &\le \int_0^1 (k+1)(1-t)^k \cdot C\,(r/\rho)^{k+1}\, dt = C\,(r/\rho)^{k+1}.\end{aligned}$$

From this we get

$$|f(x_0+h) - P_{x_0,\infty}(h)| \le |f(x_0+h) - P_{x_0,k}(h)| + |P_{x_0,k}(h) - P_{x_0,\infty}(h)| \le C\,(r/\rho)^{k+1} + \frac{C\,(r/\rho)^k}{1 - r/\rho}$$

for any $k \in \mathbb{N}$. Sending $k \to \infty$, both terms on RHS go to zero and $f(x_0+h) = P_{x_0,\infty}(h)$ in $B_r(x_0)$ for all $r < \rho$. So the equality holds on the entire ball $B_\rho(x_0)$. □

Corollary 2.35: Unique continuation principle

Let $U \subset \mathbb{R}^n$ open connected and f, g satisfy analytic estimates in U. If there exists a point $x_0 \in U$ such that

$$\partial^\alpha f(x_0) = \partial^\alpha g(x_0) \quad \forall \alpha \in \mathbb{N}^n,$$

then $f = g$ in U. In particular, if $V \subset U$ is a nonempty open subset where f and g coincide, then $f = g$ in all of U.

Proof. We define

$$G = \{x \in U \mid \partial^\alpha f(x) = \partial^\alpha g(x)\ \forall \alpha \in \mathbb{N}^n\}.$$

We will prove that $U \setminus G$ is open and G is open. Since U is connected and G is non-empty, this will force $U \setminus G = \emptyset$, as often with proofs about connected spaces.

Notice that

$$G = \bigcap_{\alpha \in \mathbb{N}^n} \{x \in U \mid \partial^\alpha (f-g)(x) = 0\} \implies U \setminus G = \bigcup_{\alpha \in \mathbb{N}^n} [\partial^\alpha (f-g)]^{-1}(\mathbb{R} \setminus \{0\})$$

is a an arbitrary union of preimages of open sets under a continuous map and so $U \setminus G$ is open.

But if $x \in G \subset U$, by Theorem 2.34, there exists a small enough ball around x, namely $\rho > 0$ from the analytic estimates for f and g (take the minimum of both of course) such that

$$f(x+h) = P^f_{x,\infty}(h), \quad g(x+h) = P^g_{x,\infty}(h)$$

for $h \in B_\rho(0)$ and $B_\rho(x)$ is contained in U. But since $x \in G$, $\partial^\alpha f(x) = \partial^\alpha g(x)$, and so the Taylor polynomials coincide and $f(x+h) = g(x+h)$. And so $B_\rho(x) \subset G$ and G is open. $\square$

Chapter 3

Optimization problems

3.1 Basic notions

Now that we have established a strong basis for manipulating and understanding real multi-valued, multi-argument functions, let's look at applications in problems. Roughly speaking, the goal of optimisation is to take a function $f(x_1, \dots, x_n)$ with constraints $g_k(x_1, \dots, x_n) = 0$ and find for example the minimum of f. For instance, one could be looking for the minimum of

$$f(x_1, x_2, x_3, x_4) = x_1^2 - 2x_3^2 + x_4 + x_2^3$$

under the constraint

$$g(x) = x_1^2 + x_2^2 + x_3^2 + x_4^2 - 1 = 0.$$

In Analysis, this task usually came in the form of finding the minimum of a function like $f(x) = x^2 + x^3 - 4x$ on an interval $[a, b]$. Let's establish the tools and theorems to do this in multiple dimensions.

Definition 3.1: Local extremum

Let $U \subset \mathbb{R}^n$ be open, $f \in C^1(U)$. We say that $x_0 \in U$ is a local minimum of f if $\exists r > 0$ such that

$$f(x_0) \leq f(x) \quad \forall x \in B_r(x_0).$$

x_0 is a maximum if the inequality goes the other way. It is an extremum if it is either a minimum or maximum. We call it strict if the inequalities are strict.

Definition 3.2: Critical point

Let $U \subset \mathbb{R}^n$ be open and $f \in C^1(U)$. A point $x_0 \in U$ is called a critical point of f if $\nabla f(x_0) = 0$.

Proposition 3.3: Gradient vanishes at extremum

Let $U \subset \mathbb{R}^n$ be open and $f : U \to \mathbb{R}$ be a class $C^1(U)$ function. Then, if for $x_0 \in U$, f is differentiable at x_0 and assumes a local extremum, $\nabla f(x_0) = 0$ and x_0 is a critical point.

Proof. Assume wlog that x_0 is a local minimum. That means $\exists r > 0$ such that $f(x) \geq f(x_0)$ for all $x \in B_r(x_0)$. Fix some $1 \leq i \leq n$. We have

$$\partial_i f(x_0) = \lim_{t \to 0} \frac{f(x_0 + te_i) - f(x_0)}{t} = \lim_{t \to 0^+} \frac{f(x_0 + te_i) - f(x_0)}{t} \geq 0,$$

as $t > 0$ and $f(x_0 + te_i) \geq f(x_0)$ if we're approaching from positive t. On the other hand, if we approach from negative t, we get

$$\partial_i f(x_0) = \lim_{t \to 0^-} \frac{f(x_0 + te_i) - f(x_0)}{t} \leq 0.$$

But that forces $\partial_i f(x_0) = 0$. If we do this for all i, we get that the gradient is zero. $\square$

3.2 Constraint optimisation

Here's a good interpretation I found for the following theorem: The Lagrange multiplier theorem states that at any local maximum (or minimum) of the function evaluated under the equality constraints, then, roughly speaking, the gradient of the function (at that point) can be expressed as a linear combination of the gradients of the constraints (at that point), with the Lagrange multipliers acting as coefficients. This is equivalent to saying that any direction perpendicular to all gradients of the constraints is also perpendicular to the gradient of the function. Or still, saying that the directional derivative of the function is 0 in every feasible direction.[1]

Theorem 3.4: Lagrange multiplier

Let $U \subset \mathbb{R}^n$ open, $f, g_j \in C^1(U)$ be functions for $1 \leq j \leq k$. Define the set

$$M = \{x \in U \mid g_1(x) = g_2(x) = \cdots = g_k(x) = 0\} \neq \emptyset.$$

Assume further that $x_0 \in M$ is a local minimum of $f|_M$ (on the ball with radius $r > 0$). Then there are real numbers $\lambda_0, \ldots, \lambda_k$ such that

$$\lambda_0 \nabla f(x_0) + \sum_{i=1}^{k} \lambda_i \nabla g_i(x_0) = 0, \quad \lambda_0^2 + \cdots + \lambda_k^2 = 1.$$

The numbers λ_i are called the Lagrange multipliers and we see that $\nabla f(x_0), \nabla g_j(x_0)$ are linearly dependent vectors.

Proof. Assume without loss of generality that f is non-negative on the ball $\overline{B_r(x_0)}$. By replacing $f(x)$ by $\tilde{f}(x) = f(x) + |x - x_0|^2$, we get $\nabla \tilde{f}(x_0) = \nabla f(x_0)$ and so we can assume without loss of generality that the minimum is strict.

Given $\varepsilon > 0$, we will minimise the "penalised function":

$$f_\varepsilon(x) = f(x) + \frac{1}{2\varepsilon} \sum_{j=1}^{k} g_i^2(x)$$

defined for $x \in \overline{B_r(x_0)}$. We will take a sequence $(\varepsilon_\ell)_\ell^\infty$ that goes to zero, for instance $\varepsilon_\ell = \frac{1}{\ell}$. For each $\ell \in \mathbb{N}$, let x_ℓ be the point of minimum of $f_{\varepsilon_\ell}(x)$ in $\overline{B_r(x_0)}$.

Notice that $(x_\ell)_\ell^\infty$ is a sequence in a compact set and so we can extract a convergent subsequence $x_{\ell_m} \to \bar{x} \in \overline{B_r(x_0)}$. We claim that $\bar{x} \in M$. Indeed, $f_{\varepsilon_\ell}(x_\ell) \leq f_{\varepsilon_\ell}(x_0) = f(x_0)$, because we picked x_ℓ to be the minimum and x_0 has all $g_j(x_0) = 0$. And so, using positivity of f,

$$\sum_{j=1}^{k} g_j^2(x_\ell) \leq 2\varepsilon_\ell f(x_\ell) + \sum_{j=1}^{k} g_j^2(x_\ell) = 2\varepsilon_\ell f_{\varepsilon_\ell}(x_\ell) \leq 2\varepsilon_\ell \cdot f(x_0).$$

[1] Lagrange multiplier - Wikipedia

And as $\ell \to \infty$, the RHS gets smaller and smaller, hence in particular for our subsequence x_{ℓ_m},

$$\lim_{m\to\infty} \sum_{j=1}^{k} g_j^2(x_{\ell_m}) = \sum_{j=1}^{k} g_j^2(\bar{x}) = 0$$

and so $\bar{x} \in M \cap \overline{B_r(x_0)}$. Now consider that

$$f(x_{\ell_m}) \leq f_{\varepsilon_{\ell_m}}(x_{\ell_m}) \leq f(x_0).$$

Taking $m \to \infty$, we get

$$f(\bar{x}) = \lim_{m\to\infty} f(x_{\ell_m}) \leq f(x_0).$$

But we assumed x_0 was a strict minimum, which forces $\bar{x} = x_0$. And so we found a subsequence $x_{\ell_m} \to x_0$ that is always the minimum of $f_{\varepsilon_{\ell_m}}$ and converges to x_0.

Use that to define $y_m = x_{\ell_m}$ and $\tilde{\varepsilon}_m = \varepsilon_{\ell_m}$. Since eventually, $y_m \in B_r(x_0)$, it is an interior minimiser of $f_{\tilde{\varepsilon}_m}$, so we have by 3.3

$$0 = \nabla f_{\tilde{\varepsilon}_m}(y_m) \implies 0 = \tilde{\varepsilon}_m \nabla f_{\tilde{\varepsilon}_m}(y_m) = \tilde{\varepsilon}_m \nabla f(y_m) + \sum_{j=1}^{k} g_j(y_m)\nabla g_j(y_m)$$

since $\nabla(g_j^2) = 2g_j\nabla g_j$ can be verified easily. But this means, for any m, the above vectors are linearly dependent ($\tilde{\varepsilon}_m$ is a non-trivial factor) i.e. $\exists\lambda^m \in \mathbb{R}^{k+1}$ that has (wlog) norm $|\lambda^m| = 1$ and satisfies

$$0 = \lambda_0^m \nabla f(y_m) + \sum_{j=1}^{k} \lambda_j^m \nabla g_j(y_m).$$

But λ^m is a sequence in the unit sphere $\mathbb{S}^k \subset \mathbb{R}^{k+1}$, which is closed and bounded, and so compact (1.41). Hence it has a convergent subsequence with limit $\lambda \in \mathbb{S}^k$, and passing the entire above equation to the limit (all functions involved are continuous),

$$\lambda_0 \nabla f(x_0) + \sum_{i=1}^{k} \lambda_i \nabla g_i(x_0) = 0, \quad \lambda_0^2 + \cdots + \lambda_k^2 = 1.$$

$\square$

Remark 3.5

We haven't done it in Linear Algebra yet, but an orthonormal basis of $\mathbb{R}^n$ is a basis $\mathcal{B} = (v_1, \ldots, v_n)$ such that $v_i \cdot v_j = \delta_{ij}$. In particular, for an orthonormal basis, if you take the change of basis matrix $O = [\mathrm{id}_{\mathbb{R}^n}]_{\mathcal{B}}^{\mathcal{E}}$ from the standard basis into $\mathcal{B}$, then $O^\top = O^{-1}$.

Lemma 3.6: Eigenvalue minimisation

Assume $v_1, \ldots, v_k \in \mathbb{R}^n$ for $k \in \{1, \ldots, n-1\}$ are linearly independent vectors and A is an $n \times n$ symmetric real matrix with eigenvectors v_i, so $Av_i = \mu_i v_i$ for $\mu_i \in \mathbb{R}$. Then $\exists w \in \mathbb{R}^n$ such that $|w| = 1$, $w \cdot v_i = 0$ and

$$Aw = \lambda w \quad \text{for } \lambda \in \mathbb{R}.$$

Proof. We will be looking to minimise the function $f : \mathbb{R}^n \to \mathbb{R}$ given by $f(x) = x \cdot Ax$ under the constraints

$$\begin{cases} g_*(x) = |x|^2 - 1 = 0 \\ g_1(x) = v_1 \cdot x = 0 \\ \vdots \\ g_k(x) = v_k \cdot x = 0. \end{cases}$$

Define M to be the set of points in $\mathbb{R}^n$ where all of the constraints are satisfied, which is closed. Let $w \in M$ be a point of minimum of $f|_M$, existing by Extreme Value Theorem 1.39. By Lagrange Multiplier Theorem 3.4, $\exists \lambda_0, \lambda_*, \lambda_1, \dots, \lambda_k$ (with sum of squares equal to 1) such that

$$\lambda_0 \nabla f(w) + \lambda_* \nabla g_*(w) + \sum_{j=1}^{k} \lambda_k \nabla g_k(w) = 0.$$

We remark that for vectors $v, w \in \mathbb{R}^n$, $v \cdot w = v^\top w$ and in particular, if A is symmetric, $v \cdot Aw = (Aw) \cdot v = (Aw)^\top v = w^\top A^\top v = w^\top Av = w \cdot Av$.

Using this, we can compute the gradients in the equation above. Consider that

$$\begin{aligned} \partial_i f(x) = \lim_{t \to 0} \frac{f(x + te_i) - f(x)}{t} &= \lim_{t \to 0} \frac{(x + te_i) \cdot A(x + te_i) - x \cdot Ax}{t} \\ &= \lim_{t \to 0} \frac{x \cdot Ate_i + te_i \cdot Ax + te_i \cdot Ate_i}{t} \\ &= 2e_i \cdot Ax \end{aligned}$$

and so $e_i \cdot \nabla f(x) = \partial_i f(x) = 2e_i \cdot Ax$. Now, since $g_*(x) = x \cdot Ix - 1$, we get $\nabla g_*(x) = 2x$. And for the rest,

$$\partial_i g_j(x) = \lim_{t \to 0} \frac{v_j \cdot (x + te_i) - v_j \cdot x}{t} = v_i e_i,$$

and so $\nabla g_j(x) = v_j$. Substituting into the Lagrange Multipliers equation,

$$\underbrace{2\lambda_0 Aw + 2\lambda_* w}_{w_1} + \underbrace{\sum_{j=1}^{k} \lambda_j v_j}_{w_2} = 0.$$

By assumption, for every $1 \leq j \leq k$,

$$v_j \cdot Aw = w \cdot Av_j = w \cdot \mu_j v_j = \mu_j w \cdot v_i = 0,$$

hence w_1 and w_2 are orthogonal and sum to 0, so they must each be 0. And so

$$\lambda_0 Aw + \lambda_* w = 0, \quad \lambda_1 v_1 + \cdots + \lambda_k v_k = 0$$

which implies $\lambda_1 = \cdots = \lambda_k = 0$, as $(v_1, \dots, v_k)$ are linearly independent. But then $\lambda_0^2 + \lambda_*^2 = 1$ and $Aw = \lambda w$ with $\lambda = -\lambda_*/\lambda_0$. And we can't have $\lambda_0 = 0$, because that would imply $w = 0$ and we had $|w| = 1$. □

Theorem 3.7: Spectral theorem for symmetric matrices

If $A \in \mathrm{M}_{n \times n}(\mathbb{R})$ is a *symmetric* $n \times n$ real matrix, then A diagonalises in some orthonormal basis.

Proof. You fix n and proceed by induction. The first time you apply the Lemma 3.6, you get the statement because you have no constraints except g_*. And then you keep applying the Lemma and getting eigenvectors that are always perpendicular to the old set of eigenvectors. This way, at some point, you get the Theorem. □

3.3 Second order optimality conditions

Now we will expand the Taylor series up to order 2 to get more information about the extrema.

Definition 3.8: Hessian matrix

Let $U \subset \mathbb{R}^n$ be open and $f \in C^2(U)$. We define the Hessian matrix of f at $x_0 \in U$

$$Hf(x_0) = D^2 f(x_0) := \begin{pmatrix} \partial_1\partial_1 f(x_0) & \dots & \partial_1\partial_n f(x_0) \\ \vdots & \ddots & \vdots \\ \partial_n\partial_1 f(x_0) & \dots & \partial_n\partial_n f(x_0) \end{pmatrix}$$

Notice that by Schwarz Theorem 2.25, $Hf(x_0)^\top = Hf(x_0)$ and this matrix is always diagonalisable.

Definition 3.9: Sign of a square symmetric matrix

Given a symmetric matrix $n \times n$ matrix with real coefficients A, the Spectral Theorem 3.7 shows the existence of an orthogonal matrix O such that $A = O\Lambda O^\top$. We say A is:
(1) *Positive definite* if all its eigenvalues are positive (> 0).
(2) *Nonnegative definite* if all its eigenvalues are nonnegative (≥ 0).
(3) *Negative definite* if all eigenvalues are negative (< 0).
(4) *Nonpositive definite* if all its eigenvalues are nonpositive (≤ 0).
(5) *Indefinite* if at least one eigenvalue is positive and at least one is negative.
(6) *Degenerate* if zero is an eigenvalue.

Proposition 3.10: Hessian and critical points

Let $U \subset \mathbb{R}^n$ be open and $f \in C^2(U)$ with $x_0 \in U$, a critical point. For $H = Hf(x_0)$:

(1) x_0 local minimum $\implies$ H non-negative definite.
(2) H positive definite $\implies$ x_0 local minimum.
(3) x_0 local max $\implies$ H non-positive definite.
(4) H negative definite $\implies$ x_0 is local max.
(5) H indefinite and not degenerate $\implies$ x_0 is neither max or min and x_0 is called a *saddle point*.

Proof. We prove some of the variations, the rest is the same.

(2): We use the Taylor expansion and via diagonalisation of H by the orthogonal invertible matrix O, substitute $x = Oy$, to get

$$f(x_0 + Oy) = f(x_0) + \frac{1}{2}\sum_{i=1}^{n} \lambda_i y_i^2 + O(|y|^3).$$

If we define $\lambda = \min \lambda_i > 0$ and then

$$f(x_0 + Oy) \geq f(x_0) + \frac{1}{2}\sum_{i=1}^{n} \lambda y_i^2 - C\,|y|^3 \geq f(x_0) + \frac{1}{2}\lambda\,|y|^2 - C\,|y|^3 \geq f(x_0),$$

for $|y|$ small enough.

(3): We expand in some arbitrary direction e_j,

$$f(x_0 + tOe_j) = f(x_0) + \frac{1}{2}\lambda_j t^2 + O(t^3).$$

If even a single eigenvalue λ_j were positive, we would be able to violate the maximality of $f(x_0)$ by taking t^2 small enough. Hence all eigenvalues are non-positive. $\square$

Theorem 3.11: Fundamental theorem of algebra

Let $P \in \mathbb{C}[x]$, of degree n. Then P has n complex roots (with repetition).

Proof. We will find 1 single root and then proceed inductively (assuming we know how to divide polynomials).

We write

$$P(x) = a_0 + a_1 z + \cdots + a_n z^n,$$

where $a_n \neq 0$ and $a_k \in \mathbb{C}$. Without loss of generality, we divide by a_n. We wish to minimize $|P(z)|$ on $\mathbb{C}$. Clearly, that's impossible as $\mathbb{C}$ is not compact. So we have to somehow reduce our domain of possible zeroes to a compact set.

We define $\beta = \inf_{z \in \mathbb{C}} |P(z)|$. We have

$$|P(z)| \geq |z|^n - (|a_0| + |a_1| |z| + \cdots + |a_{n-1}| |z|^{n-1}) \geq R^n - CR^{n-1}$$

if $|z| \geq R$ for some $R > 1$ big enough and where $C = |a_0| + \cdots + |a_{n-1}|$. But for R large enough, this is far above β. So *if* the value β is ever attainable, it will be inside of $\overline{B_R(0)}$. By Extreme Value Theorem 1.39, $\exists z_0 \in \overline{B_R(0)}$ such that $\beta = |P(z_0)|$.

It's also quite clear that $z_0 \in B_R(0)$ is in the interior, and hence is a local minimum. If $\beta = 0$, then we are simply done. So assume $\beta > 0$.

We center the polynomial at z_0, meaning we consider

$$P(z_0 + z) = \sum_{k=1}^{n} a_k (x_0 + z)^k = \sum_{k=1}^{n} b_k z^k,$$

as a polynomial of complex coefficients b_k and variable $z \in \mathbb{C}$. Note that $|b_0| = |P(z_0)| = \beta > 0$. We define $\ell = \min \{k \geq 1 \mid b_k \neq 0\}$ the index of the first non-constant non-zero coefficient. And so

$$P(z_0 + z) = b_0 + b_\ell z^\ell + \mathcal{O}(|z|^{\ell+1}) = \beta \left(1 + c_\ell z^\ell + \mathcal{O}(|z|^{\ell+1})\right),$$

where $c_k = b_k / b_0 = b_k / \beta$.

We write $c_\ell z^\ell = \rho e^{i\alpha} \cdot |z|^\ell e^{i\ell \cdot \arg(z)}$ for some $\rho, \alpha \in \mathbb{R}$. If we can achieve $\alpha + \ell \cdot \arg(z) = \pi$, by tweaking the value of z, we are done. And this can of course be done always, so once we get that, we're left with

$$|P(z_0 + z)| = \beta \cdot \left|1 + \rho r^\ell e^{i\pi} + \mathcal{O}(r^{\ell+1})\right|$$

where $r = |z|$. If you take r very small, you get

$$\beta = |P(z_0)| \leq |P(z_0 + z)| = |\beta| \, |1 - \rho r^\ell + \mathcal{O}(r^{\ell+1})| < \beta$$

which is of course a contradiction. $\square$

3.4 Convex functions

There are similar relations between convex functions and hessian matrices to those we had between convex functions and second derivatives in Analysis 1.

Definition 3.12: Convex

Let $f : A \to \mathbb{R}$ be a function defined on a convex set $A \subset \mathbb{R}^n$. We say f is convex if

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y) \quad \forall x, y \in A, t \in [0,1].$$

Proposition 3.13: Convexity and hessian

Let $U \subset \mathbb{R}^n$ be open, convex. A function $f \in C^2(U)$ is convex if and only if one of the following equivalent conditions hold:

(1) $Hf(x)$ is non-negative definite for all $x \in U$.

(2) $\forall x, y \in U : f(y) - f(x) \geq Df_x(y - x)$.

It can be proven as an exercise that $f \in C^1(U)$ is convex if and only if (2) holds.

Proof. f convex $\implies$ (1). Fix $x, y \in U$. We define $g : [0,1] \to \mathbb{R}$ by

$$g(s) = f((1-s)x + sy) = f(x + sv)$$

for $v = y - x$, which is a convex function of one variable. Since $g \in C^2([0,1])$, we get that $g'' \geq 0$. By the same computation as in the Taylor Theorem (but we're actually doing the chain rule)

$$g''(s) = \frac{1}{2} v \cdot Hf(x + sv)v \geq 0 \implies \forall v \in \mathbb{R}^n : \frac{1}{2} v \cdot Hf(x)v \geq 0$$

setting $s = 0$. But this is just an alternate characterisation of $Hf(x)$ being non-negative definite.

(1) $\implies$ (2). Fix $x, y \in U$ and define $v = y - x$. We have by Taylor Formula for the function $g(t) = f(x + tv)$ on $[0,1]$ that

$$f(y) - f(x) = g(1) - g(0) = g'(0) + \underbrace{\int_0^1 (1-s)g''(s)\,ds}_{\geq 0} \geq Df_x(v) = Df_x(y - x).$$

The integral is non-negative because one can show, like in (a) that $g''(s) \geq 0$ and by chain rule, $g'(t) = Df_{x+vt}(v)$.

(2) $\implies$ f convex. See below, the Jensen inequality proves actually a stronger version of this claim. □

Proposition 3.14: Jensen inequality

Let $U \subset \mathbb{R}^n$ be open convex and $f : U \to \mathbb{R}$ convex. For any set of weights $w_1, \ldots, w_N \in [0,1]$ such that $\sum_{i=1}^N w_i = 1$ and vectors $x_1, \ldots, x_N \in U$, we have

$$f\left(\sum_{i=1}^N w_i x_i\right) \leq \sum_{i=1}^N w_i f(x_i).$$

Proof. We show the statement under the assumption $f \in C^2(U)$ to conclude the previous proposition. The general case can be done with induction, completely without any differentiability. Fix $x_0 = \sum_{i=1}^N w_i x_i$. By assumption f is convex, so f satisfies characterisation (2) of previous proposition and we have

$$f(x_i) \geq f(x_0) + Df_{x_0}(x_i - x_0) \implies w_i f(x_i) \geq w_i f(x_0) + Df_{x_0}(w_i(x_i - x_0)).$$

We can sum this over i to get the Jensen inequality. □

Chapter 4

Inverse, implicit function theorems and submanifolds

We will now get to the hard part of the course (it's so over). For functions of class $C^1(I)$ where $I \subset \mathbb{R}$ is an interval, around any point, where $f'(x_0) \neq 0$, we can take a neighborhood small enough around x_0 such that f is monotone on that neighborhood. The inverse function theorem and proposition about the derivative of the inverse guarantee us the existence and properties of an inverse to f on the small neighborhood.

The goal of this chapter is to find a similar theorem for multivariable functions. This time instead of $f'(x_0) \neq 0$ we will require that $\det Jf(x_0) \neq 0$, a very natural generalisation. We begin with a very special case.

4.1 Small perturbations of the identity

We begin by proving forms of the inverse function theorem and auxiliary lemmas for special cases which we will employ later to prove the general ones. If a function from $\mathbb{R}^n$ to $\mathbb{R}^n$ is very close to the identity, is it invertible then?

Lemma 4.1: Small Lipschitz perturbations of the identity

Let $U \subset \mathbb{R}^n$ be open and assume that $F : U \to \mathbb{R}^m$ is a function of the form $F(x) = x + \phi(x)$, where $\phi : U \to \mathbb{R}^m$ is λ-Lipschitz with $\lambda < 1$.

(1) For any $x_0 \in U$ s.t. $B_r(x_0) \subset U$ with some radius $r > 0$, we have

$$B_{(1-\lambda)r}(F(x_0)) \subset F(B_r(x_0)) \subset F(U).$$

In particular, $F(U)$ is open.

(2) $F : U \to F(U)$ is bijective and F^{-1} is $\frac{1}{1-\lambda}$-Lipschitz.

Proof. To prove (1), given $y \in B_{(1-\lambda)r}(F(x_0))$, we will show that $\exists x \in B_r(x_0)$ s.t. $F(x) = y$, which holds if and only if $x + \phi(x) = y$. So we want to find the x such that $x = y - \phi(x) = T(x)$, for the map $T(x) = y - \phi(x)$. So we're looking for fixed points of T. But T is a contraction map, as for any $x, x' \in U$

$$|T(x) - T(x')| = |y - \phi(x) - y + \phi(x')| = |\phi(x) - \phi(x')| \leq \lambda |x - x'|.$$

And so we would love to apply the Banach Fixed Point Theorem 1.32. But we can't apply it exactly, as we're not sure that the elements of our sequence remain in the domain. We can however copy parts of the proof.

So fix $x_0 \in U$, some radius $r > 0$ and take any $y \in B_{(1-\lambda)r}(F(x_0))$. Consider the sequence defined by

$$x_{k+1} = T(x_k) = y - \phi(x_k).$$

This is of course a priori not well-defined, as we still need to show $x_k \in U$ and once we know that, we wish to show that the sequence has a limit. Let us now do that.

Notice first that

$$|x_1 - x_0| = |y - \phi(x_0) - x_0| = |y - F(x_0)| < (1-\lambda)r.$$

This shows $|x_0 - x_1| < r$, and we can proceed to define x_2. We have

$$|x_2 - x_1| = |T(x_1) - T(x_0)| \leq \lambda \, |x_1 - x_0| < \lambda(1-\lambda)r$$

and one checks that $x_2 \in B_r(x_0)$ as well, by triangle inequality. Doing this over and over, we arrive at

$$|x_{k+1} - x_k| \leq \lambda^k \, |x_1 - x_0| \, ,$$

which implies

$$|x_{k+1} - x_0| \leq \sum_{i=0}^{k} |x_{i+1} - x_i| \leq |x_1 - x_0| \sum_{i=0}^{\infty} \lambda^i < (1-\lambda)r \sum_{i=0}^{\infty} \lambda^i = r$$

hence $x_k \in B_r(x_0)$ for all k. By the same argument as in the Banach Fixed Point Theorem, we get a point $x \in \mathbb{R}^n$ that satisfies $|x - x_0| < r$, is hence in the ball $B_r(x_0)$ and that is a fixed point of the map T, so satisfies $\phi(x) + x = y$.

Now we are in position to prove (2). Assume $F(x) = y$ and $F(x') = y$ for some $x, x' \in U$. This happens if and only if

$$x + \phi(x) = x' + \phi(x') \iff |x - x'| = |\phi(x) - \phi(x')| \leq \lambda \, |x - x'| \, ,$$

and as $\lambda < 1$, this implies $x = x'$.

Now we prove the Lipschitz bound of the inverse. We have $F(x) = y$ and $F(x') = y'$, so

$$\begin{aligned} & y - y' = x - x' + \phi(x) - \phi(x') \\ \implies & |y - y'| \geq |x - x'| - |\phi(x) - \phi(x')| \leq |x - x'| - \lambda \, |x - x'| = (1-\lambda) \, |x - x'| \, . \end{aligned}$$

But this means

$$\left|F^{-1}(y) - F^{-1}(y')\right| = |x - x'| \leq \frac{1}{1-\lambda} \, |y - y'|$$

and F^{-1} is $\frac{1}{1-\lambda}$-Lipschitz. $\square$

Lemma 4.2: Automatic differential of the inverse

Let $U, V \subset \mathbb{R}^n$ be open and $f : U \to V$ and $g : V \to U$ be bijective maps such that $f(g(y)) = y$ for any $y \in V$. Assume that f is differentiable at some point $x_0 \in U$, the differential $Df_{x_0} : \mathbb{R}^n \to \mathbb{R}^n$ is *invertible* and g is Λ-Lipschitz. Then, g is differentiable at the point $y_0 = f(x_0)$ and

$$Dg_{y_0} = (Df_{x_0})^{-1}.$$

Proof. If f is differentiable at x_0, we have

$$f(x) - f(x_0) = Df_{x_0}(x - x_0) + o(x - x_0),$$

as $x \to x_0$. Plugging in $x = g(y)$ for some $y \in V$

$$y - y_0 = f(g(y)) - f(g(y_0)) = Df_{x_0}\Big(g(f(y)) - g(f(y_0))\Big) + o(g(y) - g(y_0))$$

as $y \to y_0$. We can apply $Df_{x_0}^{-1}$ on both sides of this and rearrange to get

$$g(y) - g(y_0) = Df_{x_0}^{-1}(y - y_0) + Df_{x_0}^{-1}\Big(o(g(y) - g(y_0))\Big).$$

And so to prove the theorem, it remains to show that the last term is $o(y - y_0)$. But

$$Df_{x_0}^{-1}\Big(o(g(y) - g(y_0))\Big) = o\Big(Df_{x_0}^{-1}(g(y) - g(y_0))\Big) = o\Big(\|Df_{x_0}^{-1}\|_2 \cdot \Lambda\,|y - y_0|\Big) = o(y - y_0),$$

using where $\|\cdot\|_2$ is the Hilbert-Schmidt norm, we applied Lemma 1.60 and that g is Λ-Lipschitz. Also note that it's not hard to prove that o and linear maps commute (again, using properties of the HS norm). □

Lemma 4.3: Smoothness of the inverse

Let $U \subset \mathbb{R}^{n\times n}$ be the set of invertible real $n \times n$ matrices and let $\Theta : U \to U : X \mapsto X^{-1}$ be the inverse map. Then Θ is $C^\infty(U, \mathbb{R}^{n\times n})$.

Proof. Consider that the inverse of X is given by the classical adjoint divided by the determinant $\det(X) \neq 0$. Since component-wise, X^{-1} is hence expressed as a quotient, product, sum and composition of polynomials of components of X, it is indeed infinitely differentiable, using in particular that no division by zero will ever occur ($\det(X) \neq 0$). □

4.2 Inverse function theorem

As discussed, we will now introduce the generalisation of the inverse function theorem for one variable functions. Note that we're only meaningfully inverting functions from $\mathbb{R}^n$ to $\mathbb{R}^n$.

Theorem 4.4: Inverse function theorem

Let $U \subset \mathbb{R}^n$ open and $f : U \to \mathbb{R}^n$ of class C^1, $x_0 \in U$ such that Df_{x_0} is invertible. Then $\exists U_0 \subset U$ open with $x_0 \in U_0$ such that

(1) $f|_{U_0}$ is injective
(2) $V = f(U_0)$ is open and we get the inverse $g = (f|U_0)^{-1}$
(3) $\forall x \in U_0 : Dg_{f(x)} = (Df_x)^{-1}$
(4) $g = (f|_{U_0})^{-1} : V \to U$ is of class C^1.

Moreover, if $f \in C^k$, then $g \in C^k$.

Proof. We consider without loss of generality the case $x_0 = 0$ and $f(x_0) = f(0) = 0$. Indeed, we would otherwise just be able to redefine $\tilde{f}(x) = f(x - x_0) - f(x_0)$.

We take $L = Df_0$ and define the function $F = L^{-1} \circ f$. Using the chain rule, the differential of this is

$$DF_0 = L^{-1}_{f(0)} \circ Df_0 = \mathrm{id},$$

hence if we define $\phi = F - \mathrm{id}$, we have

$$D\phi_0 = DF_0 - D\mathrm{id}_0 = 0,$$

so $\|J\phi(0)\|_2 = 0$. By continuity of the components of the Jacobian of ϕ, $\exists r > 0$ such that $\|J\phi(x)\|_2 \leq \frac{1}{2}$ in $B_r(0)$. By Proposition 2.21, it hence follows that ϕ is $\frac{1}{2}$-Lipschitz on this small ball. F is hence what we called a "small perturbation of the identity".

Apply Lemma 4.1 on $U_0 = B_r(0)$, telling us that $F|_{U_0}$ is injective, $F(U_0)$ is open and F^{-1} is 2-Lipschitz. Hence $V = f(U_0) = L(F(U_0))$ is open, as L and F continuous, proving (2). And since L is injective and F is injective, we get that $f = L \circ F$ is injective, proving (1).

To prove (3), we want to apply 4.2 at every point $x \in U_0$. For that, we need to show that Df_x is invertible for every $x \in U_0$ and $g = F^{-1} \circ L^{-1}$ is Lipschitz. We have $f = L \circ F = L \circ (\mathrm{id} + \phi)$, hence

$$Df_x = L \circ (\mathrm{id} + D\phi_x).$$

And $L \circ (\mathrm{id} + D\phi_x)$ is invertible because for any $y \in U_0$,

$$|y| - |D\phi_x y| \geq |y| - \|D\phi_x\|_2 \, |y| \geq \frac{1}{2} \, |y| \, ,$$

as we had $\|D\phi_x\|_2 \leq \frac{1}{2}$. But that means that $(\mathrm{id} + D\phi_x)$ will never have a non-trivial kernel, so it's invertible, as we're mapping from $\mathbb{R}^n$ to $\mathbb{R}^n$, which have the same dimension. Hence Df_x becomes the composition of invertible maps. Applying 4.2, we get at every point $x \in U_0$

$$Dg_{f(x)} = (Df_x)^{-1},$$

proving (3).

Since $g = F^{-1} \circ L^{-1}$ is a composition of Lipschitz maps, it itself is Lipschitz and 4.2 applies.

For (4), we need to check is that $g \in C^1(V)$. But $Jg(y) = (Jf(g(y)))^{-1}$, and $\Theta : X \to X^{-1}$ is by 4.3 C^∞, we get that $g \in C^1(V)$, by 2.24. The case for C^k is proved by induction. □

Definition 4.5: Diffeomorphism

Let $U, V \subset \mathbb{R}^n$ be open sets and $f : U \to V$ a function that is C^1, bijective and f^{-1} is C^1 as well. Then we call f a diffeomorphism. f is a C^k-diffeomorphism if f and f^{-1} are both class C^k.

4.3 Implicit function theorem

Let's motivate what we're doing a little better. Consider the circle in 2 dimensions given by the equation $x^2 + y^2 - 1 = 0$. Locally, on certain intervals, the curve of this "shape" can be given by the function $y = \sqrt{1 - x^2}$. This is however not the case for all points, so we will have to take special precautions and assumptions, but that's roughly the point of this.

Theorem 4.6: Implicit function theorem

Let $U \subset \mathbb{R}^n$ be open and $0 < d < n$. Let $f \in C^k(U, \mathbb{R}^{n-d})$ and $(x_0, y_0) \in U$ with $x_0 \in \mathbb{R}^d, y_0 \in \mathbb{R}^{n-d}$, and assume that in the block matrix

$$Jf(x_0, y_0) = \Big(J_x f(x_0, y_0) \mid J_y f(x_0, y_0) \Big) \in \mathbb{R}^{(n-d)\times n},$$

where $J_x f(x_0, y_0) \in \mathbb{R}^{(n-d)\times d}$ and $J_y f(x_0, y_0) \in \mathbb{R}^{(n-d)\times(n-d)}$, we know that $J_y f(x_0, y_0)$ is invertible.
Then $\exists U_0 = B_r(x_0) \times B_s(y_0) \subset U$ and there is a function $g : B_r(x_0) \to B_s(y_0)$ such that,

$$\{(x, y) \in U_0 \mid f(x, y) = 0\} = \{(x, y) \in U_0 \mid y = g(x)\}$$

and g is class C^k. So the level set of 0 is the graph of a C^k function g.
And finally, for all $x \in B_r(x_0)$, the Jacobi of g is given by

$$Jg(x) = -J_y f(x, g(x))^{-1} \cdot J_x f(x, g(x)).$$

Proof. Fix x_0 and y_0 as in the statement. Define the map $\Phi : U \to \mathbb{R}^n$ such that for $x \in \mathbb{R}^d, y \in \mathbb{R}^{n-d}$,

$$\Phi(x, y) = (x, f(x, y)).$$

We compute the block matrix

$$J\Phi(x_0, y_0) = \left(\begin{array}{c|c} I_{d\times d} & 0 \\ \hline J_x f(x_0, y_0) & J_y f(x_0, y_0) \end{array} \right).$$

By various rules of linear algebra, this is invertible and we can apply the Inverse Function Theorem 4.4 to get $U_0 \subset U$ such that $\Phi|_{U_0}$ is invertible, $\Phi(U_0)$ is open and $(\Phi|_{U_0})^{-1}$ is class C^k. Without loss of generality, we reduce U_0 to a "cylinder" set of the form

$$U_0 = B_r(x_0) \times B_s(y_0) \subset U.$$

Call $\Psi = \Phi^{-1} : V \to U_0$, with $V = \Phi(U_0)$. Using bijectivity of Ψ, Φ, define for $(\xi, \eta) \in V$:

$$(\xi, \eta) = \Phi(x, y) = (x, f(x, y)) \iff (x, y) = \Psi(\xi, \eta)$$

with $\xi \in \mathbb{R}^d, \eta \in \mathbb{R}^{n-d}$. We see that $\xi = x$ and we can express Ψ in the form

$$\Psi(\xi, \eta) = (\xi, G(\xi, \eta))$$

for some map $G : V \to B_s(y_0)$ that is of course C^k.

Because we defined the variables in such a way,

$$\eta = f(x, y) = 0 \iff (x, y) = \Psi(\xi, \eta) = (\xi, G(\xi, \eta)) = (x, G(x, 0)).$$

This shows that

$$f(x,y) = 0 \iff y = G(x,0).$$

In other words, $g(x) = G(x,0)$ gives the function g we were searching for.

To get the formula for $Dg(x)$, we differentiate the identity $f(x,g(x)) = 0$ using the chain rule (check component-wise, it's a pain)

$$J_x f(x,g(x)) + J_y f(x,g(x)) \cdot Jg(x) = 0,$$

where $J_y f(x,g(x))$ is invertible, since it's close enough to $J_y f(x_0,y_0)$[1]. $\square$

4.4 Submanifolds of $\mathbb{R}^n$

While the initial definition might be a little out of place, we will see what it really means.

DEFINITION 4.7: SUBMANIFOLD

A set $M \subset \mathbb{R}^n$ is called a d-dimensional submanifold of class C^k if $\forall p \in M$ $\exists U \subset \mathbb{R}^n$ such that $p \in U$, $\exists V \subset \mathbb{R}^n$ open and $\Psi : U \to V$, a C^k-diffeomorphism called the *submanifold chart* such that

$$\Psi(M \cap U) = \{y \in V \mid y_{d+1} = \cdots = y_n = 0\}.$$

We want to first heuristically present the different ways to describe a manifold. Besides the original definition, there are the following characterisations, which we will prove are equivalent. In almost each one, at every point $p \in M$, we require the existence of an open set U that contains p and satisfies one of the following criteria.

- Diffeomorphic: see definition above.
- Implicit: $M \cap U = \{x \in U \mid f(x) = 0\}$, where $f \in C^k(U, \mathbb{R}^{n-d})$ such that the rank of $Df(p)$ is $n-d$.
- Parametric: $M \cap U = \left\{G(y) \mid y \in V \subset \mathbb{R}^d\right\}$ for a map $G \in C^k(V, \mathbb{R}^n)$, where $V \subset \mathbb{R}^d$ is open and $DG(y)$ is of rank d, with $G(y) = p$.
- Graphical: $M \cap U = \left\{(y, g(y)) \mid y \in V \subset \mathbb{R}^d\right\}$ for some open set $V \subset \mathbb{R}^d$ and $g \in C^k(V, \mathbb{R}^{n-d})$.

What does any of this mean? Ignore for now all the additional qualifiers about rank of the differential and the details. Those only tell us the "dimension" of the manifold.

The diffeomorphic definition might be the least intuitive and understandable one. But what it basically says is that M is a manifold if around any point, you can map the neighborhood of that point smoothly and bijectively into a set V where there is only d non-zero coordinates, essentially reducing the set $M \cap U$ to something which is smoothly describable by just d parameters and "flattening" it out.

The implicit representation is the one people are usually familiar with, so just "solutions to an equation", which often define nice shapes (examples below).

The parametric representation is also pretty intuitive. Because the manifold is d-dimensional, we have d parameters to tune and the manifold becomes just the image of some special function G that assigns points to parameters.

[1] I know it's a little heuristic, but it's actually true, you can estimate the Hilbert-Schmidt norm from below like in the Inverse Function Theorem, left as an exercise. But most of the time more that half of this stuff doesn't even come up in computation, it's just very technical.

And finally, the graphical representation is probably actually the easiest. It is literally just the set of points $(x, f(x))$, like in $\mathbb{R}^2$, when graphing functions. Note also that we will allow permutations of this, so the pairs might not be so nicely ordered, but it remains the same in essence.

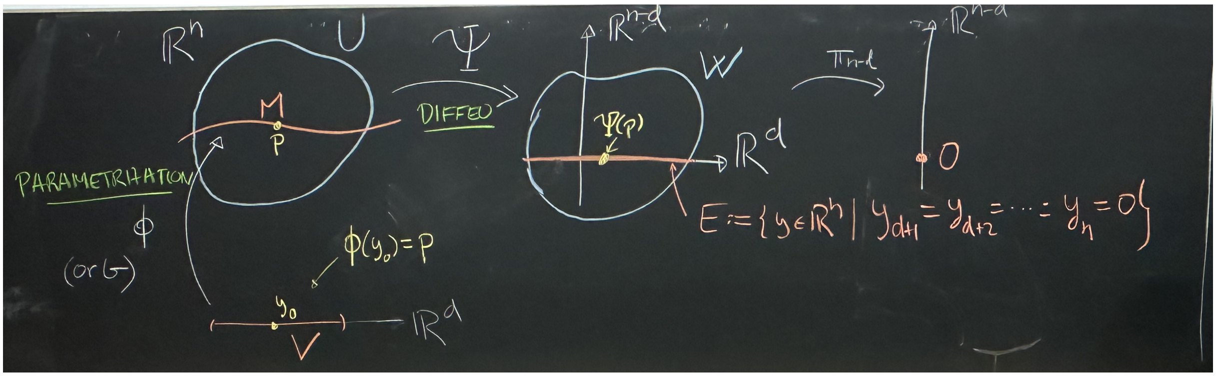


Figure 4.1: Graphical relation between the parametric and diffeomorphic representation of submanifold.

EXAMPLE 4.8: EXAMPLES OF SUBMANIFOLDS IN DIFFERENT REPRESENTATIONS

Plane: Consider a plane in $\mathbb{R}^3$. The implicit representation would be via $f(x, y, z) = ax + by + cz - h = 0$.
The parametric would be $(s, t) \mapsto sv_1 + tv_2 + p$ for vectors v_1, v_2 and a point p. Notice that here we see why the plane is a 2-dimensional manifold.
The graphical representation is for instance $z = -\frac{a}{c}x - \frac{b}{c}y + \frac{h}{c}$, essentially assigning each x, y a coordinate z, as if graphing in Desmos.
Sphere: Consider a sphere in $\mathbb{R}^3$ with fixed radius r. The implicit representation would be via $f(x, y, z) = x^2 + y^2 + z^2 - r^2 = 0$.
The parametric would be $(\varphi, \theta) \mapsto (r \sin\varphi \cos\theta,\ r \sin\varphi \sin\theta,\ r \cos\varphi)$ for angles $\varphi \in [0, \pi], \theta \in [0, 2\pi)$. Notice that here we see why the sphere is a 2-dimensional manifold. (Ignore details about whether the parameter set is open).
The graphical representation is for instance $z = \pm\sqrt{r^2 - x^2 - y^2}$, essentially assigning each x, y a coordinate z locally, as if graphing in Desmos. Notice that it's not always possible to express the whole manifold as one function, which is why we will require most of the properties to be very local.
Torus: Consider a torus in $\mathbb{R}^3$. The implicit representation would be via $f(x, y, z) = \left(\sqrt{x^2 + y^2} - R\right)^2 + z^2 - r^2 = 0$.
The parametric would be $(\varphi, \theta) \mapsto ((R + r\cos\varphi)\cos\theta,\ (R + r\cos\varphi)\sin\theta,\ r\sin\varphi)$ for angles $\varphi, \theta \in [0, 2\pi)$. Notice that here we see why the torus is a 2-dimensional manifold.
The graphical representation is not globally possible in the form $z = g(x, y)$, but locally one can solve $z = \pm\sqrt{r^2 - \left(\sqrt{x^2 + y^2} - R\right)^2}$, assigning each x, y a coordinate z only in suitable regions.

DEFINITION 4.9: PERMUTATION OF COORDINATES

A map $P : \mathbb{R}^n \to \mathbb{R}^n$ is called a permutation of coordinates if

$$P(x_1, \dots, x_n) = (x_{\sigma 1}, \dots, x_{\sigma n}),$$

where σ is a permutation.

Now we will state the 4 equivalent criteria. They may seem very arbitrary at first, but if you go back and read the simple descriptions (you will notice these are a little more nuanced), it should be easier to understand why they are all getting at the same thing.

PROPOSITION 4.10: EQUIVALENT CHARACTERISATIONS OF SUBMANIFOLDS

The following 4 characterisation for $M \subset \mathbb{R}^n$ being a submanifold are equivalent:

(1) *Diffeomorphic*: M is a d-dimensional C^k-submanifold.

(2) *Implicit*: For all $p \in M$, there are: open set $U \subset \mathbb{R}^n$ such that $p \in U$, a map $f \in C^k(U, \mathbb{R}^{n-d})$ such that $Df(p)$ has rank $n-d$ and

$$M \cap U = \{x \in U \mid f(x) = 0\}.$$

(3) *Parametric*: For all $p \in M$, there is $V \subset \mathbb{R}^d$, $y \in V$ and a map $G \in C^k(V, \mathbb{R}^n)$ such that $G(y) = p$, $DG(y)$ has rank d and for any $V' \subset V$, there is $U' \subset \mathbb{R}^n$ open such that

$$M \cap U' = G(V').$$

(4) *Graphical*: For any $p \in M$, there is $U \subset \mathbb{R}^n$ open such that $p \in U$, a set $V \subset \mathbb{R}^d$ open and a map $g \in C^k(V, \mathbb{R}^{n-d})$ such that

$$M \cap U = P(\mathrm{graph}(g))$$

such that $P : \mathbb{R}^n \to \mathbb{R}^n$ is a permutation of coordinates and

$$\mathrm{graph}(g) = \{x \in \mathbb{R}^n \mid (x_{d+1}, \ldots, x_n) = g(x_1, \ldots, x_d)\}.$$

Proof. (1) $\implies$ (2). Let $p \in M$. Since M is a submanifold, we have the diffeomorphism $\Psi : U \to V$ such that U is open, contains p and

$$x \in M \cap U \iff \Psi(x_{d+1}) = \cdots = \Psi(x_n) = 0.$$

Hence we construct $f : U \to \mathbb{R}^{n-d}$ as $f(x) = (\Psi_{d+1}(x), \ldots, \Psi_n(x))$. Then $f \in C^k(U, \mathbb{R}^{n-d})$ and $Df(p)$ has rank $n-d$ because it has the columns $D\Psi(p)$, which is entirely invertible. Then (2) trivially holds.

(2) $\implies$ (4): Let $p \in M$. If $Df(p)$ has maximal rank $n-d$, then for a suitable permutation of the variables P, when we consider the map $\tilde{f} = f \circ P^{-1}$, the $(n-d) \times (n-d)$ matrix

$$\left(\partial_i \tilde{f}_j(q)\right)_{1 \le j \le n-d,\, d+1 \le i \le n} = \begin{pmatrix} \partial_{d+1}\tilde{f}_1(q) & \partial_{d+2}\tilde{f}_1(q) & \cdots & \partial_n \tilde{f}_1(q) \\ \partial_{d+1}\tilde{f}_2(q) & \partial_{d+2}\tilde{f}_2(q) & \cdots & \partial_n \tilde{f}_2(q) \\ \vdots & \vdots & \ddots & \vdots \\ \partial_{d+1}\tilde{f}_{n-d}(q) & \partial_{d+2}\tilde{f}_{n-d}(q) & \cdots & \partial_n \tilde{f}_{n-d}(q) \end{pmatrix}$$

is invertible at $q = P(p)$. This is because we know that in the matrix $Df(p)$, there are $n-d$ linearly independent columns. And $D\tilde{f}(q) = D(f \circ P^{-1})(q) = Df(p) \cdot P^{-1}$, so $D\tilde{f}(q)$ is just $Df(p)$, but with permuted columns. And for a suitable permutation, this will be such that the last $n-d$ columns are linearly independent, hence see above.

Hence $\tilde{f}$ satisfies the assumptions of the Implicit Function Theorem 4.6 and therefore there exist an open cylinder $U_0 \subset U$ centered at q, $V \subset \mathbb{R}^d$ open, and a map $g : V \to \mathbb{R}^{n-d}$ of class C^k such that for all $x \in U_0$,

$$\tilde{f}(x) = 0 \iff (x_{d+1}, \ldots, x_n) = g(x_1, \ldots, x_d) \iff P^{-1}(x) \in M.$$

Note the last equivalence comes from the assumption (2). Choosing $x = Pz$ for z in a neighborhood of p we conclude.

(4) $\implies$ (3). Let $p \in M$. By assumption (4), we get a map $U \subset \mathbb{R}^n$ open that contains p, $V \subset \mathbb{R}^d$ open and $g : V \to \mathbb{R}^{n-d}$ such that $M \cap U = P(\text{graph}(g))$. For $y \in V$, define $\tilde{G}(y) = (y, g(y))$ s.t. $M \cap U = (P \circ \tilde{G})(V)$, simply the image. Take $G = P \circ \tilde{G}$. By the differentiation laws 2.24, $G \in C^k(V, \mathbb{R}^n)$.

Find the $y \in V$ such that $p = G(y)$, which exists by assumption $p \in U$. Now we show that $DG(y)$ is of rank d. Since

$$G(y_1, \ldots, y_d) = (y_1, \ldots, y_d, g(y_1, \ldots, y_d)) \quad \implies \quad D\tilde{G}(y) = \begin{pmatrix} I_d \\ Dg(y) \end{pmatrix},$$

its columns are linearly independent, hence $\text{rank}(D\tilde{G}(y)) = d$ and as P is a linear iso, it follows that $\text{rank}(DG(y)) = d$ as well.

Finally, for $V' \subset V$, let $U' = P(V' \times \mathbb{R}^{n-d}) \cap U \subset U$, such that

$$M \cap U' = P(\text{graph}(g)) \cap P(V' \times \mathbb{R}^{n-d}) = G(V').$$

(3) $\implies$ (1). Let $p \in M$. Let $y_0 \in V$ such that $G(y_0) = p$. Since $DG(y_0)$ has rank d, we can extend its columns to a basis of $\mathbb{R}^n$, i.e. there exist vectors $v_{d+1}, \ldots, v_n \in \mathbb{R}^n$ such that

$$\Big(DG(y_0) \mid v_{d+1} \mid \cdots \mid v_n\Big)$$

is invertible. Define the map $\Phi \in C^k(V \times \mathbb{R}^{n-d}, \mathbb{R}^n)$ by

$$\Phi(y_1, \ldots, y_n) = G(y_1, \ldots, y_d) + \sum_{j=d+1}^{n} v_j y_j.$$

Then

$$D\Phi(y_0, 0) = \Big(DG(y_0) \mid v_{d+1} \mid \cdots \mid v_n\Big)$$

is invertible. Hence by the Inverse Function Theorem 4.4, there exist open sets $W \subset V \times \mathbb{R}^{n-d}$ containing $(y_0, 0)$ and $U = \Phi(W)$ open containing $p = \Phi((y_0, 0))$, and a map $\Psi \in C^k(U, W)$ such that

$$\forall x \in U : \quad x = \Phi(\Psi(x)).$$

Now define

$$V' := \{y \in V \mid (y, 0) \in W\}.$$

Then V' is open in $\mathbb{R}^d$ and $y_0 \in V'$. By (3), there exists an open set $U' \subset \mathbb{R}^n$ such that

$$M \cap U' = G(V').$$

Replacing U by $U \cap U'$, we may assume

$$M \cap U = G(V').$$

Then for $x \in U$ we have the following. If $x \in M$, then $x \in G(V')$, so there exists $y \in V'$ such that $x = G(y) = \Phi((y, 0))$. Since Φ is bijective on W, it follows that $\Psi(x) = (y, 0)$, hence $\Psi_{d+1}(x) = \cdots = \Psi_n(x) = 0$.

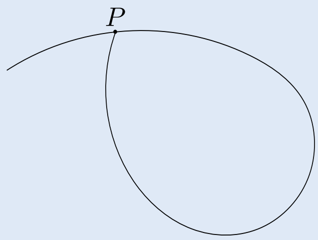
Conversely, if $\Psi_{d+1}(x) = \cdots = \Psi_n(x) = 0$, then $\Psi(x) = (y, 0)$ for some $y \in V'$ (since $\Psi(x) \in W$), and thus

$$x = \Phi(\Psi(x)) = \Phi((y, 0)) = G(y) \in G(V') = M.$$

□

REMARK 4.11: THE ISSUE WITH PARAMETRISATION

In the parametric characterisation, why do we require $M \cap U' = G(V')$ for any $V' \subset V$ and not just $M \cap U = G(V)$ as a whole? This has a good reason when loops enter the game. Consider the following curve.



Now, it's apparently possible to parametrise this figure on an open set around P by walking along the curve and infinitesimally approaching the point P from below. But this curve can clearly not be a manifold because precisely at P, any chart Ψ will *not* straighten out the curve sufficiently, because it is a 3-fold intersection. Rigorously, if you zoom in close enough to P and remove the point P, Ψ will essentially have to map 3 disconnected components in $\mathbb{R}^2$ to 3 disconnected components in $\mathbb{R} \times \{0\}$, but where all of those 3 disconnected components would have to be in contact with the image of P, which is obviously impossible.
Definitely also see Problem Sheet 7 and 10 for hands-on understanding and application of the theorem.

DEFINITION 4.12: TANGENT AND NORMAL VECTOR

Let $M \subset \mathbb{R}^n$, a d-dimensional submanifold. Then $\tau \in \mathbb{R}^n$ is tangent to M at $p \in M$ if $\exists p_k \in M, r_k > 0$ sequences such that

$$\lim_{k\to\infty} \frac{p_k - p}{r_k} = \tau.$$

We call the set of all tangent vectors T_pM the tangent space.
A vector $\nu \in \mathbb{R}^n$ is normal (perpendicular) to M at p if $\nu \cdot \tau = 0$ for all $\tau \in T_pM$.

PROPOSITION 4.13: TANGENT SPACE

Let $M \subset \mathbb{R}^n$ be a d-dimensional manifold. Let $p \in M$ and let $\Psi : U \to V$ be the submanifold chart and $E = \mathbb{R}^d \times \{0_{n-d}\} \subset \mathbb{R}^n$ such that $\Psi(M \cap U) \subset E$. For all $\tau \in \mathbb{R}^n$, $\tau \in T_pM$ if and only if $\tau \in D\Psi^{-1}_{\Psi(p)}(E)$. In particular,

$$T_pM = D\Psi^{-1}_{\Psi(p)}(E)$$

is a d-dimensional subspace of $\mathbb{R}^n$.

Proof. We're proving that any tangent vector $\tau \in T_pM$ belongs to $\Psi^{-1}_{\Psi(p)}(E)$. Indeed, consider that for some sequence p_k and r_k giving a tangent vector $\tau \in T_pM$,

$$\lim_{k\to\infty} \frac{\Psi(p_k) - \Psi(p)}{r_k} = \lim_{k\to\infty} \frac{D\Psi_p(p_k - p) + o(|p_k - p|)}{r_k} = \lim_{k\to\infty} D\Psi_p\left(\frac{p_k - p}{r_k}\right) = D\Psi_p(\tau) \in E,$$

hence $\tau \in D\Psi^{-1}_{\Psi(p)}(E)$. Notice that $\frac{|p_k - p|}{r_k}$ is bounded, hence $\frac{o(|p_k-p|)}{r_k} \leq \varepsilon \frac{|p_k-p|}{r_k}$ (for any $\varepsilon > 0$, eventually), so it goes to zero and we can drop that term and get the result.

Conversely, fix $p \in M$ take $\tau \in D\Psi^{-1}_{\Psi(p)}(E)$. Because the image of Ψ is open and E is a subspace, we can take $t > 0$ small enough, $v \in E$ so that $\tau = D\Psi^{-1}_{\Psi(p)}(v)$ and get

$$\Psi(p) + tv \in E \cap V = \Psi(M \cap U),$$

defining $p_t = \Psi^{-1}(\Psi(p) + tv) \in M \cap U$. Since $\Psi^{-1} \in C^1$, we have

$$p_t - p = \Psi^{-1}(\Psi(p) + tv) - \Psi^{-1}(\Psi(p)) = D\Psi^{-1}_{\Psi(p)}(tv) + o(|tv|) = t \cdot \tau + o(t).$$

Taking any sequence $t_k \to 0$, we get $\frac{p_{t_k} - p}{t_k} \to \tau$, proving $\tau \in T_pM$.

To see that T_pM has dimension d, notice that $D\Psi^{-1}_{\Psi(p)}$ is a linear isomorphism and T_pM is the image of that map applied to a d-dimensional subspace E. $\square$

REMARK 4.14: TANGENT SPACE IS DIFFERENTIAL OF PARAMETRISATION

We notice that for a parametrisation ϕ of M around $p = \phi(y_0) \in M$,

$$T_pM = D\phi_{y_0}(\mathbb{R}^d).$$

Why? Let $v \in \mathbb{R}^d$. For $t > 0$ let $\phi(y_0 + tv) = p_t \in M$, then

$$D\phi_{y_0}(v) = \lim_{t \to 0} \frac{\phi(y_0 + tv) - \phi(y_0)}{t} = \lim_{t \to 0} \frac{p_t - p}{t} \to \tau \in T_pM.$$

Hence $D\phi_{y_0}(\mathbb{R}^d) \subset T_pM \subset \mathbb{R}^n$. And the map $D\phi_{y_0}$ has full rank d, and its image is a subspace of T_pM, which is d-dimensional by Proposition 4.13. Hence they must coincide.

Let's maybe apply this to see what we can do in $\mathbb{R}^3$ before we finish off this chapter.

DEFINITION 4.15: PARAMETRISED SURFACE

Let $V \subset \mathbb{R}^2$ be open. A map $\phi : V \to \mathbb{R}^3$ of class C^k such that rank $D\Phi_y = 2$ for all $y \in V$ and

$$\forall V' \subset V \text{ open } \exists U' \subset \mathbb{R}^3 \text{ open s.t. } \phi(V) \cap U' = \phi(V')$$

is called a parametrisation of the surface $M = \phi(V)$. Notice that this is the characterisation (3) in Proposition 4.10, making M into an actual 2-dimensional submanifold of $\mathbb{R}^3$.

EXAMPLE 4.16

Let $u, v \in \mathbb{R}^2$. We define $\phi : \mathbb{R}^2 \to \mathbb{R}^3$ by

$$\phi(u, v) = (\cos v \cdot u, \sin v \cdot u, v)$$

and this gives you a so-called helicoid. It is left as an exercise to check that the Jacobian $J\phi(u, v)$ has rank 2. Then, $\tau_1 = \partial_u\phi(u, v)$ and $\tau_2 = \partial_v\phi(u, v)$ are both tangent vectors, by Remark 4.14, and since the rank of the Jacobian is 2, these are linearly independent and at $p = \phi(u, v)$,

$$T_pM = \operatorname{Sp}(\tau_1, \tau_2), \quad M = \phi(V).$$

And when you take the cross product

$$\nu(u, v) = \partial_u\phi \times \partial_v\phi,$$

you actually get a normal vector at $\phi(u, v) = p$.

Chapter 5

Multidimensional integration

5.1 Jordan measure in $\mathbb{R}^n$

Now that we have learned about differentiation and surfaces and manifolds in $\mathbb{R}^n$, we can move onto integration. But before we get there, we need to find out what is volume.

We introduce convenient notation for this chapter. If $X \subset \mathbb{R}^n$ is a set and $\rho > 0$ and $a \in \mathbb{R}^n$, we write

$$\rho X + a = \{\rho x + a \mid x \in X\}.$$

Visually, this corresponds to scaling by ρ and translation by a.

Definition 5.1: Dyadic cube

Given $p \in \mathbb{N}$, we say that $Q \subset \mathbb{R}^n$ is a dyadic cube of length 2^{-p} if

$$Q_p(a) = 2^{-p}(a + [0,1)^n)$$

for some $a \in \mathbb{R}^n$.

Definition 5.2: Dyadic subset

We call $F \subset \mathbb{R}^n$ a dyadic set if it is a finite union of disjoint dyadic cubes, i.e.

$$F = \bigcup_{\ell=1}^{N} 2^{-p}(a_\ell + [0,1)^n)$$

such that all the sets we are unioning over are disjoint (so $a_i \in \mathbb{Z}^n$ and $a_i \neq a_j$ for $i \neq j$).

Definition 5.3: Minecraft measure

For $F \subset \mathbb{R}^n$ dyadic, so $F = \bigcup_{\ell=1}^{N} 2^{-p}(a_\ell + [0,1)^n)$, we define the volume of F as

$$\mu(F) = N \cdot 2^{-pn}.$$

One easily checks that this formula is well-defined (size of p doesn't matter).

Proposition 5.4: Properties of μ

Unions, intersections and differences of dyadic sets are dyadic and μ for dyadic sets satisfies the following:

(1) Additivity: $\mu(E \cup F) = \mu(E) + \mu(F)$ for E, F dyadic and disjoint.

(2) Normalised: $\mu([0,1)^n) = 1$.

(3) Translation: $\mu(E + \tau) = \mu(E)$ for all dyadic E and $\tau \in 2^{-p}\mathbb{Z}^n$.

Proof. Trivial. □

Remark 5.5

For deeper reasons, μ is the only measure defined on dyadic sets that satisfies (1), (2) and (3). One can show it as an exercise. But for sure, these are the three properties we would definitely expect of a function that returns volumes of Minecraft cubes.

Now we can move onto measuring the volumes of general sets.

Definition 5.6: Outer, inner volume

Let $E \subset \mathbb{R}^n$. Define

$$\mu_{\text{out}}(E) = \inf \{\mu(G) \mid G \supset E \text{ is dyadic}\}, \quad \mu_{\text{in}}(E) = \sup \{\mu(F) \mid F \subset E \text{ is dyadic}\}.$$

Definition 5.7: Jordan measurable sets

Let $E \subset \mathbb{R}^n$ be bounded. E is called Jordan measurable if $\mu_{\text{in}}(E) = \mu_{\text{out}}(E)$ and then we write

$$\text{vol}_n(E) = \mu(E) := \mu_{\text{in}}(E).$$

Lemma 5.8: Properties of Jordan measure

The following properties hold for the inner and outer volumes.

(1) μ_{out} is subadditive:

$$E \subset \bigcup_{i=1}^N E_i \implies \mu_{\text{out}} \leq \sum_{i=1}^N \mu_{\text{out}}(E_i).$$

(2) μ_{in} is superadditive:

$$E \supset \bigcup_{i=1}^N E_i \text{ and } E_i \text{ are pairwise disjoint} \implies \mu_{\text{in}}(E) \geq \sum_{i=1}^N \mu_{\text{in}}(E_i).$$

Proof. It's enough to show this as an exercise for 2 sets, it's fairly conceptually simple, formally annoying at most. Then do induction. □

Lemma 5.9: Volume of boxes

Let $E = (a_1, b_1) \times \cdots \times (a_n, b_n) \subset \mathbb{R}^n$ where $a_i < b_i$. Then $E, \overline{E}$ are Jordan measurable and

$$\mu(E) = \mu(\overline{E}) = \prod_{i=1}^n (b_i - a_i).$$

Proof. For $p \in \mathbb{N}$ and $t \in \mathbb{R}$, we define

$$\underline{t} = \max\left\{2^{-p}k \mid k \in \mathbb{Z} \text{ s.t. } 2^{-p}k < t\right\}, \quad \overline{t} = \min\left\{2^{-p}k \mid k \in \mathbb{Z} \text{ s.t. } 2^{-p}k > t\right\}.$$

Notice that always $t - \underline{t}, \overline{t} - t \leq 2^{-p}$, $t < \overline{t}$ and $t > \underline{t}$. From this we get that for any $1 \leq i \leq n$,

$$[a_i, b_i] \subset [\underline{a_i}, \overline{b_i}), \quad [\overline{a_i}, \underline{b_i}) \subset (a_i, b_i).$$

Then it's clear that

$$\overline{E} \subset \prod_{i=1}^{n} [\underline{a_i}, \overline{b_i}) = G_p, \quad E \supset \prod_{i=1}^{n} [\overline{a_i}, \underline{b_i}) = F_p$$

and G_p, F_p are dyadic sets. Notice that also

$$\underline{b_i} - \overline{a_i} \geq b_i - a_i, \quad \overline{b_i} - \underline{a_i} \leq b_i + 2^{-p} - (a_i - 2^{-p}) = b_i - a_i + 2^{1-p}.$$

Hence[1], we get for sufficiently big p

$$\mu(F_p) = \prod_{i=1}^{n} (\underline{b_i} - \overline{a_i}) \geq \prod_{i=1}^{n} (b_i - a_i), \quad \mu(G_p) = \prod_{i=1}^{n} (\overline{b_i} - \underline{a_i}) \leq \prod_{i=1}^{n} (b_i - a_i + 2^{1-p}).$$

Letting $p \to \infty$, using $\mu(F_p) \leq \mu_{\text{in}}(E) \leq \mu_{\text{out}}(\overline{E}) \leq \mu(G_p)$, we get the statement. □

Proposition 5.10: Unions, intersections, differences of Jordan measurable sets

Let E_1 and E_2 be Jordan measurable. Then $E_1 \cup E_2$, $E_1 \cap E_2$, $E_1 \setminus E_2$ are all Jordan measurable and μ is additive, so if E_1, E_2 are Jordan measurable and disjoint then $\mu(E_1 + E_2) = \mu(E_1) + \mu(E_2)$.

Proof. First we prove additivity and measurability of the disjoint union. So assume that E_1, E_2 are disjoint. By superadditivity and subadditivity 5.8,

$$\mu_{\text{in}}(E_1) + \mu_{\text{in}}(E_2) \leq \mu_{\text{in}}(E_1 \cup E_2) \leq \mu_{\text{out}}(E_1 \cup E_2) \leq \mu_{\text{out}}(E_1) + \mu_{\text{out}}(E_2).$$

But since the last and first terms equal, the inequality collapses to an equality.

Now we inspect the difference. By assumption, for any $\varepsilon > 0$, $\exists G_i, F_i$ dyadic such that $F_i \subset E_i \subset G_i$ and

$$\mu(G_i) - \mu(F_i) < \frac{\varepsilon}{2}.$$

By inclusion,

$$F_1 \setminus G_2 \subset E_1 \setminus E_2 \subset G_1 \setminus F_2$$

and we know that $F_1 \setminus G_2$ and $G_1 \setminus F_2$ are dyadic. Purely set theoretically,

$$(G_1 \setminus F_2) \setminus (F_1 \setminus G_2) \subset (G_1 \setminus F_1) \cup (G_2 \setminus F_2).$$

Hence by subadditivity 5.8,

$$\mu((G_1 \setminus F_2) \setminus (F_1 \setminus G_2)) \leq \mu(G_1 \setminus F_1) + \mu(G_2 \setminus F_2) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Now we look at the intersection. Set theoretically, one writes

$$E_1 \cap E_2 = E_2 \setminus (E_2 \setminus E_1)$$

and so $E_1 \cap E_2$ is Jordan measurable. And finally, we express

$$E_1 \cup E_2 = (E_1 \setminus E_2) \cup (E_2 \setminus E_1) \cup (E_1 \cap E_2),$$

where the second union is disjoint, so the general union is Jordan measurable as well. □

[1] If you find a dyadic decomposition and compute the volume, everybody knows this is true, we just had to get the boxes to have coordinates that are powers of two so it's easy to fit dyadic cubes into it.

Lemma 5.11: Sandwich

Let $E \subset \mathbb{R}^n$. Assume for any $\varepsilon > 0$, there exist $F, G \subset \mathbb{R}^n$ Jordan measurable such that $F \subset E \subset G$ and $\mu(G) - \mu(F) < \varepsilon$. Then E is Jordan measurable.

Proof. For any $\varepsilon > 0$, there exist Jordan $F_\varepsilon \subset E \subset G_\varepsilon$ such that

$$\mu(F_\varepsilon) = \mu_{\text{in}}(F_\varepsilon) \le \mu_{\text{in}}(E) \le \mu_{\text{out}}(E) \le \mu_{\text{out}}(G_\varepsilon) = \mu(G_\varepsilon).$$

This means in particular that $\mu_{\text{out}}(E) - \mu_{\text{in}}(E) \le \mu(G_\varepsilon) - \mu(F_\varepsilon) < \varepsilon$. This forces $\mu_{\text{out}}(E) = \mu_{\text{in}}(E)$. $\square$

Definition 5.12: Jordan null set

A set $E \subset \mathbb{R}^n$ is called null if $\mu_{\text{out}}(E) = 0$.

Lemma 5.13: Closure, interior and the Jordan measure

Let $E \subset \mathbb{R}^n$ be Jordan measurable. Then $E^\circ, \overline{E}$ are both Jordan measurable and $\mu(E^\circ) = \mu(E) = \mu(\overline{E})$ give all the same volume.

Proof. Let $\varepsilon > 0$. Since E is Jordan, $\exists F_\varepsilon, G_\varepsilon$ dyadic such that $F_\varepsilon \subset E \subset G_\varepsilon$ and $\mu(G_\varepsilon) - \mu(F_\varepsilon) < \varepsilon$. We have

$$F_\varepsilon^\circ \subset E^\circ \subset E \subset \overline{E} \subset \overline{G}_\varepsilon.$$

Since F_ε is dyadic, it can be written as a disjoint union of Minecraft cubes

$$F_\varepsilon = \bigcup_{\ell=1}^{N} (a_\ell + [0,1)^n)2^{-p} \implies F_\varepsilon^\circ \subset \bigcup_{\ell=1}^{N} (a_\ell + (0,1)^n)2^{-p}.$$

And we proved that the volume of the open cube is the same as the volume of the closed cube in 5.9. Hence, for the dyadic $\mu(F_\varepsilon^\circ) = \mu(F_\varepsilon)$ and the same holds for $\mu(\overline{G}_\varepsilon) = \mu(G_\varepsilon)$. Hence $\mu(\overline{G}_\varepsilon) - \mu(F_\varepsilon^\circ) < \varepsilon$ and passing this into the Sandwich, we get the Lemma. $\square$

Theorem 5.14: Jordan measurability and boundary

$E \subset \mathbb{R}^n$ is Jordan measurable if and only if E is bounded and ∂E is null.

Proof. $\implies$: If E is Jordan measurable, then E is clearly bounded. If it was unbounded, you'd need infinitely many cubes to approximate its volume from the outside. Also, by Lemma 5.13,

$$\mu(\overline{E}) = \mu(E^\circ) = \mu(E).$$

By 5.10, $\partial E = \overline{E} \setminus E^\circ$ is Jordan measurable and by additivity (the interior and boundary are disjoint)

$$\mu(\overline{E}) = \mu(E^\circ) + \mu(\partial E) \implies \mu(\partial E) = 0.$$

$\impliedby$: Assume that E is bounded and $\mu_{\text{out}}(\partial E) = 0$. Given $p \in \mathbb{N}$, let

$$F_p = \bigcup \left\{ Q \text{ dyadic cube of size } 2^{-p} \mid Q \subset E^\circ \right\}$$

and

$$G_p = \bigcup \left\{ Q \text{ dyadic cube of size } 2^{-p} \mid Q \cap \overline{E} \neq \emptyset \right\}.$$

Then we clearly have $F_p \subset E^\circ \subset \overline{E} \subset G_p$. Since ∂E is null, it is Jordan measurable and for $\varepsilon > 0$, $\exists H_\varepsilon$ such that $\partial E \subset H_\varepsilon$ is dyadic and $\mu(H_\varepsilon) < \varepsilon$.

Let p be the pixel size of H_ε. Then

$$G_p \setminus F_p \subset H_\varepsilon \implies \mu(G_p) - \mu(F_p) < \varepsilon.$$

Hence E is Jordan measurable by 5.11. And so it remains to justify the inclusion. Indeed, if $z \in G_p \setminus F_p$, then there exists a cube of size 2^{-p}, call it Q_p, such that $z \in Q_p$. But this Q_p must satisfy $Q_p \not\subset E^\circ$ and $Q_p \cap \overline{E} \neq \emptyset$ at the same time. Hence there exist points $x, y \in Q_p$ such that $y \in \mathbb{R}^n \setminus E^\circ$ and $x \in \overline{E}$. Because Q_p is convex, we can draw a line between x and y and define

$$t_* = \sup \{t \in (0,1) \mid (1-t)x + ty \in E\}.$$

But this means that the point $z_* = (1-t_*)x + t_*y$ can be approached by sequences both from $\mathbb{R}^n \setminus E \subset \mathbb{R}^n \setminus E^\circ$ and from $E \subset \overline{E}$. But since both of those sets are closed, z_* must lie in the intersection, so ∂E. But that means that Q_p has non-empty intersection with the dyadic set H_ε of pixel size 2^{-p}, hence is entirely contained in it (easy to check), so $z \in H_\varepsilon$. □

LEMMA 5.15: LIPSCHITZ MAPS PRESERVE NULL SETS

Let $E \subset \mathbb{R}^n$ be null and $f : E \to \mathbb{R}^n$ be Λ-Lipschitz. Then $f(E)$ is null.

Proof. If E is null, then for any $\varepsilon > 0$, there exists dyadic $G = \bigcup_{\ell=1}^N Q_p(a_\ell)$ with volume $N \cdot 2^{-pn}$ for some $p, N \in \mathbb{N}$ such that $\mu(G) = N \cdot 2^{-pn} < \varepsilon$ and $E \subset G$.

Since f is Λ-Lipschitz, we get

$$x \in Q_p(a_\ell) \implies |f(x) - f(a_\ell)| \leq \Lambda |x - a_\ell| \leq \Lambda 2^{-p}\sqrt{n},$$

hence (draw picture to visualise) for $x \in Q_p(a_\ell)$, $f(x) \in f(a_\ell) + \Lambda 2^{-p}\sqrt{n}\,[-1,1]^n$.

By simple inclusion principles,

$$f(E) \subset \bigcup_{\ell=1}^N f(Q_p(a_\ell)) \subset \bigcup_{\ell=1}^N f(a_\ell) + \Lambda 2^{-p}\sqrt{n}\,[-1,1]^n.$$

But we can estimate (implicitly using 5.8 and 5.9) the measure of $f(E)$

$$\mu_{\text{out}}(f(E)) \leq N \cdot (\Lambda 2^{-p}\sqrt{n}2)^n \leq (2\Lambda\sqrt{n})^n \cdot N 2^{-pn} < (2\Lambda\sqrt{n})^n \varepsilon.$$

Letting $\varepsilon \to 0$, we get the statement. □

LEMMA 5.16: GRAPHS ARE NULL

Let $E \subset \mathbb{R}^{n-1}$ and $f : E \to \mathbb{R}$ be uniformly continuous. Then the graph

$$\Gamma = \{(x, f(x)) \in \mathbb{R}^n \mid x \in E\} \subset \mathbb{R}^n$$

is null.

Proof. If E is null then it is measurable hence by Theorem 5.14 bounded inside $E \subset [-N_0, N_0)^{n-1}$ and we have for arbitrary $p \in \mathbb{N}$ a cover

$$E \subset \bigcup_{\ell=1}^N Q_p(a_\ell).$$

Because of the bound for E, this must be possible for $N \leq (2N_0 \cdot 2^p)^{n-1}$, so that we do not exceed the size of $[-N_0, N_0)^{n-1}$.

Assume wlog that $Q_p(a_\ell) \cap E \neq \emptyset$ and for each $1 \leq \ell \leq N$ let $x_\ell \in E \cap Q_p(a_\ell)$ and notice that it in particular holds

$$x \in E \cap Q_p(a_\ell) \implies |x - x_\ell| < \sqrt{n} \cdot 2^{-p}.$$

Now, given $\varepsilon > 0$, by uniform continuity $\exists \delta > 0$ such that values of f are very close, but there $\exists p \in \mathbb{N}$ such that $\sqrt{n} \cdot 2^{-p} < \delta$. So if we let p be large enough,

$$\Gamma \subset \bigcup_{\ell=1}^{N} Q_p(a_\ell) \times [f(x_\ell) - \varepsilon, f(x_\ell) + \varepsilon)$$

hence we can estimate using Lemma 5.9

$$\mu_{\text{out}}(\Gamma) \leq \sum_{\ell=1}^{N} 2^{-pn} \cdot 2\varepsilon = N 2^{-pn} \cdot 2\varepsilon \leq (2N_0)^{n-1} 2^{p+1} \cdot 2^{-p} = (2N_0)^{n-1} \cdot 2\varepsilon.$$

Sending $\varepsilon \to 0$, we get the Lemma. □

5.2 The Jordan measure under linear transformations

We inspect how the Jordan measure behaves under linear transformations. First we do it for simple affine maps.

Proposition 5.17: Translation and dilation invariance

Let $E \subset \mathbb{R}^n$ be Jordan measurable. For $\rho > 0$, $a \in \mathbb{R}^n$, we have that $\rho E + a$ is also Jordan measurable and

$$\mu(\rho E + a) = \rho^n \cdot \mu(E).$$

Proof. First we prove the formula for a dyadic cube. Let $Q = 2^{-p} \cdot [0, 1)^n$ is a dyadic cube. Then $\rho Q + a$ is a box and by Lemma 5.9, $\mu(\rho Q + a) = \rho^n \mu(Q)$. Since this holds for individual cubes, and dyadic sets S are disjoint unions of dyadic cubes, the formula also holds for dyadic sets.

For E any Jordan measurable set, for any $\varepsilon > 0$, $\exists F_\varepsilon, G_\varepsilon$ dyadic such that $F_\varepsilon \subset E \subset G_\varepsilon$ and $\mu(G_\varepsilon) - \mu(F_\varepsilon) < \varepsilon$.

Hence by inclusion,

$$\rho F_\varepsilon + a \subset \rho E + a \subset \rho G_\varepsilon + a,$$

and

$$\mu(\rho G_\varepsilon + a) - \mu(\rho F_\varepsilon + a) = \rho^n \mu(G_\varepsilon) - \rho^n \mu(F_\varepsilon) = \rho^n \varepsilon.$$

Letting $\varepsilon \to 0$, we get the statement. □

Now we extend this to general linear maps.

Lemma 5.18: Measure under linear maps

Let $L : \mathbb{R}^n \to \mathbb{R}^n$ be linear and invertible. Then, for all $E \subset \mathbb{R}^n$ Jordan measurable, $L(E)$ is Jordan measurable and

$$\mu(L(E)) = \lambda(L) \cdot \mu(E)$$

for some $\lambda(L) \in \mathbb{R}$.

Proof. If E is Jordan, then it is bounded and ∂E is null, and since L is a Lipschitz map, $L(\partial E) = \partial L(E)$ (exercise: prove, use continuity) is null. Since $L(E)$ is bounded and has null boundary, it is measurable by Theorem 5.14.

If we let $L([0,1)^n) = A$, using linearity of L,

$$L(a + 2^{-p}[0,1)^n) = L(a) + 2^{-p}L([0,1)^n) = L(a) + 2^{-p}A.$$

By Lemma 5.17, we then must have

$$\mu(2^{-p}A + a) = 2^{-pn}\mu(A).$$

Hence for any dyadic set F, which is just a union of these small boxes, we conclude that $\mu(L(F)) = \mu(F) \cdot \mu(A)$.

For E Jordan measurable, for any $\varepsilon > 0$, there exist $F_\varepsilon, G_\varepsilon$ with $F_\varepsilon \subset E \subset G_\varepsilon$ and $\mu(G_\varepsilon) - \mu(F_\varepsilon) < \varepsilon$ and by inclusions,

$$L(F_\varepsilon) \subset L(E) \subset L(G_\varepsilon).$$

By what we found, we have

$$L(G_\varepsilon) - L(F_\varepsilon) < \mu(A) \cdot \varepsilon.$$

Sending $\varepsilon \to 0$, we again get the claim. $\square$

Lemma 5.19: Special stretching Lemma

Let $L : \mathbb{R}^n \to \mathbb{R}^n$ be invertible with diagonal matrix representation in the standard basis. Then $\lambda(L)$ from Lemma 5.18 is equal to

$$\lambda(L) = \lambda_1 \cdots \lambda_n = \det(L).$$

Proof. Follows from the formula for boxes, since $\lambda(L) = \mu(L([0,1)^n))$, and the scaling of $[0,1)^n$ by the diagonal map L is trivial. $\square$

Remark 5.20: $\mathrm{O}(n)$

We denote by $\mathrm{O}(n)$ the group of $n \times n$ real orthogonal matrices, so matrices O such that $O \cdot O^\top = I$. Also, get ready for playing loose with using linear maps as abstract maps or matrices. For us, it doesn't matter, because we have a canonical basis.

Lemma 5.21: Decomposition

Let $L : \mathbb{R}^n \to \mathbb{R}^n$, there exist $R_1, R_2 \in \mathrm{O}(n)$ and S diagonal such that $L = R_2 S R_1$.

Proof. This is a linear algebra theorem, but here is the idea: We take $A = L^\top L$, which is a symmetric matrix. We can apply the Spectral Theorem 3.7 to get $O^\top AO = D$, where $O \in \mathrm{SO}(n)$ and D is diagonal. Now notice that since $v \cdot Av = v^\top L^\top \cdot Lv = (Lv)^2 \geq 0$, A is non-negative definite and D has only positive entries.

Then it's easy to find S diagonal (choose all entries positive) such that $S^2 = D$, so $S^{-1}O^\top L^\top LOS^{-1} = (LOS^{-1})^\top LOS^{-1} = I_n$, because S^{-1} is diagonal and hence equals its own transpose. And so $LOS^{-1} = R \in \mathrm{SO}(n)$ and $L = RSO^\top$, as desired. ☐

Lemma 5.22: Balls are measurable

Any ball $B_r(x) \subset \mathbb{R}^n$ is measurable.

Proof. Notice that the upper and lower boundary hemispheres respectively are the graphs of uniformly continuous functions (continuous on a compact disk) so Lemma 5.16 applies, $\partial B_r(x)$ is null and the ball is obviously bounded, so by Theorem 5.14 $B_r(x)$ is measurable. ☐

Theorem 5.23: Jordan measure under linear maps

Let $L : \mathbb{R}^n \to \mathbb{R}^n$ be linear and invertible. If E is Jordan measurable, then $L(E)$ is Jordan measurable and its measure is

$$\mu(L(E)) = |\det(L)| \cdot \mu(E).$$

Proof. Using Lemma 5.21, we express $L = R_2SR_1$ and we get from Lemma 5.18 that

$$\lambda(L) = \lambda(R_2)\lambda(S)\lambda(R_1) = 1 \cdot \det(S) \cdot 1 = |\det(R_2)| \cdot \det(S) \cdot |\det(R_1)| = |\det(L)|.$$

To justify why orthogonal matrices R have $\lambda(R) = 1$, consider that by Lemma 5.22, $\mu(B_1(0)) = \mu(R(B_1(0))) = \lambda(R) \cdot \mu(B_1(0))$, since orthogonal transformations leave the unit ball unchanged (check both inclusions), and hence $\lambda(R) = 1$. ☐

Remark 5.24

We keep assuming invertibility of the linear maps because the box volume Lemma 5.9 doesn't quite apply if one of the intervals is degenerate. One can shows as an exercise that non-invertible linear maps map any measurable set to a null set. We won't be working with them, I think.

5.3 The Riemann integral

Now we can define integrals, proceeding similarly to Analysis 1.

Definition 5.25: Indicator function

Let $A \subset \mathbb{R}^n$. The indicator/characteristic function for A is a function $\mathbf{1}_A : \mathbb{R}^n \to \mathbb{R}$ defined by

$$\mathbf{1}_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \in \mathbb{R}^n \setminus A. \end{cases}$$

DEFINITION 5.26: DYADIC STEP FUNCTION

A function $g : \mathbb{R}^n \to \mathbb{R}^n$ is called a dyadic step function if

$$g(x) = \sum_{\ell=1}^{N} g_\ell \cdot \mathbf{1}_{Q_p(a_\ell)},$$

where $Q_p(a_\ell)$ are disjoint dyadic cubes of pixel size 2^{-p} and g_ℓ are values in $\mathbb{R}$.

DEFINITION 5.27: INTEGRALS OF STEP FUNCTIONS

Let $A \subset \mathbb{R}^n$. If $g : A \to \mathbb{R}^n$ is a dyadic step function with N blocks, then we define the integral of g as

$$\int_A g = \sum_{\ell=1}^{N} g_\ell \cdot \mu(Q_p(a_\ell)) = \left(\sum_{\ell=1}^{N} g_\ell \right) 2^{-pn}.$$

DEFINITION 5.28: RIEMANN INTEGRABILITY

Let $A \subset \mathbb{R}^n$. We say $f : A \to \mathbb{R}$ is Riemann integrable over A if the upper and lower sums, defined below, coincide.

$$\underline{\int}_A f = \sup \left\{ \int_A g \mid g \leq f \cdot \mathbf{1}_A \text{ and } g \text{ is a dyadic step function} \right\}$$
$$\overline{\int}_A f = \inf \left\{ \int_A h \mid h \geq f \cdot \mathbf{1}_A \text{ and } h \text{ is a dyadic step function} \right\}.$$

If they coincide, we write $\overline{\int}_A f = \underline{\int}_A f = \int_A f = \int_A f \, d\mu = \int_A f(x) \, dx$, where one can imagine $dx = dx_1 \cdots dx_n$ as the volume of an infinitesimal cube. But note that in terms of rigor, all of these are just notation to us.

LEMMA 5.29: INTEGRAL OF THE CONSTANT FUNCTION

Let $A \subset \mathbb{R}^n$ be Jordan measurable and $c : A \to \mathbb{R}$ be the constant function with value c. Then $c \cdot \mathbf{1}_A$ is Riemann integrable and $\int c \cdot \mathbf{1}_A = c \int \mathbf{1}_A = c \cdot \mu(A)$.

Proof. The proof is left as an exercise, it is basically a Jordan measure type proof. □

LEMMA 5.30: LINEARITY OF THE RIEMANN INTEGRAL

Let $f_1, f_2 : A \to \mathbb{R}$ be Riemann integrable functions and $c_1, c_2 \in \mathbb{R}$. Then $c_1 f_1 + c_2 f_2$ is Riemann integrable and

$$\int_A (c_1 f_1 + c_2 f_2) = c_1 \int_A f_1 + c_2 \int_A f_2.$$

Proof. Exercise. First prove it for step functions, where it's easy and then do the usual approximation. □

DEFINITION 5.31: POSITIVE AND NEGATIVE PART OF A FUNCTION

Let X be a set and $f : X \to \mathbb{R}$ a function. Then we define $f^+ : X \to \mathbb{R}$ and $f^- : X \to \mathbb{R}$ by

$$f^+(x) = \max\{f(x), 0\}, \quad f^-(x) = \max\{-f(x), 0\}.$$

Notice that $f = f^+ - f^-$.

LEMMA 5.32: INTEGRABILITY OF POSITIVE AND NEGATIVE PART

Let $A \subset \mathbb{R}^n$. Then $f : A \to \mathbb{R}$ is Riemann integrable over A if and only if f^+ and f^- are. In that case,

$$\int_A f = \int_A f^+ - \int_A f^-.$$

From this you also get the triangle inequality as $|f| = f^+ + f^-$.

Proof. Exercise. The only real difficulty here is proving $\implies$. □

REMARK 5.33

We will from now on assume almost every time that the functions are positive-valued, since any result can be extended to arbitrary functions using the Lemma 5.32.

LEMMA 5.34: JORDAN MEASURE AND RIEMANN INTEGRAL

Let $A \subset \mathbb{R}^n$ and $f : A \to [0, \infty)$. Consider the *hypograph*

$$\Gamma_A(f) = \left\{(x_1, \dots, x_n, x_{n+1}) \in \mathbb{R}^{n+1} \mid (x_1, \dots, x_n) \in A, 0 \leq x_{n+1} \leq f(x_1, \dots, x_n)\right\}.$$

Then f is Riemann integrable if and only if $\Gamma_A(f)$ is measurable. Moreover,

$$\int_A f = \mu_{n+1}(\Gamma_A(f)).$$

Proof. Exercise, notice the similarity in definitions. Indeed, for a lower step function $g \leq f \cdot \mathbf{1}_A$, the union of boxes

$$G = \bigcup_{\ell=1}^{N} Q_p(a_\ell) \times [0, g_\ell)$$

has volume exactly equal to the value of the step function integral, so $\mu_{n+1}(G) = \int_A g$. Doing the same for an upper step function $h \geq f \cdot \mathbf{1}_A$ and the analogously defined set of boxes H, we get $\mu_{n+1}(H) = \int_A h$. Then it's apparent that the lower and upper sums align if and only if the inner and outer measures of $\Gamma_A(f)$ align. □

LEMMA 5.35: UNIFORMLY CONTINUOUS FUNCTIONS ARE INTEGRABLE

Let $A \subset \mathbb{R}^n$ be Jordan measurable and $f : \overline{A} \to [0, \infty)$ be continuous. Then f is Riemann integrable over A.

Proof. By Lemma 5.34, we know that f is integrable over A if and only if $\Gamma_A(f)$ is Jordan measurable, but by Theorem 5.14, this is if and only if $\mu_{n+1}(\partial\Gamma_A(f)) = 0$ and $\Gamma_A(f)$ is bounded. Using the fact that A has null boundary and is bounded, because it is measurable, we wish to show the same for $\Gamma_A(f)$. Notice that

$$\partial\Gamma_A(f) \subset A_1 \cup A_2 \cup A_3,$$

where

$$
\begin{aligned}
A_1 &= \{(x, x_{n+1}) \in \mathbb{R}^n \times \mathbb{R} \mid x_{n+1} = f(x), x \in A\} \\
A_2 &= \partial A \times [0, \max_{\overline{A}} f] \\
A_3 &= A \times \{0\}\,.
\end{aligned}
$$

Since ∂A is null, one can show that A_2 has measure 0. Since A_1 is the graph of a uniformly continuous map, we have by Lemma 5.16 that it is also null. And finally, since A is bounded, it's very simple to cover A_3 with smaller and smaller cubes, hence it is also null.

Now, why does the inclusion $\partial\Gamma_A(f) \subset A_1 \cup A_2 \cup A_3$ hold? Well, take some point $(x, x_{n+1}) \in \overline{\Gamma_A(f)}$ and assume it is in none of the sets A_1, A_2, A_3. We will show that it has to be in the interior, which will make it impossible for a point to be on the boundary and not to be in $A_1 \cup A_2 \cup A_3$. Notice that any (x, x_{n+1}) on the closure of $\overline{\Gamma_A(f)}$ must satisfy $x \in \overline{A}$ and $0 \leq f(x) \leq x_{n+1}$. Then, we have two options. Either $x \in A^\circ$ or $x \in \partial A$.

We tackle the second case first, so $x \in \partial A$. But because $(x, x_{n+1}) \notin A_2$, we get $x_{n+1} > \max_{\overline{A}} f$, because ∂A was forced by assumption, contradiction.

For the second case, assume $x \in A^\circ$, forcing $x \notin \partial A$. And then, $(x, x_{n+1}) \notin A_1$ and $(x, x_{n+1}) \notin A_3$ force $0 < x_{n+1} < f(x)$. But since $x \in A^\circ$ open and $x_{n+1} \in (0, f(x))$ open, (x, x_{n+1}) lies in an open set inside $\Gamma_A(f)$, so it's in the interior. $\square$

5.4 Change of variables

Serra called this the "Final Boss of integration". The proof is insanely long, complicated and there is no chance we can cover every last detail without going on for two days worth of lecture time. Nevertheless, here it comes. I put in the effort to write it up outside of the lecture, I mostly stuck to the lecture notes, clarified some steps that I had issue comprehending and just copied others because they were straightforward.

Theorem 5.36: Change of variables formula

Let $U, V \subset \mathbb{R}^n$ open and $\Phi : U \to V$ a diffeomorphism. Assume $A \subset U$ is Jordan measurable, $\overline{A} \subset U$ and $f : \overline{A} \to [0, \infty)$ is continuous. Then f is Riemann integrable over A and

$$\int_A f(x)\,dx = \int_{\Phi(A)} \frac{f(\Phi^{-1}(y))}{|\det J\Phi(\Phi^{-1}(y))|}\,dy.$$

Proof. First we consider the heuristic idea of the proof. The idea is to approximate A by dyadic cubes from the inside. Take one such cube $x_0 + Q_p$ with $Q_p = 2^{-p}[0,1)$. Because the differential approximates the function value well,

$$\Phi(x_0 + y) \approx \Phi(x_0) + D\Phi_{x_0}(y).$$

And when we apply the diffeomorphism to the small cube, we will get

$$\Phi(x_0 + Q_p) \approx \Phi(x_0) + D\Phi_{x_0}(Q_p).$$

But we know how the measure of sets behaves under linear maps. In particular,

$$\mu(\Phi(x_0 + Q_p)) \approx \det |J\Phi(x_0)| \cdot \mu(Q_p).$$

Hence, if we correct by the inverse of the determinant, we get the same "sum over many little cubes".

So let's get to it. The proof consists of three parts, of which really only one is technically demanding. In step 1, we will prove that the expressions in the Theorem statement are well-defined in the first place. Step 2 will be the core of the proof, establishing LHS $\leq$ RHS. And step 3 will just be a reverse application of step 2 to get RHS $\leq$ LHS and then we get the Theorem.

Step 1. Note that A is bounded, hence $\overline{A}$ is compact. Since $\Phi(\overline{A})$ is the image of limit points of A and Φ is continuous, we get that any $y \in \Phi(\overline{A})$ is a limit point of $\Phi(A)$, so in $\overline{\Phi(A)}$. And using Proposition 1.38, we get that $\Phi(\overline{A})$ is closed, hence contains $\overline{\Phi(A)}$. This proves $\Phi(\overline{A}) = \overline{\Phi(A)}$. Now consider that by 4.4, $\Phi(A^\circ)$ is open and contained in $\Phi(A)$, hence $\Phi(A^\circ) \subset \Phi(A)^\circ$. Conversely, $\Phi(A)^\circ$ is an open set, hence $\Phi^{-1}(\Phi(A)^\circ)$ is open and by bijectivity contained in A, hence $\Phi^{-1}(\Phi(A)^\circ) \subset A^\circ$, giving $\Phi(A)^\circ \subset \Phi(A^\circ)$. This gives the equality $\Phi(A^\circ) = \Phi(A)^\circ$, and using the first part also $\partial\Phi(A) = \Phi(\partial A)$.

Using this, we show that $\Phi(A)$ is Jordan measurable. Since $\overline{A} \subset U$ and U is open, for any $x \in \overline{A}$, there exists a small radius $r_x > 0$ such that

$$B_{2r_x}(x) \subset U.$$

Additionally, $\Phi|_{\overline{B_{r_x}(x)}}$ is Lipschitz (generalise Proposition 2.21 and notice that the maximum of $D\Phi(x)$ is attained on $\overline{B_{r_x}(x)}$ because Φ is C^1). Since $\overline{A}$ is compact, we can easily cover A using a finite set of balls

$$B_{2r_1}(x_1), \ldots, B_{2r_k}(x_k),$$

where $x_i \in A$. Because A is measurable, ∂A is null and so is each $\partial A \cap B_{r_i}(x_i)$. And using Lemma 5.15, we get that each image $\Phi(\partial A \cap B_{r_i}(x_i))$ is Jordan null. But since

$$\partial\Phi(A) = \Phi(\partial A) = \bigcup_{\ell=1}^{k} \Phi(\partial A \cap B_{r_i}(x_i))$$

is a union of null sets, we get that $\partial\Phi(A)$ is Jordan null. It's also fairly obvious that $\Phi(A)$ is bounded, concluding that $\Phi(A)$ is Jordan measurable. Then, Lemma 5.35 tells us that both the LHS and RHS are well-defined, i.e. the integrals exist in the first place.

Step 2. Now we prove the following inequality:

$$\int_A f(x)\,dx \leq \int_{\Phi(A)} \frac{f(\Phi^{-1}(y))}{|\det J\Phi(\Phi^{-1}(y))|}\,dy$$

We use the following notation in this proof. For $p \in \mathbb{N}$ and $\delta \in (0, \frac{1}{2})$, we let

$$Q_p = 2^{-p} \cdot [0,1)^n, \quad Q_{p,\delta} = 2^{-p} \cdot [\delta, 1-\delta]^n.$$

We will prove $\forall \delta > 0\ \exists p_\delta \in \mathbb{N}$ s.t. $\forall p \geq p_\delta$ and $x_0 \in \overline{A}$,

$$(1-2\delta)^n \cdot 2^{-pn} \cdot |\det J\Phi(x_0)| = \mu_n(D\Phi_{x_0}(Q_{p,\delta})) \leq \mu_n(\Phi(x_0 + Q_p)) \qquad (*)$$

and

$$\sup\left\{|\det J\Phi(x)| \mid x \in (x_0 + Q_p) \cap \overline{A}\right\} \leq (1+\delta)\,|\det J\Phi(x_0)|\,. \qquad (**)$$

Basically, what the first equation $(*)$ is doing is estimating the volume of the image of the cube $x_0 + Q_p$ from below by a slightly shrunk (by 2δ) cube of sufficiently small pixel size, whose volume gets distorted precisely by the determinant of Φ. The second equation $(**)$ uses the continuity of $J\Phi(x)$ to estimate the size of the determinant on that small dyadic cube, which again differs only by a small factor depending on δ.

First, the equality in $(*)$ is a direct application of 5.9 and 5.23. And so it suffices to show that

$$D\Phi_{x_0}(Q_{p,\delta}) \subset \Phi(x_0 + Q_p).$$

We define the smallest distance between points in $\Phi(\overline{A})$ and ∂V as

$$R = \operatorname{dist}(\Phi(\overline{A}), \partial V) = \inf\left\{|a-b| \mid a \in \Phi(\overline{A}), b \in \partial V\right\} > 0.$$

Since V is open and $\Phi(\overline{A})$ is compact inside of it, $R = 0$ would imply the existence of a sequence $a_n \in \Phi(\overline{A})$ and $b_n \in \partial V$ such that $|a_n - b_n| \to 0$. But then we could extract a convergent subsequence of a_n with limit necessarily in ∂V, which is impossible because V was open, hence disjoint with its boundary.

We further let

$$M = \max_{x \in \overline{A}} \|J\Phi(x)\|_2.$$

For $(x_0, h) \in \overline{A} \times B_{R/M}(0)$, we wish to define the "error"

$$\Psi(x_0, h) = \Psi_{x_0}(h) := \Phi^{-1}(\Phi(x_0) + D\Phi_{x_0}(h)) - x_0,$$

but need to check this is well-defined, i.e. $\Phi(x_0) + D\Phi_{x_0}(h) \in V$. But indeed, using 1.60,

$$|D\Phi_{x_0}(h)| \leq M\,|h| < R,$$

hence $\Phi(x_0) + D\Phi_{x_0}(h)$ is at most a distance R away from a point in $\Phi(\overline{A})$, hence in V. Define also the auxiliary function

$$\widetilde{\Psi}(x_0, h) = \widetilde{\Psi}_{x_0}(h) := \Psi_{x_0}(h) - h.$$

We claim that given any $\lambda > 0$, we can find $r_\lambda > 0$ such that at any point $x_0 \in \overline{A}$ and small perturbations $|h| < r_\lambda$,

$$\left|\widetilde{\Psi}_{x_0}(h)\right| = |\Psi_{x_0}(h) - h| \leq \lambda\,|h|\,.$$

Indeed, for a fixed $x_0 \in \overline{A}$, $\widetilde{\Psi}_{x_0}(0) = 0$. And using the Chain Rule 2.16 and denoting with J_h the Jacobi matrix of $\widetilde{\Psi}_{x_0}(h)$ (with x_0 constant),

$$J_h\widetilde{\Psi}_{x_0}(0) = J\Phi^{-1}(\Phi(x_0)) \cdot J\Phi(x_0) - I_n = I_n - I_n = 0.$$

But since $J_h\widetilde{\Psi} : \overline{A} \times \overline{B_{R/M}(0)} \to \mathbb{R}^{n\times n}$ is continuous on a compact set, so uniformly continuous, we get $r_\lambda > 0$ such that

$$|h| < r_\lambda \implies \|J_h\widetilde{\Psi}(x_0, h) - J_h\widetilde{\Psi}(x_0, 0)\|_2 = \|J_h\widetilde{\Psi}(x_0, h)\| \leq \lambda.$$

This gives the overall bound

$$\sup_{x_0 \in \overline{A}} \sup_{|\xi| < r_\lambda} \|J_h\widetilde{\Psi}(x_0, \xi)\|_2 \leq \lambda.$$

Applying the MVT 2.18, we get the desired

$$|\Psi_{x_0}(h) - h| = \left|\widetilde{\Psi}_{x_0}(h)\right| = \left|\widetilde{\Psi}_{x_0}(h) - \widetilde{\Psi}_{x_0}(0)\right| \leq \lambda\,|h|\,.$$

Let $\delta > 0$ and choose $\lambda < \delta/\sqrt{n}$ and p_δ large enough that $\sqrt{n} \cdot 2^{-p_\delta} < r_\lambda$. Then for $p \geq p_\delta$, we get

$$h \in Q_{p,\delta} \implies |h| \leq \sqrt{n} \cdot 2^{-p} < r_\lambda \implies |\Psi_{x_0}(h) - h| \leq \lambda\,|h| < \delta \cdot 2^{-p}.$$

Since every point of $Q_{p,\delta}$ has at least distance $\delta \cdot 2^{-p}$ from the boundary of Q_p, we get

$$\Psi_{x_0}(Q_{p,\delta}) \subset Q_p,$$

or equivalently,

$$\Phi^{-1}(\Phi(x_0) + D\Phi_{x_0}(Q_{p,\delta})) \subset x_0 + Q_p,$$

to which we can apply the bijection Φ to get $(*)$.

Now we prove $(**)$. For that notice that $c := \min_{x\in\overline{A}} |\det J\Phi(x)|$ is strictly positive. Since $|\det J\Phi(x)|$ defines a continuous function on the compact set $\overline{A}$, we get for any $\delta > 0$ an $r_\delta > 0$ such that

$$\Big| |\det J\Phi(x)| - |\det J\Phi(x_0)| \Big| < \delta c \quad \forall\, |x - x_0| < r_\delta.$$

If we just shrink p_δ so that $\sqrt{n} \cdot 2^{-p_\delta} < r_\delta$, we get

$$|\det J\Phi(x)| \le |\det J\Phi(x_0)| + \delta c \le (1+\delta)\, |\det J\Phi(x_0)|$$

for $x \in (x_0 + Q_p) \cap \overline{A}$. This proves $(**)$.

Let $g : \mathbb{R}^n \to \mathbb{R}$ be a dyadic step function approximating $f \cdot \mathbf{1}_A$ from below. More precisely, for $\varepsilon > 0$, choose

$$g = \sum_{\ell=1}^{N} g_\ell \cdot \mathbf{1}_{Q_p(a_\ell)}, \quad Q_p(a_\ell) := a_\ell + Q_p,$$

with $g_\ell > 0$, such that the cubes $Q_p(a_\ell)$ are pairwise disjoint,

$$g \le f \cdot \mathbf{1}_A \quad \text{in } \mathbb{R}^n,$$
$$\int_A f - \varepsilon \le \int_A g.$$

Since $g_\ell > 0$ and $g \le f \cdot \mathbf{1}_A$, each cube $Q_p(a_\ell) \subset A$.

Applying $(*)$ and $(**)$ with $x_0 = a_\ell$, we obtain for all $p \ge p_\delta$

$$2^{-pn} \le \frac{1+\delta}{(1-2\delta)^n} \frac{\mu_n\big(\Phi(Q_p(a_\ell))\big)}{\sup\{|\det J\Phi(x)| \mid x \in Q_p(a_\ell)\}}.$$

Moreover,

$$g_\ell \le \inf_{x\in Q_p(a_\ell)} f(x) = \inf_{y\in\Phi(Q_p(a_\ell))} f\big(\Phi^{-1}(y)\big).$$

Combining the two equations, we obtain

$$g_\ell\, 2^{-pn} \le \frac{1+\delta}{(1-2\delta)^n} \inf_{y\in\Phi(Q_p(a_\ell))} \frac{f\big(\Phi^{-1}(y)\big)}{|\det J\Phi(\Phi^{-1}(y))|}\, \mu_n\big(\Phi(Q_p(a_\ell))\big).$$

Since the sets $\Phi(Q_p(a_\ell))$ are pairwise disjoint and contained in $\Phi(A)$, summing over ℓ gives

$$\begin{aligned}
\int_A f - \varepsilon \le \int_A g &= \sum_{\ell=1}^{N} g_\ell\, 2^{-pn} \\
&\le \frac{1+\delta}{(1-2\delta)^n} \sum_{\ell=1}^{N} \inf_{y\in\Phi(Q_p(a_\ell))} \frac{f\big(\Phi^{-1}(y)\big)}{|\det J\Phi(\Phi^{-1}(y))|}\, \mu_n\big(\Phi(Q_p(a_\ell))\big) \\
&\le \frac{1+\delta}{(1-2\delta)^n} \int_{\Phi(A)} \frac{f\big(\Phi^{-1}(y)\big)}{|\det J\Phi(\Phi^{-1}(y))|}\, dy.
\end{aligned}$$

Since $\varepsilon > 0$ and $\delta > 0$ are arbitrary, we obtain the inequality stated right at the beginning of the step.

Step 3. This will be very straightforward. Apply the obtained inequality to the diffeomorphism

$$\tilde{\Phi} := \Phi^{-1} : \Phi(A) \to A$$

and to the continuous function

$$\tilde{f}(y) := \frac{f(\Phi^{-1}(y))}{|\det J\Phi(\Phi^{-1}(y))|}, \quad y \in \Phi(A).$$

This yields

$$\int_{\Phi(A)} \tilde{f}(y)\, dy \leq \int_A \frac{\tilde{f}(\tilde{\Phi}^{-1}(x))}{\left|\det J\tilde{\Phi}(\tilde{\Phi}^{-1}(x))\right|}\, dx.$$

Now $\tilde{\Phi}^{-1} = \Phi$, so

$$\frac{\tilde{f}(\tilde{\Phi}^{-1}(x))}{\left|\det J\tilde{\Phi}(\tilde{\Phi}^{-1}(x))\right|} = \frac{\tilde{f}(\Phi(x))}{|\det J\Phi^{-1}(\Phi(x))|} = \frac{f(x)}{|\det J\Phi(x)| \cdot |\det J\Phi^{-1}(\Phi(x))|} = f(x),$$

because $\det J\Phi(x) \cdot \det J\Phi^{-1}(\Phi(x)) = 1$. Therefore

$$\int_{\Phi(A)} \frac{f(\Phi^{-1}(y))}{|\det J\Phi(\Phi^{-1}(y))|}\, dy \leq \int_A f(x)\, dx.$$

This proves the other inequality and hence the statement. $\square$

Remark 5.37: Reverse change of variables

If we let $\Psi : V \to U$ a diffeomorphism and $f : \overline{A} \to \mathbb{R}$ continuous with $\overline{A} \subset U$ Jordan measurable, Theorem 5.36 gives us

$$\int_A f(x)\, dx = \int_{\Psi^{-1}(A)} f(\Psi(y)) \cdot |\det J\Psi(y)|\, dy.$$

Let's put the change of variables to work.

Example 5.38: Computing the area of the disc

We consider the disk $B_1 = B_1(0) \subset \mathbb{R}^2$ and wish to compute $\mu_2(B_1) = \pi$. We define

$$\Psi(y_1, y_2) = \begin{pmatrix} y_1 \cos y_2 \\ y_1 \sin y_2 \end{pmatrix}$$

defined on $D = (0,1) \times (0, 2\pi)$ and mapping into B_1. And notice that although $B_1 \neq \Psi(D)$, the part of B_1 that is not covered by Ψ has measure zero, hence $\mu_2(B_1) = \mu_2(\Psi(D))$. But by Lemma 5.29,

$$\mu_2(\Psi(D)) = \int_{\Psi(D)} 1\, dx = \int_D |\det J\Psi(y)|\, dy.$$

And actually, one computes $\det J\Psi(y_1, y_2) = y_1 > 0$, so we're looking at

$$\int_{(0,1)\times(0,2\pi)} y_1\, dy.$$

But, using Lemma 5.9, the volume of this (consider a drawing of the hypograph) is $\frac{1}{2} \cdot \mu_3((0,1) \times (0, 2\pi) \times (0,1)) = \frac{1}{2} \cdot 2\pi = \pi$. And so we know the area of the disk.

Example 5.39: Computing the volume of the sphere

We wish to compute $\mu_3(B_1) = \frac{4}{3}\pi$ for $B_1 \subset \mathbb{R}^3$. For that, we have the polar coordinates again

$$\Psi(y_1, y_2, y_3) = \begin{pmatrix} y_1 \cos y_2 \cos y_3 \\ y_1 \sin y_2 \cos y_3 \\ y_1 \sin y_3 \end{pmatrix}$$

with domain $D = (0,1) \times (0, 2\pi) \times (-\frac{\pi}{2}, \frac{\pi}{2})$. One computes $|\det J\Psi| = y_1^2 \cos y_3$ an so we have to compute

$$\mu_3(B_1) = \mu_3(\Psi(D)) = \int_{(0,1)\times(0,2\pi)\times(-\frac{\pi}{2},\frac{\pi}{2})} y_1^2 \cos y_3 \, dy.$$

Now, to compute this integral, we have to introduce the slicing formula.

5.5 Cavalieri's principle and Fubini's theorem

The Cavalieri principle in 3D says roughly, that for any measurable set $A \subset \mathbb{R}^n$, you can compute the volume of A by partitioning it into very thin slices, and to obtain the total volume, you just have to sum over all the little slices.

For the following proofs, split the notation of points in $\mathbb{R}^n$ as points in $\mathbb{R}^k \times \mathbb{R}^{n-k}$. So writing $(x, y) \in \mathbb{R}^n$ implies that $x \in \mathbb{R}^k$ and $y \in \mathbb{R}^{n-k}$.

Theorem 5.40: Cavalieri

Let $A \subset \mathbb{R}^n$ be Jordan measurable, $A \subset (-C, C)^n$. Define the slices for $x \in (-C, C)^k$ as

$$A_x = \left\{ y \in \mathbb{R}^{n-k} \mid (x, y) \in A \right\} \subset (-C, C)^{n-k}$$

and their inner and outer volumes in $\mathbb{R}^{n-k}$

$$\overline{\varphi}(x) = \mu_{n-k,\text{out}}(A_x), \quad \underline{\varphi}(x) = \mu_{n-k,\text{in}}(A_x).$$

Then $\overline{\varphi}, \underline{\varphi} : (-C, C)^k \to \mathbb{R}$ are Riemann integrable and

$$\int_{(-C,C)^k} \overline{\varphi}(x) \, dx = \int_{(-C,C)^k} \underline{\varphi}(x) \, dx = \mu_n(A).$$

Proof. Given a dyadic cube of pixel size 2^{-p} in $\mathbb{R}^n$, we write

$$Q_p^n(z) = z + 2^{-p}[0,1)^n.$$

With $z = (x, y) \in \mathbb{R}^k \times \mathbb{R}^{n-k}$, we can rewrite the cube as

$$Q_p^n(z) = Q_p^k(x) \times Q_p^{n-k}(y).$$

Since A is measurable, given $\varepsilon > 0$, let F, G dyadic sets of size 2^{-p} such that $F \subset A \subset G$ and $\mu(G) - \mu(F) < \varepsilon$. For $x \in (-C, C)^k$ define

$$G_x = \left\{ y \in \mathbb{R}^{n-k} \mid (x, y) \in G \right\}, \quad F_x = \left\{ y \in \mathbb{R}^{n-k} \mid (x, y) \in F \right\}.$$

Now observe that $F_x \subset A_x \subset G_x$ and $F_x = F_{x'}$ for all $x, x' \in Q_p^k(a)$, because the cubes of F can be "factored" as $Q_p^n((x, y)) = Q_p^k(x) \times Q_p^{n-k}(y)$. Same holds for G_x.

So $f(x) = \mu(F_x)$ and $g(x) = \mu(G_x)$ are both constant on each cube $Q_p^k(a)$, hence dyadic step functions for which it holds that $f(x) \leq \underline{\varphi}(x) \leq \overline{\varphi}(x) \leq g(x)$. Now, each cube $Q_p^k(a) \times Q_p^{n-k}(b)$ contributes function value $2^{-p(n-k)}$ times domain base length 2^{-pk}, so

$$f(x) = \mu(F_x) = 2^{-pn} \cdot \# \text{ of dyadic cubes in } F_x.$$

and if we take the integral over all possible $Q_p^k(a)$ cubes,

$$\int_{\mathbb{R}^k} f(x)\,dx = 2^{-pn} \cdot \# \text{ of dyadic cubes in } F = \mu(F),$$

and analogously,

$$\int_{\mathbb{R}^k} g(x)\,dx = 2^{-pn} \cdot \# \text{ of dyadic cubes in } G = \mu(G).$$

Hence

$$\mu(F) = \int f(x)\,dx \leq \underline{\int} \underline{\varphi}(x)\,dx \leq \overline{\int} \overline{\varphi}(x)\,dx \leq \int g(x)\,dx = \mu(G).$$

But since $\mu(G) - \mu(F) < \varepsilon$, $\mu(F) \leq \mu(A) \leq \mu(G)$ and $\varepsilon > 0$ was arbitrary, we get the entire equality. □

Theorem 5.41: Fubini

Let $A \subset \mathbb{R}^n$ be bounded in $(-C, C)^n$ and $f : A \to \mathbb{R}$ be Riemann integrable over A. Denote $\tilde{f} = f \cdot \mathbf{1}_A$ defined on $\mathbb{R}^n$. For $x \in \mathbb{R}^k$, denote $\tilde{f}_x(y) = \tilde{f}(x, y)$, which is a function $(-C, C)^{n-k} \to \mathbb{R}$ and define the marginals $\underline{\varphi}, \overline{\varphi} : (-C, C)^k \to \mathbb{R}$ as

$$\underline{\varphi}(x) = \underline{\int}_{(-C,C)^{n-k}} \tilde{f}_x(y)\,dy, \quad \overline{\varphi}(x) = \overline{\int}_{(-C,C)^{n-k}} \tilde{f}_x(y)\,dy.$$

Then both these functions are integrable and

$$\int_A f = \int_{(-C,C)^k} \underline{\varphi}(x)\,dx = \int_{(-C,C)^k} \overline{\varphi}(x)\,dx.$$

Proof. Similar to Cavlieri, either exercise or read the official lecture notes. □

Corollary 5.42: Integral over a compact box

Let $K = [a_1, b_1] \times \cdots \times [a_n, b_n] \subset \mathbb{R}^n$ and $f : K \to \mathbb{R}$ continuous. Then

$$\int_K f(x)\,dx = \int_{a_1}^{b_1} \int_{a_2}^{b_2} \cdots \int_{a_n}^{b_n} f(x_1, \ldots, x_n)\,dx_n \cdots dx_2\,dx_1.$$

Furthermore, the order in which you integrate does not matter.

Proof. We apply Fubini 5.41 with $k = 1$ to get

$$\int_A f(x)\,dx = \int_{[a_1,b_1]} \overline{\varphi}(x)\,dx,$$

where

$$\overline{\varphi}(x_1, \ldots, x_n) = \overline{\int}_{[a_2,b_2]\times\cdots\times[a_n,b_n]} f(x_1, \ldots, x_n)\,dx_2 \cdots dx_n.$$

And setting $y = (x_2, \dots, x_n)$, we let $f_{x_1}(y) = f(x_1, y)$ which is a continuous function from $[a_2, b_2] \times \cdots \times [a_n, b_n]$, so integrable and we can write instead

$$\overline{\varphi}(x_1, \dots, x_n) = \int_{[a_2,b_2]\times\cdots\times[a_n,b_n]} \overline{\varphi}(x_1, y)\, dy = \int_{[a_2,b_2]\times\cdots\times[a_n,b_n]} f_{x_1}(y)\, dy.$$

And so we get

$$\int_A f(x)\, dx = \int_{[a_1,b_1]} \left(\int_{[a_2,b_2]\times\cdots\times[a_n,b_n]} f(x_1, y)\, dy \right) dx_1.$$

Apply this repeatedly to compute the entire integral. Notice that you could have done this with any other coordinate, just using the function $f \circ P$ with P a permutation of coordinates ($\det P = \pm 1$) and using change of variables 5.36 to get $\int_A f = \int_{P^{-1}(A)} f \circ P$. You can write it down and see. This actually holds in general, you don't need Fubini. $\square$

Example 5.43: Finishing the volume of the sphere

Now we can finish computing the volume of the sphere. If we take $A = (0, 1) \times (0, 2\pi) \times (-\frac{\pi}{2}, \frac{\pi}{2})$ and close it to $K = \overline{A}$ (won't alter the volume as the boundary ∂A is null, so the hypograph volume at those points disappears – see 5.35), we get

$$\int_0^1 \int_0^{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} y_1^2 \cos y_3\, dy_3\, dy_2\, dy_1.$$

One computes one after another, the integrals w.r.t. to variables y_1, y_2, y_3, treating all others as a constant, giving $\frac{4}{3}\pi$.

5.6 Differentiation under the integral sign

Important physics theorem alert! We can exchange integrals and derivatives under special conditions.

Theorem 5.44: Differentiation under the integral

Let $U \subset \mathbb{R}^{n+1}$ be open, $f : U \to \mathbb{R}$ and $K \times [a, b] \subset U$ for $K \subset \mathbb{R}^n$ compact. We will write $f(x, t)$ for $x \in K$ and $t \in [a, b]$. Assume $f, \partial_t f$ are continuous and define the function $F : [a, b] \to \mathbb{R}$ as $F(t) = \int_K f(x, t)\, dx$. Then it holds $\forall t_0 \in (a, b)$ that

$$F'(t_0) = \int_K \partial_t f(x, t_0)\, dx.$$

More compactly,

$$\frac{d}{dt}\bigg|_{t=t_0} \int_K f(x, t)\, dx = \int_K \partial_t f(x, t_0)\, dx.$$

Proof. We have

$$F'(t_0) = \lim_{h\to 0} \frac{\int_K f(x, t_0 + h)\, dx - \int_K f(x, t_0)\, dx}{h}.$$

Using linearity of the integral 5.30, we get

$$F'(t_0) = \lim_{h\to 0} \int_K \frac{f(x, t_0 + h) - f(x, t_0)}{h}\, dt.$$

By the mean value theorem, there exists $\xi_x \in (t_0, t_0+h)$ such that

$$\frac{f(x,t_0+h)-f(x,t_0)}{h} = \partial_t f(x,\xi_x).$$

But because $\partial_t f$ is continuous on $K \times [a,b]$, it is uniformly continuous (i.e. its restriction to that set is). Hence $\forall \varepsilon > 0\ \exists \delta > 0$ such that

$$h < \delta \implies \xi_x - t_0 < \delta \implies |\partial_t f(x,\xi_i) - \partial_t f(x,t_0)| < \varepsilon.$$

Using this, if we let $E(x) = \partial_t f(x,\xi_i) - \partial_t f(x,t_0)$, we get

$$\int_K \frac{f(x,t_0+h)-f(x,t_0)}{h}\,dx = \int_K \partial_t f(x,t_0) + E(x)\,dx = \int_K \partial_t f(x,t_0)\,dx + \mathcal{O}(\varepsilon).$$

Why is this $\mathcal{O}(\varepsilon)$? Because we have the monotonicity bound $\left|\int_K E\right| \leq \int_K |\partial_t f(x,\xi_i) - \partial_t f(x,t_0)|\,dx \leq \mu_n(K)\cdot\varepsilon$. But now, let $h \to 0$ and you get the equality. $\square$

5.7 Improper integrals

We can also expand the definition of improper integrals from Analysis 1.

Definition 5.45: Improper integral

Let $U \subset \mathbb{R}^n$ open, $f : U \to [0,\infty)$ continuous. Define the *improper* integral

$$\int_U f = \sup\left\{\int_K f \mid K \subset U \text{ compact and Jordan measurable}\right\}.$$

Remark 5.46: Improper integral as sequence of integrals

If we have a sequence of nested compact Jordan measurable sets $K_0 \subset K_1 \subset \cdots$ such that $U = \bigcup_{\ell \geq 0} K_\ell^\circ$ and each $K_\ell \subset U$, then

$$\int_U f = \lim_{\ell\to\infty} \int_{K_\ell} f.$$

Why? First notice that the RHS limit exists because $\int_{K_\ell} f$ is monotone increasing so has a limit in $[0,\infty]$. Let $K \subset U$ compact and Jordan measurable. Because $\bigcup_{\ell\geq 0} K_\ell$ is a cover for K and K is compact, there exists a K_{ℓ_0} such that $K \subset K_{\ell_0}$. But then

$$\int_K f \leq \int_{K_{\ell_0}} f \leq \lim_{\ell\to\infty}\int_{K_\ell} f.$$

Hence the same bound holds for the improper integral $\int_U f \leq \lim\limits_{\ell\to\infty} \int_{K_\ell} f$.
And the other inequality is trivial because each K_ℓ is in the set which we are taking the supremum of the receive $\int_U f$.
In particular, because dyadic sets can be closed to be compact, the definition of improper integral doesn't clash with the original definition of integral.

5.8 Integration over parametrised submanifolds

Let's remind ourselves why we are doing all of this. Among the "final bosses" of Analysis 2 is for instance the Divergence Theorem, stating

$$\int_\Omega \sum_{i=1}^n \partial_i F_i \, dx = \int_{\partial\Omega} F \cdot \nu \, dS,$$

for $\Omega \subset \mathbb{R}^n$ open, $\partial\Omega$ a C^1 submanifold and $F \in C^1(\overline{\Omega}, \mathbb{R}^n)$. Now, we already know a big portion of this expression. We know open sets, boundaries, submanifolds, partial derivatives and integrals over sets with respect to the Jordan measure. The last part missing is what it means to take the integral over the submanifold $\partial\Omega$ with the surface measure dS.

So that's what we're working on now. First, we need to define volumes of parametrised d-dimensional submanifolds. We start with the easiest case, i.e. 1-dimensional manifolds. We will call them lines and their volume lengths. Also, no, not every manifold is parametrised, we will extend the definitions in the next chapter.

Definition 5.47: Parametrised 1-manifold (curve)

We call $M \subset \mathbb{R}^n$ a curve if it is a 1-dimensional submanifold. Usually, we attribute to M a parametrisation (in the sense of 4.10 (4), but G is global) $\phi : V \to \mathbb{R}^n$ with $V \subset \mathbb{R}$ open such that $M \subset \phi(V)$, which we also call a curve.

Definition 5.48: Length of a curve

Let $V \subset \mathbb{R}$ open, such that $[a, b] \subset V$ and $\phi : V \to \mathbb{R}^n$ a curve. Then we define the length of the curve from a to b as

$$L(\phi([a, b])) = \int_a^b |\phi'(t)| \, dt.$$

What do we imagine this length of a curve should satisfy?

(1) Independence of a reparametrisation $\phi \to \phi \circ \psi$,

(2) Additivity,

(3) Invariance under Euclidean isometry,

(4) Normalisation, so the length of $[0, 1] \times \{0\} \in \mathbb{R} \times \mathbb{R}^{n-1}$ should be 1.

Now, for 1, we will prove a general variant soon, 2 follows the additivity of the one-dimensional integral. A Euclidean isometry is a map of the form $F(x) = Rx + b$, where R is an orthogonal $n \times n$ matrix. But then

$$(F \circ \phi)'(t) = D(F \circ \phi)(t) = DF(\phi(t)) \cdot D\phi(t) = R \cdot \phi'(t),$$

and in particular since $\|R\|_2 = 1$ (Linear Algebra), we get $|(F \circ \phi)'(t)| = |\phi'(t)|$. Normalisation can be verified by computation.

Now this we can extend to general dimensions. Notation-wise, when I say M is *parametrised by ϕ*, I'm referring to the characterisation of a submanifold from Proposition 4.10 (4) and assuming that the parametrisation ϕ suffices for all points on M (we will later hopefully introduce partitions of unity which take care of this). When I asked the professor about this ambiguity, he argued this is just how it is in differential geometry. So henceforth a curve can be either, and it will have to be clear from the context. The same

DEFINITION 5.49: d-VOLUME OF PARAMETRISED d-SUBMANIFOLD

Let $V \subset \mathbb{R}^d$ open and $M \subset \mathbb{R}^n$ a d-dimensional manifold parametrised by $\phi : V \to \mathbb{R}^n$. Let $E \subset V$ Jordan measurable with $\overline{E} \subset V$, then we define the d-volume of $\phi(E)$ as

$$\mathrm{vol}_d(\phi(E)) = \int_E \sqrt{\det(D\phi^\top D\phi)}(x)\, dx.$$

We call the quantity $\sqrt{\det(D\phi^\top D\phi)}$ the *Gram Determinant* and it is a continuous function $\overline{E} \to \mathbb{R}$, hence the integral is well-defined by 5.35.

REMARK 5.50: GRAM DETERMINANT

Let L be an $n \times d$ matrix with $d \leq n$ and rank d. Then $L^\top L$ is a $d \times d$ matrix of full rank (see Linalg 2 exercise sheet 8) and $\sqrt{\det(L^\top L)}$ is the Gram determinant. To check that the square root exists, consider that $L^\top L$ is diagonalisable and its eigenvalues are all positive (see proof of 5.21).
For $d = 2$, one checks as an exercise that if $L = (v, w)$ for $v, w \in \mathbb{R}^3$, then

$$\det(L^\top L) = |v \times w|^2 = |v|^2 |w|^2 - (v \cdot w)^2.$$

In particular, if $\phi : V \to \mathbb{R}^3$ is a parametrisation of the surface M, then for $E \subset V$ Jordan measurable such that $\overline{E} \subset V$,

$$A(\phi(E)) := \mathrm{vol}_2(\phi(E)) = \int_E |\partial_1 \phi \times \partial_2 \phi|\,.$$

See also slides from the interactive presentation by Serra, I think it provides a very different and much more intuitive understanding on why this is a *good* definition for d-volume.

EXAMPLE 5.51: AREA OF THE SPHERE

We parametrise the unit ball $B_1(0) \subset \mathbb{R}^3$ with

$$\Psi(\theta, \varphi) = \begin{pmatrix} \cos\varphi \cos\theta \\ \sin\varphi \cos\theta \\ \sin\theta \end{pmatrix}.$$

Then the area (so 2-volume) of B_1 is given by (ignore sets of measure 0 as always)

$$A\Big(\Psi\Big((0, 2\pi) \times (-\tfrac{\pi}{2}, \tfrac{\pi}{2})\Big)\Big) = \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} \sqrt{\det D\Psi^\top D\Psi}\, d\varphi d\theta.$$

One computes

$$D\Psi^\top D\Psi = \begin{pmatrix} 1 & 0 \\ 0 & \cos^2\theta \end{pmatrix} \implies \det(D\Phi^\top D\Phi) = \cos^2\theta,$$

which we can bring back to compute

$$A(B_1) = \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} \cos\theta\, d\theta d\varphi = 4\pi.$$

Lemma 5.52: Invariance under reparametrisation

Let $V, W \subset \mathbb{R}^d$ open and $M \subset \mathbb{R}^n$ a d-dimensional manifold parametrised by $\phi : V \to \mathbb{R}^n$. Assume that $\psi : W \to V$ is a C^1 diffeomorphism and let $E \subset V$ Jordan measurable such that $\overline{E} \subset V$. Then

$$\mathrm{vol}_d(\phi(E)) = \mathrm{vol}_d(\phi \circ \psi(\psi^{-1}(E))).$$

Proof. We compute

$$D(\phi \circ \psi)(x) = D\phi(\psi(x)) \cdot D\psi(x) \implies D(\phi \circ \psi)(x)^\top = D\psi(x)^\top \cdot D\phi(\psi(x))^\top.$$

This we use to find the Gram determinant

$$\det(D(\phi \circ \psi)^\top D(\phi \circ \psi)) = \det(D\psi^\top D(\phi \circ \psi)^\top D(\phi \circ \psi) D\psi) = \det(D\psi)^2 \cdot \det(D\phi D\phi^\top) \circ \psi.$$

We integrate over this using change of variables 5.36

$$\int_E \sqrt{\det(D\phi^T D\phi)}(x)\, dx = \int_{\psi^{-1}(E)} \sqrt{\det(D\phi^\top D\phi)}(\psi(x))\, |\det D\psi(y)|\, dy$$

and plug in the Gram determinant we computed to get

$$\int_{\psi^{-1}(E)} \sqrt{\det(D\psi)^2 \det(D\phi D\phi^\top) \circ \psi}(y)\, dy.$$

But then the respective definitions of the d-volumes match. □

Remark 5.53: The issue with reparametrisation

Assume we are given multiple parametrisations for parts of a manifold that might intersect. In particular, assume we have some manifold M and that we have $\phi_i : V_i \to M \cap W_i$ for $i = 1, 2$. It is possible that $\widetilde{M} := M \cap (W_1 \cap W_2) \neq \emptyset$. In general, the d-volume of ϕ and ψ will not be the same, because they don't even look at the same parts of a manifold. What really matters is that they align on the part of the manifold where they intersect.
We check this by redefining the parametrisation $\phi_i : \widetilde{V}_i \to \widetilde{M}$ with $\widetilde{V}_i = \phi_i^{-1}(\widetilde{M})$. Then we define $\psi = \phi_2^{-1} \circ \phi_1 : \widetilde{V}_1 \to \widetilde{V}_2$, which is of class C^1 (check course notes). And so it remains to show the d-volume is invariant under the reparametrisation with this ψ, which we know is true by 5.52.

Lemma 5.54: Invariance under orthogonal transformations

Let $V \subset \mathbb{R}^d$ open and $M \subset \mathbb{R}^n$ a d-dimensional manifold parametrised by $\phi : V \to \mathbb{R}^n$. Let $R \in O(n)$. Then for $E \subset V$ Jordan measurable such that $\overline{E} \subset V$,

$$\mathrm{vol}_d(R\phi(E)) = \mathrm{vol}_d(\phi(E)).$$

Proof. Use the fact that for any linear map $L : \mathbb{R}^d \to \mathbb{R}^n$, we gave

$$\det((RL)^\top RL) = \det(L^\top R^\top RL) = \det(L^\top L).$$

The rest is obvious. □

Remark 5.55: More intuition about the Gram determinant

Let L be an $n \times d$ matrix of rank d. We know $L(\mathbb{R}^d)$ is a d-dimensional subspace. By Linear Algebra, there exists $R \in O(n)$ such that $RL = \begin{pmatrix} \widetilde{L} \\ 0 \end{pmatrix}$ for $\widetilde{L}$ a $d \times d$ matrix. But then,

$$\sqrt{\det((RL)^\top RL)} = \sqrt{\det(\widetilde{L}^\top \widetilde{L})} = \left|\det(\widetilde{L})\right| = \left|\det(\widetilde{L}[0,1]^d)\right| = \left|\det(L[0,1]^d)\right|,$$

so the Gram determinant gives us precisely the stretching of d-dimensional volume. So like $L([0,1]^d)$ would be some parallelepiped in $\mathbb{R}^n$, $\widetilde{L}([0,1]^d)$ would be the projection of that onto a flat subspace, where $\det \widetilde{L}$ tells us exactly how the d-volume is stretched.

Definition 5.56: Support of a function

Let $U \subset \mathbb{R}^n$ be open and $f : U \to \mathbb{R}$ continuous. Define the support of f as

$$\operatorname{spt}(f) = \overline{\{x \in U \mid f(x) \neq 0\}}.$$

If $\operatorname{spt}(f)$ is bounded, we call f compactly supported in U.

Definition 5.57: Integral over parametrised submanifold

Let $V \subset \mathbb{R}^d$ open, $M \subset \mathbb{R}^n$ a d-dimensional submanifold parametrised by $\phi : V \to \mathbb{R}^n$ such that $\phi(V) = M$ and $f : M \to \mathbb{R}^n$ continuous such that $\operatorname{spt}(f \circ \phi) \subset V$. Then

$$\int_M f(p)\, d\mathrm{vol}_M(p) = \int_M f\, d\mathrm{vol}_M := \int_V \left((f \circ \phi) \cdot \sqrt{\det(D\phi^\top D\phi)}\right)(x)\, dx.$$

By similar computations like for the d-volume, this is invariant under isometries and reparametrisation. We also often write $d\mathrm{vol}_d = d\mathrm{vol}_M$.

Remark 5.58: Integral is well-defined

The integral exists because V is open and the support $\operatorname{spt}(f)$ is covered by finitely many balls $B_{r_i}(x_i) \subset V$ such that $A := \bigcup_{i=1}^N B_{r_i/2}(x_i) \subset V$ and $\overline{A} \subset V$. But since f is zero on $V \setminus \overline{A}$, it suffices to take the integral not over V, but over $\overline{A}$, so we know it exists by 5.35.

Proposition 5.59: Integration over graphical submanifolds

Let $V \subset \mathbb{R}^{n-1}$ open and let $g : V \to \mathbb{R}$ class C^1. Consider the submanifold (the fact this is one was on a problem sheet) given by

$$M = \left\{x \in \mathbb{R}^{n-1} \times \mathbb{R} \mid x_n = g(x_1, \dots, x_{n-1})\right\}.$$

Then

$$\int_M f\, d\mathrm{vol}_{n-1} = \int_V (f \circ \phi)(x_1, \dots, x_{n-1}) \sqrt{1 + |\nabla g(x_1, \dots, x_{n-1})|^2}\, dx.$$

In particular, for $f \equiv 1$, we get the $n-1$-volume

$$\mathrm{vol}_{n-1}(M) = \int_V \sqrt{1 + |\nabla g(x_1, \dots, x_{n-1})|^2}\, dx$$

Proof. It suffices to compute the Gram determinant. It requires a trick, but it's not instructive. ☐

Chapter 6

Global integral theorems

The goal now is to show a lot of theorems for small open balls around individual points on a manifold. Then on *compact* manifolds, which we will define later, we will be able to generalise the theorems easily.

6.1 Parametrised graphical submanifolds

We need to prepare definitions and Lemmas that will help us formulate the divergence theorem.

Definition 6.1: Bounded C^k domain

A bounded open set $\Omega \subset \mathbb{R}^n$ is called a bounded C^k domain if $M = \partial\Omega$ is an $(n-1)$-dimensional manifold of class C^k.

Definition 6.2: Exterior and interior normal map

Let $\Omega \subset \mathbb{R}^n$ be a bounded C^1 domain. A normal vector ν at $p \in \partial\Omega$ is called exterior if

$$p + h\nu \in \mathbb{R}^n \setminus \overline{\Omega} \quad \text{for all sufficiently small } h > 0$$

and interior if

$$p + h\nu \in \mathbb{R}^n \in \Omega^\circ \quad \text{for all sufficiently small } h > 0.$$

Lemma 6.3: Exterior unit normal

Let Ω be a bounded C^1 domain, $p_0 \in \partial\Omega$ the point we're focusing on and assume $\phi : V \to \mathbb{R}^n$ defined by $\phi(y) = (y, g(y))$ where $g : V \to \mathbb{R}$, with $V \subset \mathbb{R}^{n-1}$ open, is a graphical C^1 parametrisation of $M := \partial\Omega \cap B_r(p_0)$ (including the technical note about the intersection, see Proposition 4.10).
Suppose further that

$$\Omega \cap B_r(p_0) = \{(y, x_n) \in \mathbb{R}^n \mid x_n < g(y)\} \cap B_r(p_0)$$

and define for $p = (y, g(y))$ the vector

$$\nu(p) = \nu(y, g(y)) := \frac{(-\nabla g(y), 1)}{\sqrt{1 + |\nabla g(y)|^2}}.$$

Then $\nu(p)$ is at each point $p \in M$ an exterior normal of unit length.

Proof. Recall that by Remark 4.14, since we have the parametrisation of M given by $\phi(y) = (y, g(y))$, then T_pM is an $n-1$-dimensional space spanned by $\partial_i\phi$ for $1 \leq i \leq n-1$. In our case, the partials are

$$\partial_i\phi(y) = e_i + \partial_i g(y) \cdot e_n,$$

hence up to scalar factors, for $1 \leq i \leq n-1$,

$$\partial_i\phi \cdot \nu = (e_i + \partial_i g \cdot e_n) \cdot ((-\nabla g, 0) + e_n) = -\partial_i g(x) + \partial_i g(x) = 0.$$

And so ν is perpendicular to all vectors in the tangent space. When you divide it by its length, you get a unit vector.

To prove it is exterior, consider that if we need to show $(y', y_n) = (y, g(y)) + h(-\nabla g(p), 1) \notin \overline{\Omega}$. For that, we would have to show that $y_n > g(y')$. But we can do a Taylor expansion (check details)

$$g(y') = g(y - h\nabla g(y)) = g(y) - |\nabla g(y)|^2 h + o(h) < g(y) + h.$$

So for small enough h, the normal is exterior. $\square$

I added one more thing that I thought would be useful when dealing with manifolds, which we could have introduced earlier but whatever.

Remark 6.4: Uniqueness of the tangent space

The definition of the tangent space $T_pM := \mathrm{Im}(D\varphi(u_0))$ (characterisation 4.14) is independent of the choice of chart. Indeed, if $\varphi : U \to M$ and $\psi : V \to M$ are two charts with $\varphi(u_0) = \psi(v_0) = p$, then the transition map $\psi^{-1} \circ \varphi$ is a C^1-diffeomorphism. By the chain rule,

$$D\varphi(u_0) = D\psi(v_0)\, D(\psi^{-1} \circ \varphi)(u_0).$$

Since $D(\psi^{-1} \circ \varphi)(u_0)$ is invertible, the domains ultimately coincide and the images

$$D\varphi(U) = D\psi(V).$$

Definition 6.5: Exterior unit map

If $\Omega \subset \mathbb{R}^n$ is a bounded C^1 domain, the continuous map $\nu : \partial\Omega \to \mathbb{S}^{n-1}$ assigning to each point $p \in \partial\Omega$ the exterior normal vector at p is called exterior unit normal map. Notice that since the tangent space at a point of a manifold is defined uniquely, the map is unique to each manifold.

Remark 6.6: C^k function on non-open domains

If $A \subset \mathbb{R}^n$ is not open we say that $f \in C^k(A, \mathbb{R}^n)$ if there exist $U \supset A$ open and $\widetilde{f} \in C^k(U, \mathbb{R}^n)$ such that $\widetilde{f}|_A = f$.

6.2 Local divergence theorem

We are now in a position to prove a weaker version of the divergence theorem. We will prove theorems using always one particular parametrisation, but don't forget that we put in the work last chapter and defined 5.57 and 5.49 such that they are invariant under such reparametrisations and also isometries.

This will be the setup for every Proposition in the following section. Let Ω be a bounded C^1 domain, $p_0 \in \partial\Omega$. Consider the locally graphically parametrised manifold $M := \partial\Omega \cap B_r(p_0)$ and let ν denote its unit exterior normal map. Suppose also that

$$\Omega \cap B_r(p_0) = \{(y, x_n) \in \mathbb{R}^n \mid x_n < g(y)\} \cap B_r(p_0),$$

where $g : V \subset \mathbb{R}^{n-1} \to \mathbb{R}$ is the local graphic representation, and the induced $\phi(y) = (y, g(y))$.

PROPOSITION 6.7: DIVERGENCE THEOREM REDUCED VERSION 1

Let $f : \overline{\Omega} \to \mathbb{R}$ be a C^1 function. Assume $\operatorname{spt}(f) \subset \overline{\Omega} \cap B_r(p_0)$. Then

$$\int_\Omega \partial_n f \, dx = \int_M f(p)\nu(p) \cdot e_n \, \mathrm{dvol}_M(p).$$

Proof. We make use of the graphical parametrisation $M = \phi(V)$ for $\phi(x) = (x, g(x))$. Let's inspect RHS first. Using Proposition 5.59,

$$\int_M f(p)\nu(p) \cdot e_n \, \mathrm{dvol}_M(p) = \int_V f \circ \phi \frac{(-\nabla g(x), 1)}{\sqrt{1 + |\nabla g(x)|^2}} \cdot e_n \sqrt{1 + |\nabla g(x)|^2} \, dx.$$

But we notice that after the scalar product is taken, all that remains will be $\int_V f(\phi(x)) \, dx$.

For the LHS, setting $\Gamma = \{(x, x_n) \mid x_n < g(x)\}$, because the function vanishes outside the support, we're effectively integrating over $\overline{\Omega \cap B_r(p_0)}$, so it's well-defined by 5.35, 5.58 and by assumption,

$$\int_\Omega \partial_n f \, dx = \int_\Gamma \partial_n f(x, x_n) \, dx.$$

But by Fubini, we can integrate for each vertical line (to do this rigorously gets ugly, see notes), so by Fundamental Theorem of Calculus, we get

$$\int_V \int_{-\infty}^{g(y)} \partial_n f(y, x_n) \, dx_n dy = \int_V [f(y, \cdot)]_{-\infty}^{g(y)} \, dy = \int_V f(y, g(y)) \, dy.$$

But by definition of ϕ, these are the same. $\square$

PROPOSITION 6.8: DIVERGENCE THEOREM REDUCED VERSION 2

Let $f : \overline{\Omega} \to \mathbb{R}$ be a C^1 function. Assume $\operatorname{spt}(f) \subset \overline{\Omega} \cap B_r(p_0)$. Then $\exists \varepsilon > 0$ such that $\forall w \in \mathbb{R}^n$, if $|w| = 1$ and $|w - e_n| < \varepsilon$,

$$\int_\Omega \partial_w f \, dx = \int_M f(p)\nu(p) \cdot w \, \mathrm{dvol}_M(p).$$

Proof. This is the most technical and awful part of the proof, but it's a very intuitive and easy statement. Compare with course notes, I might rewrite. $\square$

Proposition 6.9: Divergence theorem reduced version 3

Let $f : \overline{\Omega} \to \mathbb{R}$ be a C^1 function. Assume $\operatorname{spt}(f) \subset \overline{\Omega} \cap B_r(p_0)$. Then $\forall w \in \mathbb{R}^n$,

$$\int_\Omega \partial_w f \, dx = \int_M f(p)\nu(p) \cdot w \, \mathrm{dvol}_M(p).$$

Proof. If we have proven Proposition 6.8, then we can this last step is not so hard. Indeed, the equation of Proposition 6.8 is linear in w and holds for all $w \in B_\varepsilon(e_n) \cap \mathbb{S}^{n-1}$. But we know from linear algebra and Euclidean spaces that $\mathbb{R}^n = \operatorname{Sp}(B_\varepsilon(e_n) \cap \mathbb{S}^{n-1})$, so the equations holds for any $w \in \mathbb{R}^n$. □

Definition 6.10: Divergence

Let $F \in C^1(\Omega, \mathbb{R}^n)$ (we call such functions a vector field). Then we define the divergence of F at a point $x \in \Omega$ as $\operatorname{div}(F)(x) = \sum_{i=1}^n \partial_i F_i(x)$. This defines a real valued function $\operatorname{div}(F) : \Omega \to \mathbb{R}$.

Theorem 6.11: Local Divergence Theorem

Let $F \in C^1(\overline{\Omega}, \mathbb{R}^n)$ such that $\operatorname{spt}(F) \subset \overline{\Omega} \cap B_r(p_0)$. Then

$$\int_\Omega \operatorname{div}(F) = \int_M F \cdot \nu \, \mathrm{dvol}_M.$$

Proof. Use Proposition 6.9 with $w = e_i$, for each $1 \leq i \leq n$ and $f = F_i$. We get

$$\int_\Omega \partial_i F_i = \int_M F_i e_i \cdot \nu \, \mathrm{dvol}_M$$

for each $1 \leq i \leq n$. We can sum over all i to get the Theorem, apply linearity of the integral, scalar product and definition of divergence. □

6.3 Integration over compact submanifolds

But how do we extend everything we did since introducing the d-volume to submanifolds that do not admit a single parametrisation? Indeed, a some work will have to be done to define an integral for multiple parametrisation and to further also prove that it is a sound definition. Only after that can we tackle the global version of the divergence theorem.

Lemma 6.12: Paritition of unity

Let $K \subset \mathbb{R}^n$ be a compact set and for $1 \leq \ell \leq N$, $B_{p_\ell}(r_\ell) = B_\ell$ open balls that cover K. Then $\exists U$ open such that $K \subset U$ and N functions $\eta_1, \dots, \eta_N : \mathbb{R}^n \to [0, \infty)$ and a function $\tilde{\eta} : \mathbb{R}^n \to [0, \infty)$, all C^∞, such that

$$1 = \sum_{\ell=1}^N \eta_\ell(x) + \tilde{\eta}(x) \quad \forall x \in \mathbb{R}^n$$

and $\operatorname{spt}(\eta_\ell) \subset B_\ell$ and $\operatorname{spt}(\tilde{\eta}) \subset \mathbb{R}^n \setminus U$.

Proof. Define the (non-analytic btw) functions $\xi, \tilde{\xi}$ on $\mathbb{R}^n$ via

$$\xi(x) = \begin{cases} e^{-\frac{1}{1-|x|^2}} & \text{if } x \in B_1(0) \\ 0 & \text{if } x \in \mathbb{R}^n \setminus B_1(0) \end{cases} \qquad \tilde{\xi}(x) = \begin{cases} 0 & \text{if } x \in B_1(0) \\ e^{-\frac{1}{|x|^2-1}} & \text{if } x \in \mathbb{R}^n \setminus B_1(0) \end{cases}.$$

We know $\mathrm{spt}(\xi) = \overline{B_1(0)}$. For $r > 0$ and $p \in \mathbb{R}^n$, we define the scaled and shifted function $\xi_{p,r}(x) := \xi(\frac{x-p}{r})$, which has support $\mathrm{spt}(\xi_{r,p}) = \overline{B_r(p)}$. We can cover

$$K \subset \bigcup_{\theta \in (0,1)} \bigcup_{\ell=1}^{N} B_{\theta r_\ell}(p_\ell),$$

and since the sets form a chain and K is compact, $\exists \theta \in (0,1)$ s.t. $K \subset \bigcup_{\ell=1}^N B_{\theta r_\ell}(p_\ell)$. Define again

$$\xi_\ell(x) = \xi\left(\frac{x-p_\ell}{\sqrt{\theta} r_\ell}\right), \quad \tilde{\xi}_\ell(x) = \tilde{\xi}\left(\frac{x-p_\ell}{\theta r_\ell}\right)$$

and notice that the function $\tilde{\xi}_0(x) = \prod_{i=1}^N \tilde{\xi}_\ell(x)$ is 0 on the set $\bigcup_{\ell=1}^N B_{r_\ell \theta}(p_\ell) := U \supset K$. And put all of that together

$$S(x) = \sum_{\ell=1}^N \xi_\ell(x) + \tilde{\xi}_0(x),$$

which is by construction never 0. Hence define $\eta_\ell(x) := \frac{\xi_\ell(x)}{S(x)}$ and $\tilde{\eta}(x) = \frac{\tilde{\xi}_0(x)}{S(x)}$, which satisfy the statement. □

DEFINITION 6.13: GRAPHICAL COVER OF A COMPACT d-SUBMANIFOLD

Let $M \subset \mathbb{R}^n$ be a compact d-dimensional submanifold of class C^k, with $k \geq 1$. We call a collection of N maps $\phi_\ell : V_\ell \to \mathbb{R}^n, 1 \leq \ell \leq N$, of class C^k, a graphical cover of M if for each ℓ, the following properties are satisfied:

(1) V_ℓ is an open set of $\mathbb{R}^d$.

(2) There is an orthogonal $n \times n$ matrix R_ℓ such that $\phi_\ell : V_\ell \to \mathbb{R}^n$ is of the form

$$\phi_\ell(x_1, \ldots x_d) = R_\ell(x_1, \ldots, x_d, g_\ell(x_1, \ldots, x_d))$$

for some $g_\ell \in C^k\left(V_\ell, \mathbb{R}^{n-d}\right)$.

(3) There is $p_\ell \in M$ and $r_\ell > 0$ such that

$$\phi_\ell(V_\ell) \cap B_{r_\ell}(p_\ell) = M \cap B_{r_\ell}(p_\ell) \quad \text{and} \quad B_{r_\ell}(p_\ell) \subset R_\ell\left(V_\ell \times \mathbb{R}^{n-d}\right).$$

(4) The balls $B_{r_\ell}(p_\ell)$ cover M, so $M \subset \bigcup_{\ell=1}^N B_{r_\ell}(p_\ell)$.

LEMMA 6.14: GRAPHICAL COVER OF A COMPACT d-SUBMANIFOLD

Every compact d-dimensional submanifold $M \subset \mathbb{R}^n$ admits a graphical cover.

Proof. Recall Proposition 4.10, in particular characterisation (4). For any given point $p \in M$, we are guaranteed an open set U such that $M \cap U = R(\mathrm{graph}(g))$ for R a permutation of coordinates (in particular orthogonal) and $g \in C^1(V, \mathbb{R}^{n-d})$.

Then define the function $\phi : V \to \mathbb{R}^n$ by

$$\phi(x_1, \ldots, x_d) = R(x_1, \ldots, x_d, g(x_1, \ldots x_d)).$$

Finally, for some $y \in V$ such that $p = \phi(y) = R(y, g(y))$, find $r > 0$ such that $B_r(y) \subset V$ and $B_r(p) \subset U$, because then you automatically get $B_r(p) = B_r(R(y, g(y))) \subset R(V \times \mathbb{R}^{n-d})$.

Now, cover M with these balls for every single $p \in M$ and extract a finite subcover. □

Definition 6.15: Integral over a compact d-submanifold

Let $M \subset \mathbb{R}^n$ be an d-dimensional compact C^1 submanifold and $f : M \to \mathbb{R}$ a continuous function. Given a graphical cover $\phi_\ell : V_\ell \to \mathbb{R}^n, 1 \leq \ell \leq N$, let $\eta_\ell : \mathbb{R}^n \to [0, \infty)$ be a partition of unity (exists by Lemma 6.12) on M w.r.t. the balls $\{B_{r_\ell}(p_\ell)\}_{1 \leq \ell \leq N}$ associated to the graphical cover. We define:

$$\int_M f \, d\,\mathrm{vol}_d := \sum_{\ell=1}^N \int_{M \cap B_\ell(x_\ell)} f \eta_\ell \, d\,\mathrm{vol}_d,$$

where the second integral is defined by 5.57.

This definition is once again invariant under choice of: paritition of unity, parametrisations, graphical covers. The proof for that is mostly a lot of tedious details, but the "Analysis" part has already been covered when we proved invariance for locally parametrised submanifolds.

This was totally skipped in the lecture, we just moved onto the divergence theorem without covering these details. And there's good reason for that, because it's very technical only to get a result which you would expect anyway. The details of this and generalisation will definitely be covered later in the studies in topics like differential geometry and such. For now, this is good. And most of the time, we work with one single parametrisation anyway, so the computation is simplified (although one graphical representation for a manifold with one parametrisation may already be insufficient).

6.4 Global divergence theorem

Now we can prove the full correct version.

Theorem 6.16: Divergence theorem

Let $\Omega \subset \mathbb{R}^n$ be a bounded C^1 domain. Let $F \in C^1(\overline{\Omega}, \mathbb{R}^n)$ be a vector field. Then

$$\int_\Omega \operatorname{div} F = \int_{\partial\Omega} F \cdot \nu \, d\mathrm{vol}_{n-1}.$$

Proof. Because Ω is a C^1 domain, by Lemma 6.14, $\partial\Omega$ admits a graphical cover with the associated orthogonal linear maps, which we will denote L_ℓ.

Let η_ℓ be a partition of unity w.r.t. this graphical cover as in Lemma 6.12 with $\tilde{\eta}$ denoting the background function. Define

$$F_\ell := F \cdot \eta_\ell, \quad \tilde{F} = \begin{cases} F \cdot \tilde{\eta} & \text{on } \overline{\Omega} \\ 0 & \text{on } \mathbb{R}^n \setminus \overline{\Omega} \end{cases} \quad \implies \quad \sum_{\ell=1}^N F_\ell + \tilde{F} = F \text{ on } \overline{\Omega}.$$

From the Lemma, we also get $\operatorname{spt}(F_\ell) \subset B_{r_\ell}(p_\ell)$ and $\operatorname{spt}(\tilde{F}) \subset \Omega^\circ$ (because it's zero around the boundary thanks to the partition of unity and defined to be zero outside of $\overline{\Omega}$). By construction, up to the isometry L_ℓ, $\Omega \cap B_{r_\ell}(p_\ell)$ satisfies the conditions for 6.11 with the function F_ℓ (actually, $F_\ell \circ L_\ell^{-1}$) hence by change of variables,

$$\int_\Omega \operatorname{div} F_\ell = \int_{\partial\Omega} F_\ell \cdot \nu \, d\mathrm{vol}_{n-1}.$$

Moreover, since $\tilde{F} \in C^1(\Omega, \mathbb{R}^n)$ and $\operatorname{spt}(\tilde{F}) \subset \Omega^\circ$, if we could show

$$\int_\Omega \operatorname{div} \tilde{F} = 0,$$

we would be done. Indeed, we'd sum over all ℓ and get

$$\int_\Omega \operatorname{div} F = \int_\Omega \operatorname{div}\left(\sum_{\ell=1}^n F_\ell + \tilde{F}\right) = \int_{\partial\Omega}\left(\sum_{\ell=1}^n F_\ell\right)\cdot \nu \, d\mathrm{vol}_{n-1} = \int_{\partial\Omega} F\cdot\nu\, d\mathrm{vol}_{n-1}.$$

To prove the claim, we will show $\int_\Omega \partial_i \tilde{F}_i = 0$ for all $i = 1, \dots, n$ separately. Because the support of $\tilde{F}$ is compact and strictly contained in the open set Ω°, for all $t > 0$ small enough,

$$\int_\Omega \tilde{F}_i(x + te_i)\, dx = \int_{\Omega + te_i} \tilde{F}_i(x)\, dx.$$

But we can differentiate under the integral sign using 5.44,

$$\frac{d}{dt}\bigg|_{t=0} \int_\Omega \tilde{F}_i(x + te_i)\, dx = \int_{\Omega + te_i} \partial_i\tilde{F}_i(x)\, dx.$$

But the RHS doesn't actually depend on t once it's small enough, because the support of $\tilde{F}$ is contained very well in Ω (so there's always at least some ball distance to the boundary), so shifting very little doesn't even change the points we integrate over. Once you establish that, the theorem is proven. □

Example 6.17: Archimedes Principle

When a body is submerged in water, at each infinitesimal normal area to its surface, it experiences a force of Area $\cdot\, p \cdot \nu$, where p is the pressure. The pressure at a point that is submerged z distance under water is given by $p = z\rho$, where ρ is the density of water. But then, the total force is given by (physicists write dS for $d\mathrm{vol}_{n-1}$)

$$F = \int_{\partial\Omega} p\cdot\nu\, dS = \rho\int_{\partial\Omega} z\cdot\nu\, dS.$$

In particular, in the z-component, we get

$$F\cdot e_3 = \rho\int_{\partial\Omega}(ze_3)\nu\, dS = \rho\int_\Omega 1\, dx = \rho\mathrm{vol}_3(\Omega),$$

which is just the weight of the water that has been displaced by the volume of Ω. Hence we have a mathematical verification of Archimedes Principle.

Example 6.18: Area and volume of the sphere

Let $\partial B_1 \subset \mathbb{R}^n$ be the unit sphere. We know

$$\mathrm{vol}_{n-1}(\partial B_1) = \int_{\partial B_1} 1\, dS = \int_{\partial B_1} \underbrace{p\cdot\nu(p)}_{=1}\, dS.$$

But what is the divergence of $F(x) = x$? Well, you have to sum over each component once, each time the partial derivative is 1, so $\operatorname{div}(x) = n$. But this gives us

$$\mathrm{vol}_{n-1}(\partial B_1) = \int_{B_1} n\, dx = \mathrm{vol}_n(B_1)\cdot n.$$

This gives us the relation between the volume of the n-sphere and its area.

Example 6.19: Continuity equation (Fluid Dynamics)

The continuity equation appears in many variations in physics. For instance, in fluid dynamics, we assign each point and time in a fluid a velocity v and a density ρ. The mass of fluid inside a control volume Ω should satisfy

$$\frac{d}{dt}\int_\Omega \rho\, dx = \int_{\partial\Omega} \rho v \cdot \nu\, dS.$$

But we can differentiate under the integral sign and apply the divergence theorem to get

$$\int_\Omega \partial_t \rho\, dx = \int_\Omega \operatorname{div}(\rho v)\, dx.$$

And since this holds for all domains Ω and the functions are continuous (proof next year), this forces $\partial_t \rho = \operatorname{div}(\rho v)$.
Similar continuity equations hold for many other fields in physics. We saw one version of this already in electrodynamics this semester, where we have $\partial_t \rho = \operatorname{div} E$.

Definition 6.20: Rotation (curl)

Let $F \in C^1(\Omega, \mathbb{R}^2)$. We define the curl, rotation or vorticity as $\operatorname{rot} F = \partial_2 F_1 - \partial_1 F_2$.

Theorem 6.21: Green's theorem

Let $\Omega \subset \mathbb{R}^2$ be a C^1 domain. Let $F \in C^1(\overline{\Omega}, \mathbb{R}^2)$ be a vector field, and $\partial\Omega$ be a curve. If we let $\tau(p)$ be the tangent vector to the curve at a point $p \in \partial\Omega$, (by convention normalise it and make it point counterclockwise). Then we have ($dL = d\mathrm{vol}_1$)

$$\int_{\partial\Omega} F \cdot \tau\, dL = \int_\Omega \operatorname{rot}(F)\, dx.$$

Proof. Define another vector field by $G = F^\perp = (-F_2, F_1)$ and apply the divergence theorem to G. Notice that the tangent vector τ is perpendicular to the normal vector ν. Hence the relation $F \cdot \tau = -G \cdot \nu$ holds and we have by 6.16,

$$\int_\Omega \operatorname{div}(G) = \int_{\partial\Omega} G \cdot \nu\, dL = \int_{\partial\Omega} -F \cdot \tau\, dL.$$

But since $\operatorname{div} G = -\operatorname{rot} F$, we get the theorem. □

6.5 Line integrals

To start inspecting new types of integrals, we have to get into differential forms. We start with the simplest type. In what comes, we identify the covector space $\mathbb{R}^{n*}$ with the set of row vectors (they're canonically isomorphic anyway so we might as well). And I for the first time in history start paying attention to the difference between row and column vectors.

Definition 6.22: 1-forms

Let $U \subset \mathbb{R}^n$ be open. A map assigning to each point $x \in U$ a covector $x \mapsto (\alpha_1(x), \dots, \alpha_n(x)) = \alpha_x \in \mathbb{R}^{n*}$ is called a 1-form. We say that a 1-form is C^k if every $\alpha_i : U \to \mathbb{R}$ is C^k.

Example 6.23: Exterior derivative

For $f \in C^k(U, \mathbb{R})$, notice that $x \mapsto Df(x)$ is a 1-form. We will talk about this in more detail maybe later, but it makes sense to call f a 0-form. Then the map $df : U \to \mathbb{R}^{n*} : x \mapsto Df(x)$ turns the 0-form f into a 1-form df.

Definition 6.24: Line integral

If $\alpha : U \to \mathbb{R}^{n*}$ is a class C^1 1-form in $U \subset \mathbb{R}^n$ open and $\gamma : [a, b] \to U$ a C^1 curve, then we define the line integral as

$$\int_\gamma \alpha := \int_a^b \alpha_{\gamma(t)}(\gamma'(t))\, dt = \int_a^b \alpha_{\gamma(t)} \cdot \gamma'(t)\, dt.$$

Lemma 6.25: Invariance under positive reparametrisation

Let $U \subset \mathbb{R}^n$ open, $\gamma : [a, b] \to U \subset \mathbb{R}^n$ a curve and $\psi : [c, d] \to [a, b]$ a C^1 diffeomorphism such that $\psi' > 0$. Then

$$\int_\gamma \alpha = \int_{\gamma\circ\psi} \alpha.$$

Proof. We can substitute $t = \psi(s)$ and we get

$$\int_{\gamma\circ\psi} \alpha = \int_c^d \alpha_{\gamma\circ\psi(t)}((\gamma \circ \psi)'(s))\, ds = \int_c^d \alpha_{\gamma\circ\psi(s)}(\gamma'(\psi(s)))\psi'(s)\, ds = \int_a^b \alpha_{\gamma(t)}\gamma'(t)\, dt = \int_\gamma \alpha.$$

□

Definition 6.26: Conservative vector field / Exact 1-form

Let $U \subset \mathbb{R}^n$ be connected and open. A 1-form $\alpha : U \to \mathbb{R}^{n*}$ class C^1 is *exact* if $\exists f \in C^2(U)$ such that $\alpha = df$.

Remark 6.27: Parallels to physics and familiar notation

Consider a vector field F and the associated 1-form $\alpha = F^\top$. Then

$$\int_\gamma \alpha = \int_\gamma F^\top = \int_a^b F^\top(\gamma(t))\gamma'(t)\, dt = \int_a^b F(\gamma(t)) \cdot \gamma'(t)\, dt = \int_\gamma F(s) \cdot ds$$

is just the usual "work along a path" from physics.
We say a vector field $F \in C^1(U, \mathbb{R}^n)$ is *conservative* if $\exists f : U \to \mathbb{R}$ C^2 such that $\nabla f = F$, which is the analogue to "exact".

Lemma 6.28: Path independence

Let $\alpha : U \to \mathbb{R}^{n*}$ be an exact 1-form such that $\alpha = df$ for $f \in C^2(U)$. Then for any C^1 curve $\gamma : [a, b] \to U$,

$$\int_\gamma \alpha = f(\gamma(b)) - f(\gamma(a)).$$

Proof. We compute using the chain rule

$$\int_\gamma \alpha = \int_a^b \alpha_{\gamma(t)}(\gamma'(t))\,dt = \int_a^b df_{\gamma(t)}(\gamma'(t))\,dt = \int_a^b (f\circ\gamma)'\,dt = f(\gamma(b)) - f(\gamma(a)).$$

□

Lemma 6.29: Integrability conditions

Let $U \subset \mathbb{R}^n$ open, $\alpha : U \to \mathbb{R}^{n*}$ be exact ($\alpha = df$) and C^1. Then $\partial_j\alpha_i = \partial_i\alpha_j$.

Proof. Schwarz Theorem 2.25 applies: $\partial_j\alpha_i = \partial_j\partial_i f = \partial_i\partial_j f = \partial_i\alpha_j$. □

Lemma 6.30: Converse on convex domains

Assume $U \subset \mathbb{R}^n$ is open convex and let $\alpha : U \to \mathbb{R}^{n*}$ be a 1-form that satisfies the integrability conditions, i.e. $\partial_i\alpha_j = \partial_j\alpha_i$. Then $\exists f : U \to \mathbb{R}$ class C^2 such that $df = \alpha$.

Proof. Assume without loss of generality that $0 \in U$ (otherwise translate). We use the path integral of α along the straight line from 0 to $x \in U$ to define a function $f : U \to \mathbb{R}$ by

$$f(x) := \int_\gamma \alpha = \int_0^1 \alpha(tx)\cdot x\,dt$$

for $x \in U$ and $\gamma(t) = tx$. Let us prove that $\alpha = df$.

Indeed, fix $j \in \{1,\dots,n\}$ and consider, as a preparation for the computation of $\partial_j f$, for $h \in \mathbb{R}^n$

$$\begin{aligned}\partial_j(\alpha(tx)\cdot x) &= \partial_j\left(\sum_{k=1}^n \alpha_k(tx)x_k\right) = \sum_{k=1}^n \partial_j\alpha_k(tx)tx_k + \alpha_j(tx)\\ &= t\sum_{k=1}^n \partial_k\alpha_j(tx)x_k + \alpha_j(tx) = \partial_t\left(t\alpha_j(tx)\right)\end{aligned}$$

where we used the integrability conditions.

Hence, using Theorem 5.44, for $x \in U$ we can compute $\partial_j f(x)$ as

$$\partial_j f(x) = \partial_j \int_0^1 \alpha(tx)\cdot x dt = \int_0^1 \partial_t\left(t\alpha_j(tx)\right)dt = \left[t\alpha_j(tx)\right]_{t=0}^{t=1} = \alpha_j(x).$$

Thus, $\alpha = df$, and f is C^2. □

6.6 Differential forms

Let's think heuristically for a second. How do we generalise line integrals? Which quantities over parametrised submanifolds of dimension k are invariant under reparametrisation? That is the question that we will answer with differential forms. We inspect a map

$$\alpha_p(T) = \alpha(p, T)$$

that takes a point $p \in \mathbb{R}^n$ and an $n\times k$ matrix T. Then we would define something like

$$\int_M \alpha := \int_V \alpha(\phi(x), D\phi(x))\,dx,$$

for $\phi : V \subset \mathbb{R}^k \to \mathbb{R}^n$ a parametrisation of the manifold $M = \phi(V)$. Notice the similarity to the definition of the line integral. Let $\psi : W \to V$ be a C^1 diffeomorphism with $\det D\psi > 0$, which we refer to as "orientation-preserving". Using the chain rule, we get

$$\int_W \alpha(\phi \circ \psi(y), D(\phi \circ \psi)(y))\, dy = \int_W \alpha(\phi(\psi(y)), D\phi(\psi(t))D\psi(y))\, dy.$$

Similar to the line integral, we want $\alpha(p, TS) = \alpha(p, T) \det S$ for T an $n \times k$ and S an $k \times k$ matrix. Then, using change of variables, the integral would be

$$\int_W \alpha(\phi(\psi(y)), D\phi(\psi(y))) \det D\psi(y)\, dy = \int_V \alpha(\phi(x), D\phi(x))\, dx.$$

This is the characterising property of k-forms.

Example 6.31: Important example

We fix a point p and a $k \times n$ matrix A. If we define $\alpha_A(T) := \det(AT)$, then we can verify

$$\alpha_A(TS) = \det(ATS) = \det(AT)\det(S) = \alpha_A(T)\det(S).$$

More general, you can fix matrices $A_1, \dots, A_N$ that are $k \times n$ and $c_1, \dots, c_N \in \mathbb{R}$ and define

$$\alpha(T) = \sum_{\ell=1}^{N} c_\ell \det(A_\ell T),$$

which also has the desired property.

Definition 6.32: k-form in $\mathbb{R}^n$

Let $k \geq 1$ and $U \subset \mathbb{R}^n$ open. A k-form is a map α that assigns each point $p \in U$ a map $\alpha_p : \mathbb{R}^{n\times k} \to \mathbb{R}$ that satisfies

(1) $\alpha_p(TS) = \alpha_p(T)\det(S)$ for all $T \in \mathbb{R}^{n\times k}$ and $S \in \mathbb{R}^{k\times k}$,

(2) α_p multilinear in the columns (or rows) of the input matrix.

In the exercise sheet, we will prove that if α_p satisfies (1) and (2), then

$$\alpha_p(T) := \sum_{i \in \mathcal{I}_k} c_{i_1\cdots i_k} \det((e_{i_1}, \dots, e_{i_k})^\top T),$$

for the multi-index set $\mathcal{I}_k = \{(i_1, \dots, i_k) \in \mathbb{N}^k \mid i_1 < \cdots < i_k\}$, so the set of increasing multi-indices of length k and coefficients $c_i \in \mathbb{R}$. There are n over k such coefficients.

In fact, this is so important and common that we define the "simple" k-forms ω_I which assign each point $p \in U$ the map $\omega_I : T \mapsto \det(e_I^\top T)$ satisfying (1) and (2) for each $I \in \mathcal{I}_k$ and express any other k-form as a linear combination of these.

Definition 6.33: Wedge product

Let α be a k-form and β an m-form such that $k + m \leq n$. Fix α_p and β_p at a point $p \in U$. If α_p and β_p are of the form $\alpha_p(T) = \det(AT)$ and $\beta_p(S) = \det(BS)$, we define the map

$$(\alpha_p \wedge \beta_p)(X) := \det\left(\begin{pmatrix} A \\ \hline B \end{pmatrix} X\right),$$

for X a $n \times (k+m)$ matrix, $(A|B)^\top$ a $(k+m) \times n$ matrix.

Once you know the definition for the simple version, use linearity to get

$$\alpha_p \wedge \beta_p = \left(\sum_{i=1}^{M} a_i \alpha_i\right) \wedge \left(\sum_{j=1}^{N} b_j \beta_j\right) = \sum_{i=1}^{M}\sum_{j=1}^{N} a_i b_j \alpha_i \wedge \beta_j.$$

Then, for each point $p \in U$, define $(\alpha \wedge \beta)_p := \alpha_p \wedge \beta_p$.

Definition 6.34: Conventions about k-forms

By convention, 0-forms in a set $U \subset \mathbb{R}^n$ open are $C^\infty(U)$ functions.
Also, for any k-form α assigning $x \in U$ the map α_x, we write

$$\alpha_x(T) = \sum_{I \in \mathcal{I}_k} c_I(x)\omega_I(T),$$

and we demand that each $c_I(x)$ is a C^∞ function in x.
For $k > n$, the matrix $e_I^\top T$ is *always* not invertible, hence any k-form with $k > n$ is identically 0.
For $k = n$, there is only 1 multi-index in the set $\mathcal{I}_n$, hence all n-forms just look like $c(x)\cdot\det(I_n T) = c(x)\det(T)$.
We denote $\Omega^k(U)$ the set of k-forms with coefficients C^∞. In particular, $\Omega^0(U) = C^\infty(U)$.

Definition 6.35: Differential

Let $U \subset \mathbb{R}^n$ open, $0 \le k \le n$ and $\alpha \in \Omega^k(U)$ a k-form written as

$$\alpha_x(T) = \sum_{I \in \mathcal{I}_k} c_I(x)\omega_I(T).$$

Then $d\alpha$ is a $k+1$-form defined by

$$d\alpha := \sum_{I \in \mathcal{I}_k} dc_I \wedge \omega_I,$$

where we remind ourselves that $dc_I : x \mapsto (Dc_I)_x$ and it is a 1-form, so each $dc_I \wedge \omega_I$ is a $k+1$-form.

Remark 6.36: Relation to old notation

The coordinate functions $f(x) = x_i$ for $1 \le i \le n$ have $df_x = Df(x) = e_i^\top$. We denote $dx_i := df_x$.
And so we see the identity $dx_i = e_i^\top$. And we know any k-form can be written as

$$\alpha_x(T) = \sum_{I \in \mathcal{I}_k} c_I(x)\omega_I(T).$$

And from the definition of the wedge product $\omega_I = dx_{i_1} \wedge dx_{i_2} \wedge \cdots \wedge dx_{i_k} =: dx_I$.
With this observation, any k-form α in U has the expression

$$\alpha_x = \sum_{I \in \mathcal{I}_k} c_I(x)\, dx_I$$

and the dx_I are kind of like basis vectors, and they are constructed from the dx_i.
It is left as an exercise to show that for $\Psi : V \to U$ a diffeomorphism between $V, U \subset \mathbb{R}^n$,

$$d\Psi_1 \wedge d\Psi_2 \wedge \cdots \wedge d\Psi_n = \det D\Psi(x) \cdot dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n.$$

And so it becomes apparent how this notation is consistent with the change of variables formula.

From now on, denote $e_I^*(A) := \det(e_I^\top A) = \omega_I(A) = dx_I(A)$ and these form a basis for all k-forms. The wedge factors we denote $e_i^* = dx_i$ such that $dx_I = e_I^* = \bigwedge dx_i = \bigwedge e_i^*$.

Lemma 6.37: Double differential

Let $\alpha \in \Omega^k(U)$ with $U \subset \mathbb{R}^n$ open. Then $d(d\alpha) \in \Omega^{k+2}(U)$ is identically 0. Compactly, $d^2 = 0$.

Proof. It suffices to check this at the basis vectors, so compute $dd(f\omega_I)$ for $f : U \to \mathbb{R}$ smooth.

$$d(f\omega_I) = \left(\sum_{i=1}^n \partial_i f\, dx_i\right) \wedge \omega_I \implies d(df\omega_I) = \left(\sum_{i=1}^n \sum_{j=1}^n \partial_j \partial_i f\, dx_j \wedge dx_i\right) \wedge \omega_I.$$

As an exercise, one shows $dx_j \wedge dx_i = -dx_i \wedge dx_j$, and by Schwarz Theorem 2.25, the entire sum vanishes. □

Lemma 6.38: Leibniz Rule

Let $U \subset \mathbb{R}^n$ open, $\alpha \in \Omega^k(U)$ and $\beta \in \Omega^\ell(U)$. Then

$$d(\alpha \wedge \beta) = (d\alpha) \wedge \beta + (-1)^k \alpha \wedge d\beta.$$

Proof. Denote $\alpha = \sum_{I \in \mathcal{I}_k} a_I e_I^*$ and $\beta = \sum_{J \in \mathcal{I}_\ell} b_J e_J^*$. By definition,

$$d\alpha = \sum_{I \in \mathcal{I}_k} da_I \wedge e_I^*, \quad d\beta = \sum_{J \in \mathcal{I}_\ell} db_J \wedge e_J^*.$$

On the other hand, by simple product rule $d(a_I b_I) = (da_I) b_I + a_I (db_I)$,

$$d(\alpha \wedge \beta) = \sum_{I \in \mathcal{I}_k} \sum_{J \in \mathcal{I}_\ell} (da_I b_J + a_I db_J)\, e_I^* \wedge e_J^*.$$

We can split this up into

$$\sum_{I \in \mathcal{I}_k} \sum_{J \in \mathcal{I}_\ell} da_I b_I\, e_I^* \wedge e_J^* + \sum_{I \in \mathcal{I}_k} \sum_{J \in \mathcal{I}_\ell} a_I db_I\, e_I^* \wedge e_J^*$$

On the other hand,

$$d\alpha \wedge \beta + \alpha \wedge d\beta = \sum_{I \in \mathcal{I}_k} \sum_{J \in \mathcal{I}_\ell} da_I\, e_I^* \wedge (b_J\, e_J^*) + \sum_{I \in \mathcal{I}_k} \sum_{J \in \mathcal{I}_\ell} a_I\, e_I^* \wedge (db_J \wedge e_J^*).$$

Now, in the first sum, we can by definition of wedge commute $e_I^* \wedge (b_J\, e_J^*) = b_J\, e_I^* \wedge e_J^*$. Since the wedge is associative, we look for $e_I^* \wedge db_J$ and notice that

$$\det((e_{i_1}, \ldots, e_{i_k}, db_J)^\top A) = e_I^* \wedge db_J(A) = (-1)^k db_J \wedge e_I^*(A) = \det((db_I, e_{i_1}, \ldots, e_{i_k})^\top A),$$

where we used the fact that the determinant is alternating (verify as an exercise). So we commute at the cost of the sign and get the statement. □

Corollary 6.39: Differential of wedged exact forms

Let α be a k-form such that $\alpha = df_1 \wedge df_2 \wedge \cdots \wedge df_k$ for $f_i \in C^\infty(U)$. Then $d\alpha = 0$.

Proof. Apply Lemma 6.38 and 6.37 and prove by induction. □

Definition 6.40: Pullback

Let $U \subset \mathbb{R}^n, V \subset \mathbb{R}^m$ open and $\alpha \in \Omega^k(U)$. Let $F : V \to U$ be a C^∞ map. We define the pullback of α denoted $F^*\alpha \in \Omega^k(V)$ as follows:

$$(F^*\alpha)_y(A) := \alpha_{F(y)}(DF(y) \cdot A).$$

It is an exercise to show that $F^*\alpha$ is indeed a k-form on V.

Lemma 6.41: Pullback of basis form

Let $U \subset \mathbb{R}^n$ open. Assume $\alpha = e_I^* \in \Omega^k(U)$ for $I = (i_1, \ldots, i_k) \in \mathcal{I}_k$ is a k-form and $F(y) = (F_1(y), \ldots, F_n(y))$ such that $F \in C^\infty(U)$. Then

$$F^*\alpha = F^*e_I^* = dF_{i_1} \wedge \cdots \wedge dF_{i_k}.$$

Sometimes we write $F^*dx_I = dF_{i_1} \wedge \cdots \wedge dF_{i_k} =: dy_{i_1} \wedge \cdots \wedge dy_{i_k}$.

Proof. We have by definitions, noticing that $dF_i(y) = DF_i(y) = (\partial_1 F_i(y), \ldots, \partial_n F_i(y))$ and $e_I^*(DF) = DF_i$,

$$\begin{aligned}(F^*\alpha)_y(A) &= \alpha_{F(y)}(DF(y) \cdot A) = e_I^*(DF(y) \cdot A) = \det(e_I^\top DF(y) \cdot A) \\ &= \det\left((dF_{i_1}(y) \mid \cdots \mid dF_{i_k}(y))^\top \cdot A\right).\end{aligned}$$

Hence if you just split this into wedge factors, you get the statement. □

Proposition 6.42: Pullback commutes with wedge and differential

Let $U \subset \mathbb{R}^n, V \subset \mathbb{R}^m$ open and $F : V \to U$ a C^∞ map. Then

(1) $\forall \alpha \in \Omega^k(U), \beta \in \Omega^\ell(U) : F^*(\alpha \wedge \beta) = F^*\alpha \wedge F^*\beta$,

(2) $\forall \alpha \in \Omega^k(U) : d(F^*\alpha) = F^*(d\alpha)$.

Proof. It once again is enough to show the statements for $\alpha_x = f(x)\, e_I^*$ and $\beta_x = g(x)\, e_J^*$ with $f, g : U \to \mathbb{R}$ C^∞ functions and $I \in \mathcal{I}_k, J \in \mathcal{I}_\ell$.

(1) By definition of wedge product, $(\alpha \wedge \beta)_x = f(x)g(x)\, e_I^* \wedge e_J^*$. By definition of pullback,

$$F^*(\alpha \wedge \beta)_y(T) = f(F(y))g(F(y))\, (e_I^* \wedge e_J^*)(DF(y) \cdot T).$$

On the other hand,

$$(F^*\alpha \wedge F^*\beta)_y(T) = f(F(y))\, e_I^*(DF(y) \cdot T_1) \wedge g(F(y))\, e_J^*(DF(y) \cdot T_2),$$

if we split up $T \in \mathbb{R}^{n\times(k+\ell)}$ as $T = (T_1, T_2)$ for $T_1 \in \mathbb{R}^{n\times k}$ and $T_2 \in \mathbb{R}^{n\times \ell}$. Using the wedge product definition on basis forms, we obtain

$$e_I^*(DF(y)T_1) \wedge e_J^*(DF(y)T_2) = \det\begin{pmatrix} e_I^\top DF(y)T_1 \\ e_J^\top DF(y)T_2 \end{pmatrix}.$$

Since $f(F(y))g(F(y))$ is scalar-valued and factors out, we conclude

$$(F^*\alpha \wedge F^*\beta)_y(T) = f(F(y))g(F(y))\, \det\begin{pmatrix} e_I^\top DF(y)T_1 \\ e_J^\top DF(y)T_2 \end{pmatrix} = F^*(\alpha \wedge \beta)_y(T).$$

(2) We write $\alpha = f\, dx_{i_1} \wedge \cdots \wedge dx_{i_k}$. By definition of differential, $d\alpha = df \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k}$, hence by (1) and Lemma 6.41,

$$F^*(d\alpha) = F^*(df) \wedge F^*(dx_{i_1}) \wedge \cdots \wedge F^*(dx_{i_k}) = F^*(df) \wedge dF_{i_1} \wedge \cdots \wedge dF_{i_k}.$$

By the chain rule and since $F^* f = f \circ F$,

$$F^*(df)_y(T) = df_{F(y)}(DF(y) \cdot T) = Df_{F(y)} DF(x) \cdot T = D(f \circ F)_y T = d(F^* f)_y(T).$$

Thereby $F^*(d\alpha) = d(f \circ F) \wedge dF_{i_1} \wedge \cdots \wedge dF_{i_k}$. Meanwhile, splitting up the differential into its rows,

$$(F^*\alpha)_y(A) = \alpha_{F(y)}(DF(y) \cdot A) = (f \circ F\, dF_{i_1} \wedge \cdots \wedge dF_{i_k})_y(A).$$

So we conclude $d(F^*\alpha) = d(f \circ F) \wedge dF_{i_1} \wedge \cdots \wedge dF_{i_k}$ and hence $d(F^*\alpha) = F^*(d\alpha)$. ☐

Now we bring this all back to the motivation and why we started inspecting differential forms in the first place. The key desideratum was invariance under reparametrisation.

Definition 6.43: Orientable submanifold

A k-submanifold $M \subset \mathbb{R}^n$ is called orientable if it is covered by local parametrisations with transition maps that have positive Jacobi determinants. That is, for any two local parametrisations ϕ_1, ϕ_2 that cover different parts of the manifold $M \cap U_1$ and $M \cap U_2$ respectively, then the reparametrisation $\psi = \phi_2^{-1} \circ \phi_1$ for $M \cap U_1 \cap U_2$ has positive Jacobi determinant.

Remark 6.44: Criterion in 3D

In $\mathbb{R}^3$, a 2-submanifold $M \subset \mathbb{R}^3$ is orientable if and only if there exists a normal vector map $\exists \nu : M \to \mathbb{S}^2 \subset \mathbb{R}^3$, $\nu(p) \perp T_pM$ and ν is continuous. See exercise sheet.

Definition 6.45: Integral over parametrised submanifolds using k-forms

Let $V \subset \mathbb{R}^k, U \subset \mathbb{R}^n$ open and $\phi : V \to U$ be a parametrised submanifold $M = \phi(V)$ that is orientable. Let $\alpha \in \Omega^k(U)$. We define

$$\int_M \alpha := \int_V \alpha(D\phi(x))\, dx.$$